Annals of the ICRP

ICRP PUBLICATION 162

Reference Organ Absorbed and Effective Dose Coefficients for Common Radiographic Examinations

Editor-in-Chief
C.H. CLEMENT

Associate Editors
T. YASUMUNE
K. NAKAMURA
Q. FAN

Authors on behalf of ICRP
N. Petoussi-Henss, D.G. Sutton, Y.S Yeom, M. Zankl,
K. Applegate, D.L. Miller, H. Schlattl, J.T.M. Jansen, C. Lee,
J. Li, Y. Lee, J.W. Choi

PUBLISHED FOR

The International Commission on Radiological Protection

by

SAGE

Please cite this issue as 'ICRP, 2025. Reference organ absorbed and effective dose coefficients for common radiographic examinations. ICRP Publication 162. Ann. ICRP 54(1).'

CONTENTS

GUEST EDITORIAL 5

ABSTRACT 7

MAIN POINTS 9

1. INTRODUCTION 11

2. THE ICRP REFERENCE PHANTOMS 19

2.1. Adult reference computational phantoms 19
2.2. Paediatric reference phantoms 22

3. RADIOGRAPHIC PROJECTIONS CONSIDERED 27

3.1. Projection selection – adult radiography 27
3.2. Projection selection – paediatric radiography 28
3.3. Projection definition – adult and paediatric radiography 28

4. DETERMINATION OF DOSE COEFFICIENTS FOR MONOENERGETIC PHOTONS 33

4.1. Monte Carlo photon transport calculation in the anthropomorphic phantoms for monoenergetic values 33
4.2. Dose coefficients for monoenergetic photons at 1-keV intervals – description of tables 42

5. DOSE COEFFICIENTS FOR EXAMPLE X-RAY SPECTRA 43

6. APPLICATION OF DOSE COEFFICIENTS 47

6.1. Using the bremsstrahlung spectral data to estimate organ and effective doses 47
6.2. Using the convolution process for a user-defined spectrum 48
6.3. Possible application of the dose coefficients 51

7. UNCERTAINTIES AND LIMITATIONS 55

7.1. Uncertainties of dose coefficients calculated with Monte Carlo methods 55
7.2. Influence of phantom and field size on the dose coefficients 55

7.3. Uncertainties in dose estimates resulting from applications of the coefficients 57
7.4. Limitations in interpretation 58

8. CONCLUSIONS 61

REFERENCES 63

ANNEX A. DOSIMETRIC QUANTITIES USED IN PATIENT DOSIMETRY IN RADIOGRAPHY 71

A.1. Organ absorbed dose and equivalent dose 71
A.2. Effective dose 72
A.3. Air kerma and associated quantities 74
A.4. References 76

ANNEX B. DEFINITIONS OF FIELDS FOR THE PROJECTIONS CONSIDERED 77

B.1. Adult radiography 78
B.2. Paediatric radiography 89
B.3. References 108

ANNEX C. APPLICATION OF SKELETAL FLUENCE-TO-DOSE RESPONSE FUNCTIONS FOR PHOTONS IN MONTE CARLO RADIATION TRANSPORT CODES 109

C.1. Skeletal dosimetry method of *Publication 116* 109
C.2. Refinements of the original fluence-to-dose response function sets for both adult and paediatric reference phantoms 112
C.3. Implementation of fluence-to-dose response functions and dose enhancement factors to some Monte Carlo codes used in medical physics 113
C.4. References 119

ANNEX D. DESCRIPTION OF ELECTRONIC SUPPLEMENT 123

ACKNOWLEDGEMENTS 125

Guest Editorial

TEXT TO FOLLOW

SAGE ICRP Publication 162 ICRP

REFERENCE ORGAN AND EFFECTIVE DOSE COEFFICIENTS FOR COMMON RADIOGRAPHIC EXAMINATIONS

ICRP PUBLICATION 162

Approved by the Commission in November 2024

Abstract–This publication presents organ absorbed dose and effective dose coefficients for patients resulting from common diagnostic radiographic examinations, calculated using computational voxel phantoms representing the International Commission on Radiological Protection's (ICRP) reference newborn, 1-year-old, 5-year-old, 10-year-old, 15-year-old, and adult male and female. Reference imaging examinations were first defined to cover the most common projections of clinical practice. In this publication, these examinations are limited to those that are clinically consistent with imaging the body morphometry of the reference individuals using digital radiography. Monte Carlo radiation transport simulations for the reference imaging examinations were performed using the radiation transport codes Geant4 (Geometry and Tracking, Version 4) and EGSnrc (Electron Gamma Shower National Research Council of Canada) to report organ absorbed dose and effective dose coefficients for each of the reference computational phantoms and for each of the relevant reference imaging examinations. For quality assurance purposes, and following standard procedures developed for other ICRP publications presenting dose coefficients using reference phantoms, further spot-check calculations were performed. The Monte Carlo calculations were performed for monoenergetic radiation fields in 1-keV photon energy bins from 3 to 150 keV for adults and from 1 to 80, 100, or 120 keV for infants and children, allowing the reconstruction of dose coefficients for spectral beams. From the monoenergetic values, effective and organ absorbed dose coefficients for a variety of x-ray spectra for the defined radiographic projections were computed using a convolution program. These organ absorbed dose coefficients are given as dose normalised to kerma-area product, in mGy Gy^{-1} cm^{-2}, and dose per air kerma free-in-air at 1-m distance from the source, in mGy Gy^{-1}. The effective dose coefficients are given in units of mSv Gy^{-1} cm^{-2} and mSv Gy^{-1}, respectively. The present publication is the first of a series reporting organ absorbed and effective doses for various diagnostic x-ray modalities.

© 2025 ICRP. Published by SAGE.

Keywords: Radiography; Projections; Organ absorbed dose; Effective dose; Monte Carlo; Children; Adults; Optimisation

MAIN POINTS

- **This publication describes the International Commission on Radiological Protection's (ICRP) approach to developing organ absorbed dose coefficients for radiographic imaging for a range of examinations commonly performed on adults and children. This is the first set of such coefficients published by the Commission and, as such, will be of benefit to a wide range of practitioners and national authorities. This publication is the first of a series reporting organ absorbed and effective doses for various diagnostic x-ray modalities.**

- **Monte Carlo radiation transport simulations on the reference voxel computational phantoms of the ICRP male and female newborn, 1-year-old, 5-year-old, 10-year-old, 15-year-old, and adult were performed for the most frequent radiographic examinations, considering good radiographic practice and standard conditions.**

- **Calculations were carried out in 1-keV photon energy bins from 3 to 150 keV for adults, and from 1 to 80, 100, or 120 keV for infants and children, depending on the examination, phantom, and Monte Carlo code used.**

- **Data are provided in 1-keV bins as dose per fluence at 1-m distance from the source. The results can be convolved with any x-ray spectral distribution that a user wishes to choose to obtain spectral dose coefficients.**

- **Dose coefficients are provided for some 'typical' examples of spectral beams and examinations (384 for adults and 496 for infants and children) at 50–130 kV and filtration of 2.5, 3.0, and 3.5 mm Al, and also for 3.0 mm Al + 0.1 mm Cu for paediatric cases. These coefficients are reported as organ absorbed and effective doses per kerma-area product, and per air kerma free-in-air at 1-m distance from the source.**

- **All data can be found in the electronic supplement in the form of Microsoft Excel and ASCII tables as well as viewer outputs. The dose coefficients for the selected bremsstrahlung spectra are also available as part of the ICRP dose viewer, which is an application freely available via the Apple App Store and the Google Play Store.**

1. INTRODUCTION

(1) X-ray imaging procedures make the largest contribution to human exposure to ionising radiation from artificial sources. Conventional radiography (excluding dental radiography) accounts for 63% of all radiological procedures worldwide and 23% of the annual collective effective dose from medical exposures (UNSCEAR, 2022).

(2) All x-ray examinations may carry an associated radiation risk that, although not large, must be considered when patients are imaged. The benefits to the patient undergoing a properly justified and optimised radiological procedure should outweigh the risk resulting from the radiation exposure. In medicine, the requirement is to manage the radiation dose to the patient to be commensurate with the medical purpose. The goal is to use the appropriate dose to obtain the image quality required to achieve the desired medical objective (ICRP, 2023a). Thus, an important component of the justification and optimisation process is the availability of information on the organ and effective doses that patients are receiving, coupled with an understanding of whether these doses are reasonable in the context of the examination being performed (ICRP, 2007b).

(3) One way to obtain information on patient doses is to use metrics that can be easily measured directly, such as kerma-area product. However, there are many occasions in day-to-day practice in which organ or tissue doses are needed. These cannot be measured directly in patients, but they can be derived from other directly measurable quantities using appropriate dose coefficients. A dose coefficient relates a specified dosimetric quantity to some other quantity, such as the normalisation quantity, which can be readily measured or estimated in the clinical situation. For stochastic effects, the specified dosimetric quantities are the organ and tissue doses (ICRU, 2005), and the normalisation quantities are the kerma-area product, air kerma, and fluence at a defined reference plane. Therefore, the availability of prior estimates of dose coefficients based on simulations of the examination employing human computational phantoms (also referred to by many authors as 'models') and computer codes simulating the transport of radiation in different media are useful. Daily decisions for justifying individual patient imaging exposures, or for optimising protection through selecting the most appropriate technique, require approximate estimates of dose that can be used to judge risks to health. Generic values of organ absorbed dose and effective dose coefficients provide a straightforward tool with enough information about general radiation exposure levels linked to detriment for the purpose of making these everyday decisions (ICRP, 2021). One example, as highlighted in *Publication 147* (ICRP, 2021), is that when two different x-ray imaging modalities are considered, the primary factor determining the choice will be the potential benefit to the patient, but comparison of effective or organ doses can be a secondary factor in guiding a referral test selection. Another example would be the inclusion of effective or organ dose data in the evaluation of the impact of

changing a technique factor, such as tube potential, when updating a protocol. Further examples include the evaluation of the effect of added filtration in paediatric radiology, the investigation of the effect of novel filtration on organ absorbed and effective dose, and the use of organ dose data when deciding whether or not to employ a particular radiographic projection.

(4) Unintended exposures of patients in radiographic procedures can be assessed to some extent using effective and organ doses derived from simulations employing numerical methods and patient-specific exposure data. These can provide sufficient information for the incident investigation and report, and inform decisions on requirements for more detailed assessments (ICRP, 2021). A standardised methodology will aid national and international assessments, and comparisons of such doses.

(5) Variations in representative patient doses resulting from differences in equipment, radiographic technique, and other factors existing in different countries can be very instructive in radiation protection practice. However, there is no standardised method for estimating patient doses, yielding potential issues for comparisons. For example, there are differences of up to 25% in some published data, which can be traced to the different phantoms used for the estimation of the dose coefficients (ICRP, 2021). Some of the differing methods for calculation of organ and effective dose coefficients are described in Paras 8–13. A standardised methodology for such dose calculations, as offered in the present publication, will enable even more meaningful comparisons and learning.

(6) Before a research proposal involving the exposure of human beings to ionising radiation is approved, an evaluation of possible detriment to the individuals involved must be made and recorded (ICRP, 1992b). The availability of a standardised approach to the estimation of patient organ and effective doses will ensure that a consistent approach can be taken to dose estimation, bearing in mind that, when considering potential radiation-related risks in research subjects, allowance should be made for age, sex, and health status.

(7) There is a growing demand from patients, families, medical professionals, and relevant regulatory agencies to document patient exposures from diagnostic medical imaging procedures [see, for example, Council of the European Union (2014); IAEA (2018)]. Ethical and professional considerations around accountability, transparency, and patient-centredness emphasise the importance of these requirements (ICRP, 2024a). As a result, various tables and software tools have been developed by various authors, built upon extensive libraries of pre-calculated dose coefficients obtained using computational phantoms and Monte Carlo radiation transport codes.

(8) Coefficients between absorbed or equivalent dose to organs at risk and measurable quantities commonly used in x-ray radiographic diagnosis have been calculated using Monte Carlo methods since the 1980s, mainly in conjunction with mathematical MIRD-type phantoms initially developed by the Medical Internal Radiation Dose (MIRD) Committee of the Society of Nuclear Medicine in the

USA. In the mid-1970s, Rosenstein (1976) published tabulations of organ doses in the MIRD phantom (Snyder et al., 1969) for diagnostic radiography. Since then, compilations of organ and effective dose coefficients for the most common (and some other) examinations have been published using the Cristy and Eckerman (1987) MIRD-type phantoms of adults and children, the latter derived to represent infants and children of various ages, all hermaphrodite, and the sex-specific Adam and Eva adult phantoms (Kramer et al., 1982). For these phantoms, the organ masses and volumes conformed with the ICRP data of the former Reference Man (ICRP, 1975). These dose coefficients relate the quantities of interest to measurable quantities, such as incident air kerma, entrance surface air kerma, and kerma-area product.

(9) Examples of published coefficients relating to radiography include those reported in Jones and Wall (1985), Rosenstein et al. (1979), Rosenstein (1988), Drexler et al. (1985, 1990, 1993), Le Heron (1992), Hart et al. (1994a, 1996a,b), Schultz et al. (1994), and Petoussi-Henss et al. (1995).

(10) Until the mid-1990s, the results of these studies were usually published in reports containing extensive tables with conversion coefficients for the most frequent x-ray examinations. One of the earliest examples of program suites to interrogate and present this data was Xdose and Childdose, developed by Le Heron and based on tabulated coefficients in Reports 262 (Hart et al., 1994a,b) and 279 (Hart et al., 1996a,b) of the National Radiological Protection Board. Later, stand-alone software tools for absorbed dose calculation in x-ray diagnosis began to appear. One of the first was published by Rannikko et al. (1997), who developed a computer program for organ absorbed dose calculations with a size- and sex-adjustable phantom. This method utilised analytical functions based on Monte-Carlo-generated depth dose distributions in water.

(11) In 1998, Servomaa and Tapiovaara (1998) produced the program 'A Monte Carlo program for calculating patient doses in medical x-ray examinations (PCXMC)', which is distributed by the Radiation and Nuclear Safety Authority in Finland (Tapiovaara and Siiskonen, 2008). PCXMC is a Monte Carlo computer code that uses the hermaphrodite adult and paediatric MIRD-type phantoms of Cristy and Eckerman (1987), and allows for application of scaling factors to modify the size of the phantoms as required. PCXMC functionality includes the ability to calculate organ and tissue absorbed doses, effective dose as per the *Publication 103* (ICRP, 2007b) regimen, and cancer risk using the Seventh Committee on Biological Effects of Ionizing Radiation (BEIR VII) model (Committee to Assess Health Risks from Exposure to Low Levels of Ionizing Radiation, 2006). PCXMC does not use pre-calculated data but carries out a Monte Carlo calculation for the exposure conditions defined by the user. Projections are specified by the user, but could be pre-defined for batch or comparative purposes if required. Further citations on dose coefficients in conventional radiography can be found in Report 74 of the International Commission on Radiation Units and Measurements (ICRU) (2005) and in Seidenbusch et al. (2019).

(12) The development of tomographic or voxel-based phantoms began in the 1980s in order to address issues raised by the anatomical limitations of the mathematical MIRD-type phantoms, and several phantoms have been proposed over time. Studies employing voxel phantoms for Monte Carlo calculations to determine organ doses in radiography include Veit and Zankl (1992, 1993), Zankl (1993), Petoussi-Henss et al. (1995, 2002), and Winslow et al. (2004).

(13) In 2008, Kramer et al. developed a Monte-Carlo-based software tool called 'CALDose X' (Kramer et al., 2008) that contains within its functionality the ability to calculate organ and tissue absorbed doses using the voxel phantoms FAX06 and MAX06 (Kramer et al., 2006), which have standard (i.e. reference) sizes as described in *Publication 89* (ICRP, 2002). The functionality of the tool also included the assessment of cancer risk based on the factors presented in *Publication 103* (ICRP, 2007b) and BEIR VII model (Committee to Assess Health Risks from Exposure to Low Levels of Ionizing Radiation, 2006). CALDose_X 5.0 is the current version of the software program for calculating organ absorbed doses and radiological risks using the *Publication 89*-based mesh phantoms FASH and MASH (Cassola et al., 2010; Kramer et al., 2010). The online tool allows the user to select one of 10 phantoms, four adult and six paediatric, on which the calculation can be based. The adult phantoms are further subdivided to allow age and body mass index to be selected.

(14) For many years, ICRP has produced reference dose coefficients for common diagnostic nuclear medicine procedures (ICRP, 1988, 1992a, 1998, 2008, 2015b). However, to date, ICRP has not provided reference dose coefficients for x-ray imaging procedures, and consequently different methodologies and phantoms are used to convert measurements to estimates of effective dose or some surrogate of effective dose. These calculations necessarily rely on disparate published data based on the use of older stylised hermaphrodite phantoms that are not in alignment with the most recent ICRP reference phantoms. In addition, different computational methods for radiation transport have been used to report organ absorbed and equivalent doses from which effective dose is computed. The availability of ICRP reference organ and effective dose coefficients will, therefore, facilitate the work of national and international authorities responsible for such calculations.

(15) Organ doses from radiographic procedures can be assessed using Monte Carlo simulations and computational phantoms in the form of dose coefficients that relate measured kerma-area product or air kerma to organ or effective dose. As stated above, most of the existing lists of organ doses and relevant software are based on stylised, mathematical MIRD-type phantoms. In the last few years, dose coefficients using voxel computational phantoms have become available (ICRP, 2010, 2013, 2015a, 2016a,b, 2017, 2019, 2020b, 2022, 2023b). The voxel phantoms, constructed using high-resolution medical images, present a significant advance towards more realistic representation of the human body. Previous studies have shown that, for some examinations, differences in dose coefficients do occur between the different types of phantoms (voxel or MIRD), and can be attributed to the

different amount of shielding of an organ, depending on the position and mutual overlapping of these organs (Petoussi-Henss et al., 2002; Zankl et al., 2002; Bolch et al., 2010; Zankl, 2010; Lee, 2021). Individual dose differences for voxel phantoms are mainly due to the dependence of x-ray absorption on patient diameter, but may also be due to in-field organ size, shape, and positioning. Computations of dose coefficients for routine examinations have shown significant differences for individual organs between different voxel phantoms, but also between the ICRP reference computational phantoms and the MIRD-type reference phantoms (Petoussi-Henss et al., 2006).

(16) This publication describes the ICRP approach to developing organ absorbed dose coefficients for radiographic imaging. The coefficients have been produced for the adult male and female ICRP reference voxel phantoms (ICRP, 2009) and the suite of ICRP reference paediatric voxel phantoms (ICRP, 2020a). The phantoms are described briefly in Section 2 of this publication.

(17) Radiographic examinations for which coefficients have been developed were chosen based on their frequency as described by the United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR) (2010) and the National Council on Radiation Protection and Measurements (NCRP) (2019). The radiographic examinations selected for paediatric radiography are those that are either commonly performed examinations, as for adult radiography, or those that result in high exposures or exposures to the more radiosensitive organs. For each examination, relevant projections were selected. Dental and mammographic examinations were not considered for either adults or paediatrics. The methodology for examination selection and projection definition is described in Section 3.

(18) The detail of the Monte Carlo photon transport calculations used to determine organ and effective dose coefficients is presented in Section 4. Two sets of coefficients are provided for each projection. In the first set, the data are provided in 1-keV energy bins. The coefficients given in each of the monoenergetic energy bins are given as absorbed dose normalised to fluence at a reference plane defined 1 m from the source. Note that the fluence at the reference plane is always estimated in the absence of the phantom. Organ and effective dose coefficients for any user-defined x-ray spectra can be generated using a simple convolution. In the context of this work, the term 'convolution' refers to element-wise integration of dose per fluence with energy fluence. This first set is for those users who wish to tailor the results to any given beam quality (i.e. allowing for any combination of applied tube potential and filtration).

(19) For those users who do not need such fine tuning, organ absorbed dose coefficients are provided for selected bremsstrahlung spectra between 50 and 120 kV and total filtration between 2.5 and 3.5 mm Al, as well as an additional 0.1 mm Cu in the case of paediatric projections. The dose coefficients are normalised to kerma-area product (Gy cm^2), air kerma free-in-air at 1 m (Gy), and fluence at 1 m (cm^{-2}) from the x-ray tube. A description of the method used to obtain these dose coefficients, including the convolution method, can be found in Section 5.

(20) Effective dose, E, is defined in *Publication 103* (ICRP, 2007b) using the equation:

$$E = \sum w_T \left[\frac{H_T^M + H_T^F}{2} \right] \tag{1.1}$$

where H_T^M and H_T^F are the equivalent doses to the tissues or organs T of the reference male and female, respectively, and w_T is the tissue weighting factor for target tissue T, with $\sum w_T = 1$. Effective dose is thus a sex averaged quantity. This definition is valid for all ICRP reference ages. For more detail on effective dose, the reader is referred to Annex A and *Publication 103* (ICRP, 2007b). The present publication provides organ absorbed dose coefficients for both male and female phantoms. It also provides, for each examination, a value of the coefficient for either $\sum w_T \times H_T^M$ or $\sum w_T \times H_T^F$, depending on whether the examination is performed on the male or female phantom. The effective dose coefficient for an examination can be obtained by taking the average of the two. For ease of reproduction and to avoid possible confusion, the two quantities are given the nomenclature $E_{103,\,\#\#M}$ or $E_{103,\,\#\#F}$ where ## refers to the age of the phantom being used, and M or F signifies male or female. For example, $E_{103,\,AM}$ refers to the coefficient of the quantity $\sum w_T \times H_T^M$ resulting from an examination on an adult male phantom, and $E_{103,\,01F}$ refers to the coefficient of the quantity $\sum w_T \times H_T^F$ resulting from an examination on a 1-year-old female phantom.

(21) *Publication 147* (ICRP, 2021) provides additional guidance on the use of effective dose to estimate risks, particularly in evaluating exposures of patients from medical procedures. In medical applications, estimates of effective dose can be used for comparing doses from different diagnostic and interventional imaging modalities [e.g. computed tomography (CT) and nuclear medicine procedures], and exposure techniques that give different spatial distributions of radiation within the body tissues. In this context, effective dose is used to provide a generic indicator for classifying different types of medical procedure into broad risk categories for the purpose of communicating risks to clinicians and patients. Effective dose is also used to inform decisions on justification and optimisation of patient diagnostic and interventional procedures, planning requirements in research studies, and evaluation of unintended exposures. In each of these cases, effective dose provides an approximate indicator of possible risk. Thus, effective dose can be used prospectively as an indicator of radiation detriment in justification decisions, and when planning medical research studies involving radiation exposure, or retrospectively in assessments of accidental exposures.

(22) The intended use of the dose coefficients provided in this publication is to provide data that can be used when addressing issues such as those discussed in Paras 2–7 above. Potential uses include population-based dosimetry, justification and optimisation in radiography, easier comparisons of doses at national and

international levels, and the provision of information to aid in clinical decision making and communication. The dose coefficients are not intended to be, and should not be, used for individual patient dosimetry. The target audience for this publication includes all those who have an interest in patient dosimetry and optimisation, including national authorities responsible for dose calculations, medical physicists, research scientists, radiologists, radiographers, and other radiological practitioners. This publication is the first of a series reporting organ absorbed and effective dose coefficients for a range of diagnostic x-ray modalities including radiography, CT, fluoroscopically guided interventional radiology, and diagnostic paediatric fluoroscopy. The work will also be extended to consider the case of pregnant and non-reference patients.

(23) All the coefficients generated are provided in the electronic annex to this publication. Included in the electronic annex are two excel-based viewers that enable users to access the mono-energetic organ dose data and also the data for the selected bremsstrahlung spectra. A separate external dose viewer is available that allows users to quickly access the results for the selected bremsstrahlung spectra using Apple or Android mobile devices.

2. THE ICRP REFERENCE PHANTOMS

2.1. Adult reference computational phantoms

(24) Computational phantoms of the human body, together with radiation transport codes, have been employed for many years in the evaluation of organ dose coefficients in occupational, environmental, and medical radiation protection. Over the last two decades, voxel phantoms that are derived mostly from (whole-body) medical image data of real people instead of the older stylised computational body phantoms have been introduced. A voxel (volume element) is a rectangular cuboid specifying the organ or tissue – each organ/tissue consists of a large number of voxels. A voxel phantom (or model) is a three-dimensional representation of the human body in the form of an array of identification numbers, arranged in slices, rows, and columns. Each entry in this array represents a tissue voxel; organs are then represented by those voxels having the same identification number, and are arranged spatially to represent the organ volume. More information on voxel phantoms, and their development and use can be found elsewhere (Xu and Eckerman, 2010).

(25) For the computation of organ absorbed doses, the adult male and female reference computational phantoms, representing the ICRP reference adult male and reference adult female, are used in this publication. These phantoms were adopted by ICRP and ICRU as the phantoms for the computation of the ICRP reference dose coefficients, and are described extensively in *Publication 110* (ICRP, 2009). The reference computational phantoms are based on human CT data, and were constructed by modifying the voxel phantoms (Zankl and Wittmann, 2001; Zankl et al., 2005) of two individuals (Golem and Laura) whose body height and mass closely resembled the reference data given in *Publication 89* (ICRP, 2002) on the reference male and reference female. The reference values for standing height and body mass of the ICRP reference individuals can be seen in Table 2.1. The organ masses of both phantoms were adjusted to the reference data with high precision, without altering their realistic anatomy significantly. The phantoms contain all target regions relevant to the assessment of human exposure to ionising radiation for radiological protection purposes, including all tissues and organs that contribute to the protection quantity 'effective dose' (ICRP, 2007a).

Table 2.1. Reference values for standing height and body mass for the International Commission on Radiological Protection reference individuals (ICRP, 2002).

	Height (cm)		Mass (kg)	
	Male	Female	Male	Female
Newborn	51	51	3.5	3.5
1-year-old	76	76	10	10
5-year-old	109	109	19	19
10-year-old	138	138	32	32
15-year-old	167	161	56	53
Adult	176	163	73	60

(26) The male reference computational phantom consists of approximately 1.95 million tissue voxels (excluding voxels representing the surrounding vacuum), each with a slice thickness (corresponding to the voxel height) of 8.0 mm, and an in-plane resolution (i.e. voxel width and depth) of 2.137 mm, corresponding to a voxel volume of 36.54 mm^3. The number of slices is 220, resulting in a body height of 1.76 m and total body mass of 73 kg. The female reference computational phantom consists of approximately 3.89 million tissue voxels, each with a slice thickness of 4.84 mm and an in-plane resolution of 1.775 mm, corresponding to a voxel volume of 15.25 mm^3. The number of slices is 346, and thus the body height is 1.63 m and the total body mass is 60 kg. Note, that for both phantoms, 2 additionnal slices of skin at the top and bottom were added (ICRP, 2009). The number of individually segmented structures is 136 in each phantom, and 53 different tissue compositions have been assigned to them. The various tissue compositions reflect both the elemental composition of the tissue parenchyma (ICRU, 1992) and each organ's blood content (ICRP, 2002) (i.e. organ composition inclusive of blood). Fig. 2.1 shows frontal (coronal) views of the male (left) and female (right) computational phantoms.

(27) Due to the limited resolution of the source tomographic data upon which these phantoms were constructed, and the very small dimensions of some of the ICRP-defined source and target regions, not all tissues are represented explicitly. In the skeleton, for example, the target tissues of interest are the haematopoietically active bone marrow located within the marrow cavities of spongiosa, as well as the endosteal layer lining the surfaces of the bone trabeculae and the inner surfaces of the medullary cavities of the long bones (currently assumed to be 50 μm in thickness). Due to their small dimensions, these two target tissues are incorporated as homogeneous constituents of spongiosa within the reference phantoms. At lower energies of photons and neutrons, secondary charged-particle equilibrium is not fully established in these tissue regions over certain energy ranges. Consequently, more refined techniques for accounting for these effects in skeletal dosimetry are used in this publication. A detailed description of the skeletal dosimetry can be found in *Publication 116* (ICRP, 2010) and in Annex C of this publication.

(28) As the ICRP reference computational phantoms, both adult and paediatric (as described in the next section), have been constructed from CT data of real patients, they represent the reference patient in a supine position. That means that the abdomen is flatter than in a standing person, the abdominal organs are shifted upwards towards the chest, the lungs are compressed, and the curvature of the spine is slightly different compared with that of a standing person (ICRP, 2009). For the purposes of this reference modelling, an assumption must be made that the anatomy and organ location of these phantoms do not change when they are assumed to be in an upright position, such as during a chest postero-anterior examination. A study on the effects of posture on organ dose coefficients from some typical idealised external exposures for a person in a supine position and an upright position showed agreement within 2–20% for photon energies >50 keV (Sato et al., 2008). For energies <50 keV, some organs could exhibit larger differences up to a factor of

approximately 2, depending on the projection geometry (i.e. antero-posterior, postero-anterior, lateral, etc.) and the potential shift of the organ in question when the posture changes. For example, the absorbed dose to the liver and stomach for an idealised lateral projection will be higher for a phantom which is constructed from data of a person in an upright position than for a phantom constructed from data obtained in a supine position. For the brain and oesophagus, for example, these variations are mostly <10%, even at low energies. However, effective dose was found to be less sensitive to posture for all energies. Thus, it can be concluded that the dosimetric impact of the person's position is limited, and the organ position of the voxel phantoms in a supine position is acceptable for the applications intended.



Fig. 2.1. Images of the adult male (left) and adult female (right) voxel reference computational phantoms (ICRP, 2009). The following organs can be identified by different surface colours: breast, colon, eyes, lungs, liver, pancreas, salivary glands, small intestine, stomach, thyroid, urinary bladder, and teeth. Muscle and adipose tissue are semi-transparent.

2.2. Paediatric reference phantoms

(29) The series of 10 ICRP paediatric computational phantoms, described in *Publication 143* (ICRP, 2020a), are as follows:

- Newborn – male and female;
- 1-year-old – male and female;
- 5-year-old – male and female;
- 10-year-old – male and female; and
- 15-year-old – male and female.

(30) These phantoms were derived from a series of computational phantoms developed originally at the University of Florida (UF) and later in collaboration with the National Cancer Institute (NCI). Consequently, the original phantoms from which the ICRP paediatric reference phantoms were derived are presently referred to as the 'UF/NCI phantom series' (Lee et al., 2010). The UF/NCI phantoms are of a third generation of phantom technology – hybrid phantoms – in which the outer body contour and internal organ surfaces are modelled using the computer animation techniques of either polygon mesh or NURBS (non-uniform rational B-spline) surfaces depending on the complexity of anatomical structures. The polygon meshes are a cluster of adjacent triangles, while the NURBS surfaces are a cluster of three-dimensional points in space between which a surface is interpolated. Over the past few years, it has become possible to use computational phantoms in these two formats directly in some Monte Carlo transport codes as necessary. However, most transport codes still utilise a voxel format, composed of tiny rectangular cuboids. A computer script was therefore used to convert the UF/NCI hybrid phantoms from their surface format to a voxel format for Monte Carlo simulations. These ICRP paediatric reference phantoms in voxel format (ICRP, 2020a) are thus consistent with the format of the *Publication 110* adult reference phantoms (ICRP, 2009).

(31) As noted in Lee et al. (2010), the UF/NCI series of phantoms are traceable directly to real human anatomy. The newborn phantoms are based upon full-body CT imaging of a 6-day-old female cadaver, while the remainder of the paediatric series (1-year-old to 15-year-old phantoms) are based upon combinations of paediatric head and full torso CT images, and rescaled CT-based images of adult arms and legs. The latter approach was necessary as medical imaging of children rarely includes the arms within the imaging field. For CT sources employed in the development of these phantoms, the reader is referred to Table 3.1 of *Publication 143* (ICRP, 2020a). From the initial series of segmented images, various anatomic sources were used to resize both internal organ anatomy and exterior body size. The most important document used was *Publication 89* (ICRP, 2002), providing internal organ masses, and values of total mass and height (see Table 2.1). Additional reference sources were used to target various body circumferential dimensions not given as reference values in *Publication 89*. The final series of the UF/NCI hybrid phantoms

thus fully conform to reference anatomy specified by the Commission, and are fully traceable to real human CT anatomy. In this manner, the ICRP paediatric reference phantom series are fully compatible with the process used to develop the *Publication 110* (ICRP, 2009) phantoms, which were also based upon segmentation of real human CT anatomy.

(32) Another unique feature of the ICRP paediatric reference phantoms (and of the UF/NCI phantoms) is their explicit coupling to micro-CT-based models of skeletal dosimetry. As noted in Hough et al. (2011) and Johnson et al. (2011), an extensive series of cadaver bone harvests, ex-vivo skeletal CT imaging, and ex-vivo spongiosa core micro-CT imaging were used to construct models of tissue dosimetry in the skeletons of the ICRP adult reference phantoms. This work is described more formally in Annexes D and E of *Publication 116* (ICRP, 2010). The paediatric series of ICRP reference phantoms similarly have accompanying models of skeletal anatomy at both its macrostructural and microstructural dimensions. Thus, the methods proposed in *Publication 116* for external photons and neutrons, and in *Publication 133* (ICRP, 2016a) for internal beta and alpha particles, as well as photons, for the adult reference phantoms are available in reporting skeletal tissue dosimetry to paediatric reference phantoms (ICRP, 2023b).

(33) In the ICRP implementation (ICRP, 2020a), the following further refinements have been made to the UF/NCI series of paediatric phantoms (Pafundi, 2009; Wayson, 2012):

- a sub-segmented skeletal model to include regions of cortical bone, spongiosa, and medullary marrow;
- photon dose response functions (DRFs) for internal and external photon dosimetry to active marrow and endosteum;
- new age-specific regional blood distribution model (Wayson, 2012);
- corresponding model of the major blood vessels;
- separation of subcutaneous fat and skeletal muscle from what was formerly residual soft tissues; and
- inclusion of lymph nodes [see Lee et al. (2013)].

(34) The series of ICRP paediatric reference phantoms are in voxel format and fully conform to the framework established in *Publication 110* (ICRP, 2009). All organs and tissue structures modelled in the *Publication 110* reference adult male and female phantoms are included in the series of ICRP paediatric reference phantoms with consistent identification (ID) numbers (see Annex A of *Publication 110*). Representative images of the ICRP reference paediatric series are given in Fig. 2.2.

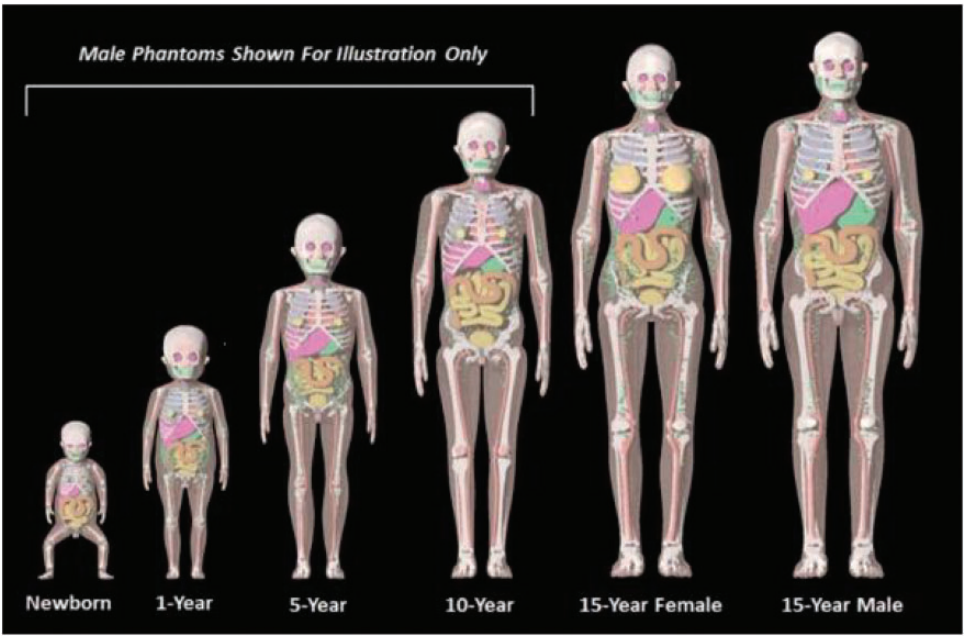


Fig. 2.2. Series of International Commission on Radiological Protection paediatric reference phantoms. The male and female newborn, 1-, 5-, and 10-year-old phantoms are anatomically identical, except for their gonads and urinary bladder (ICRP, 2020).

(35) While the ICRP paediatric reference phantoms are identical in format to the *Publication 110* (ICRP, 2009) adult reference phantoms regarding the ID numbers of the various source and target organs, one important difference is the voxel resolution. One of the main advantages of hybrid phantom technology is that one can select the voxel resolution in the conversion of the polygon mesh/NURBS format of the phantom to the voxel format of that same anatomy. Table 2.2 tabulates the voxel resolutions, array size, and total matrix size finally adopted for the ICRP paediatric reference phantoms. The phantom dimensions ensure continuous conformance with the 1% matching of reference masses, as well as conformance to reference total skin thickness as given by data in *Publication 89* (ICRP, 2002). It is noted that for the newborn phantom, the voxels are cubic (i.e. same dimensions in x, y, and z directions), while rectangular prisms with larger z dimensions than x and y dimensions were adopted for the older phantoms in order to keep the matrix size constant at approximately 55 million voxels in total. In contrast, the *Publication 110* adult male and female phantoms have total matrix sizes of 1.9 and 3.9 million voxels, respectively. The need for higher resolution is to preserve organ anatomy in the smaller anatomy of the paediatric reference individuals.

Table 2.2. Voxel resolution, voxel number, and total matrix size of the International Commission on Radiological Protection paediatric reference phantom series.

Phantom	Resolution (mm)			Array size			Matrix size (million)
	x	y	z	x	y	z	
Newborn, male and female	0.663	0.663	0.663	345	211	716	52.1
1-year-old, male and female	0.663	0.663	1.400	393	248	546	53.2
5-year-old, male and female	0.850	0.850	1.928	419	230	572	55.1
10-year-old, male and female	0.990	0.990	2.425	419	226	576	54.5
15-year-old, male	1.250	1.250	2.832	407	225	586	53.7
15-year-old, female	1.200	1.100	2.828	401	236	571	54.0

3. RADIOGRAPHIC PROJECTIONS CONSIDERED

3.1. Projection selection – adult radiography

(36) The radiographic projections used for adult radiography were selected on the basis of annual examination frequency in the UK (alone), Europe (including UK), the USA, and UNSCEAR Health Care Level 1 countries (Hart et al., 2010; UNSCEAR, 2010; European Commission, 2015; NCRP, 2019). The six most common types of adult examination were chosen, namely chest/thorax, pelvis and hip, abdomen, lumbar spine, cervical spine, and thoracic spine. Table 3.1 shows their frequency expressed in terms of examinations per 1000 population per year. Exposures involving extremities were not considered because of their low contribution to effective dose. The exact projections modelled for each of the examination types are shown in Table 3.2.

Table 3.1. Frequency of radiographic examinations per 1000 adult population per year (in decreasing order).

Examination type	UK[*]	European average[†]	USA[+]	UNSCEAR HCL 1[§]
Chest/thorax	146.7	194	341.8	168
Pelvis and hip	39.0	48.7	63.2	40
Abdomen	20.1	22.5	37.9	45
Lumbar spine (including LSJ)	14.9	33.6	34.8	31
Cervical spine	9.3	16.9	15.1	32
Thoracic spine	4.4	9.8	7.8	16

UNSCEAR, United Nations Scientific Committee on the Effects of Atomic Radiation; HCL, health care level; LSJ, lumbosacral joint.
[*]Hart et al. (2010).
[†]European Commission (2015).
[+]NCRP (2019).
[§]UNSCEAR (2010).

Table 3.2. Adult projections modelled.

Examination type	Projections modelled
Chest/thorax	PA, AP, LLat, RLat
Pelvis and hip	AP
Abdomen	AP
Lumbar spine (including LSJ)	AP, LLat, RLat, RObl
Cervical spine	AP, LLat, RLat
Thoracic spine	AP, RLat, LLat

PA, postero-anterior; AP, antero-posterior; RObl, right oblique; RLat, right lateral; LLat, left lateral; LSJ, lumbosacral joint.

3.2. Projection selection – paediatric radiography

(37) The radiographic projections selected for paediatric radiography were those that were either commonly performed examinations (Dorfman et al., 2011; European Commission, 2018) similar to adult radiography, or those that resulted in high exposures or exposures to radiosensitive organs. Therefore, skull and scoliosis full-spine radiographs are included because they are both common in infants and children and they expose these patients to higher doses than typical single body part examinations. In practice, the number of projections for each examination may vary with the age of the infant/child as well as geographic location, available resources, and clinical indication, as with adult radiography. The exact projections modelled are shown in Table 3.3.

Table 3.3. Paediatric projections modelled.

Examination type	Projections modelled
Chest	PA (5-, 10-, 15-year-old), AP
Pelvis	AP
Abdomen	AP
Lumbar spine	AP, LLat, RLat
Thoracolumbar spine (newborn, 1-year-old alone)	RLat, LLat
Scoliosis	AP (newborn, 1-year-old alone), PA (all other ages), RLat, LLat
Skull	AP, PA, RLat, LLat

PA, postero-anterior; AP, antero-posterior; RLat, right lateral; LLat, left lateral.

3.3. Projection definition – adult and paediatric radiography

(38) Whilst it is acknowledged that there is no 'definitive projection' for any examination, all projections modelled were considered to be examples of a part of the process of optimisation, or good practice and technique. The projections used in the simulations were defined using an iterative process as shown in Fig. 3.1. Typical field sizes and positioning for each projection were obtained initially from a standard textbook of radiography, intended to teach student radiographers the proper way to obtain radiographs (Whitley et al., 2015). The information was then used, in conjunction with expert clinical input, to define initial positioning of the irradiation field on the phantom.

(39) For defining the field borders with respect to the phantoms, information on the rectangular cuboids containing the organs under consideration was used (i.e. the minimum and maximum columns, rows, and slabs occupied by the respective organs, together with the voxel dimensions). If specified in the geometry description, additional margins were added. For example, in the abdominal radiograph, for a position '1–2 cm above dome of diaphragm', 1.5 cm was added to the topmost

coordinate of the liver. In this way, the minimum and maximum z coordinates to be covered by the field were evaluated, and thus the field height and centre, where the latter marks the z coordinate of the radiation source. Next, the maximum extensions of the slab were assessed where the source z coordinate is located. For antero-posterior and postero-anterior projections, the minimum and maximum rows of that slice were assessed, and for lateral projections, the minimum and maximum columns were assessed. These mark the entrance and exit points of the central beam with respect to the body. If use of an anti-scatter grid was assumed, the image receptor plane was assumed to be 10 cm away from the beam exit point; if no grid was to be considered, the exit point marked the location of the image receptor plane. The source coordinate was then assumed at the focus-to-receptor distance on the opposite side of the body. For the lateral field margins, the same method was applied as for the superior and inferior margins (i.e. use of the rectangular cuboid containing the organs considered and adding margins if needed). Thus, the field width and centre were assessed, where the centre marks the third source coordinate. The field size determined was assumed to be at the centre of the phantom, in a plane between the beam entrance and exit points into and from the body. Similarly, the irradiation fields shown on the phantoms in Annex B are at the centre of the phantoms.

(40) The position of each projection was assessed by two experienced radiologists, and positioning and field definition were amended until there was agreement between them that both positioning and field definition were representative of good technique. Whilst the radiologists had a good understanding of what the radiographic image should contain, mapping that image to the mid-coronal plane of the phantom was sometimes difficult, and was responsible for most of the iterations in the amendment phase depicted in Fig. 3.1.

(41) For graphical representation of the radiation fields and full details of their sizes and positioning, as well as anatomical landmarks, see Annex B.

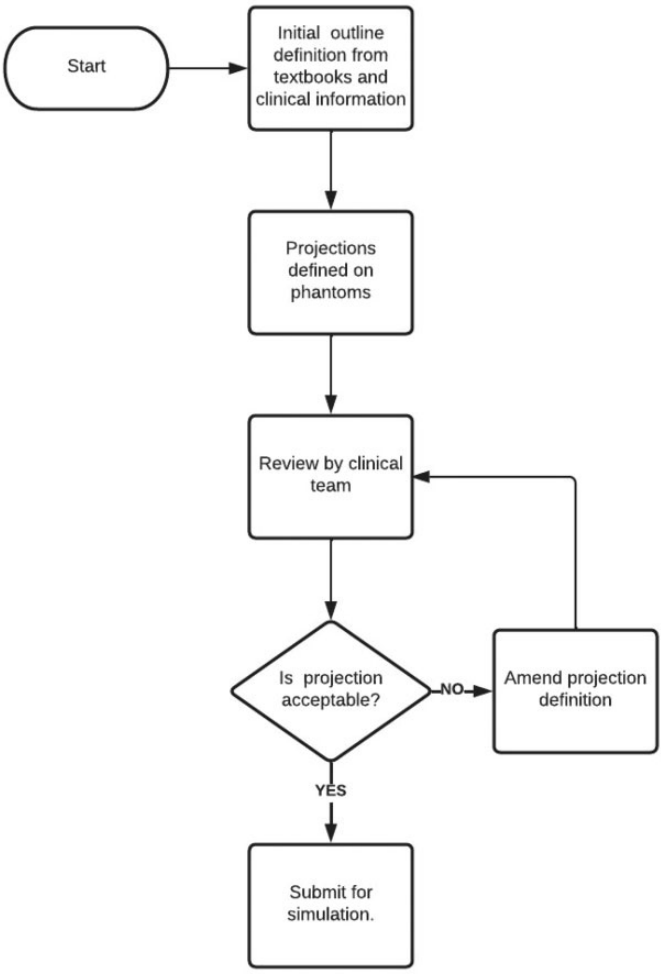


Fig. 3.1. Flow diagram showing the process involved in the definition of each radiographic projection.

(42) To obtain the dose coefficients for this publication, calculations were carried out assuming irradiation of the phantoms placed in a vacuum by point source, collimated beams. It should be noted that no collimators are simulated: the 'collimation' is completely idealised, and allows only photons coming from the isotropic point source that hit the phantom inside the rectangular field borders assumed to represent the radiographic exposures. The implication of this assumption is considered in Section 7.3. As an example, Fig. 3.2 shows a schematic representation of the exposure for a postero-anterior chest examination of the reference adult male. Details of the Monte Carlo calculation methodology can be found in Section 4.

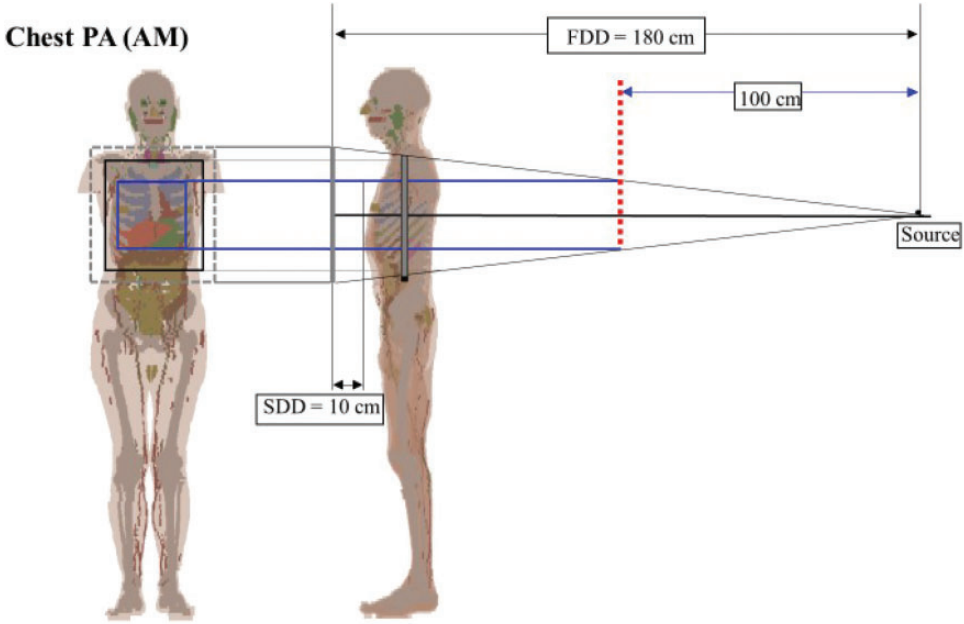


Fig. 3.2. Radiographic exposure of the chest of the adult male (AM) reference phantom from the postero-anterior (PA) direction, indicating relevant quantities. Front view: the larger rectangle (in grey, dotted) represents the field size at the detector plane, the smaller rectangle (in blue) represents the field size at the reference plane, and the mid-sized rectangle (in black, solid) represents the field size at the centre of the phantom. Lateral view: the reference plane at 100 cm from the source (red, dotted line) and the position in the centre of the phantom at which the field is defined (see Annex B) are shown. FDD, focus-to-detector distance; SDD, skin-to-detector distance.

4. DETERMINATION OF DOSE COEFFICIENTS FOR MONOENERGETIC PHOTONS

(43) A 'dose (conversion) coefficient', c, relates a dosimetric quantity to some other quantity (i.e. the normalisation quantity) which can be readily measured or calculated in the clinical situation. For radiography of adults and children, the specified dosimetric quantities are the organ absorbed dose, D_T, and effective dose, E. The air kerma-area product, KAP (P_{KA}), air kerma free-in-air at a specified distance, K_a, and incident air kerma, $K_{a,i,}$ are used as normalisation quantities. See Annex A for definitions of each of these normalisation dose metric quantities.

4.1. Monte Carlo photon transport calculation in the anthropomorphic phantoms for monoenergetic values

(44) Table 4.1 shows the examinations that were simulated. For each of these examinations, calculations involved the computation of organ absorbed dose coefficients in each sex- and age-specific phantom. Particle transport calculations were performed with the EGSnrc code for adults and Geant4 for infants and children. For validation purposes, spot-check calculations were performed with EGSnrc for a selection of paediatric cases and with Geant4 for a selection of adult reference phantoms and exposures. Table 4.2 summarises the details of the methodology used for the Monte Carlo calculations of the dose coefficients.

(45) During radiographic examinations, the arms of the patient are either alongside the body (causing attenuation and scattering of the x-ray beam), above the head, or positioned in a way that enhances the image (e.g. for a chest postero-anterior examination, the hands are placed at the patient's side at waist level, usually with palms out, which results in flexion at the elbow and moves the arms away from the chest). Newborn and 1-year-old patients cannot cooperate; thus for these two ages, the arms are only kept out of the field for lateral spine examinations, as they would obscure the structures being examined.

(46) The ICRP reference phantoms – adult and paediatric – have their arms alongside the body. These are not easily movable, but for simulations of the projections when the arms are outside the beam, they were removed by replacing the arm tissues simply with a vacuum, whereas the shoulder tissues were left in (see Fig. 4.1). The involved tissues of the arms are skin, adipose tissue, muscle, lymph nodes, blood, cartilage, and bone.

(47) Table 4.1 also shows whether the simulations were performed with or without the arms of the phantoms. It should be noted that for calculating the respective dose coefficients, the original masses of all organs of the phantoms were used (i.e. inclusive of arms). This is to allow for the fact that the arms of the phantom, although removed for the simulations, are there in reality, and the risk, for example, of the skin is associated with the tissue weighting factor (w_T) which corresponds to the whole organ (i.e. the skin over the whole body).

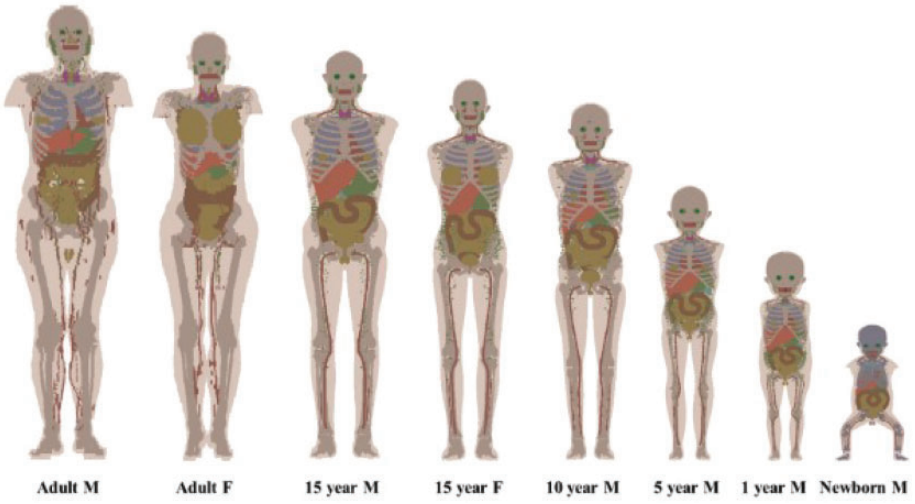


Fig. 4.1. Series of the adult and paediatric reference voxel phantoms with their arms removed. Male phantoms alone are shown for the newborn, 1-, 5-, and 10-year-old. M, male; F, female.

Table 4.1. List of the simulated examinations, with the indication if these were performed with or without the arms of the phantoms.

Examination	Newborn	1-year-old	5-year-old	10-year-old	15-year-old	Adult
Chest PA	N/S	N/S	Without	Without	Without	Without
Chest AP	With	With	With	With	With	With
Chest Lat	N/S	N/S	N/S	N/S	N/S	Without
Pelvis AP	With	With	Without	Without	Without	Without
Abdomen AP	With	With	Without	Without	Without	Without
Lumbar spine AP	With	With	With	With	With	With
Lumbar spine Lat	Without	Without	Without	Without	Without	Without
Lumbar spine Obl	N/S	N/S	N/S	N/S	N/S	Without
Cervical spine AP	N/S	N/S	N/S	N/S	N/S	With
Cervical spine Lat	N/S	N/S	N/S	N/S	N/S	Without
Thoracic spine AP	N/S	N/S	N/S	N/S	N/S	With
Thoracic spine Lat	N/S	N/S	N/S	N/S	N/S	Without
Thoracolumbar spine Lat	Without	Without	N/S	N/S	N/S	N/S
Scoliosis AP	With	With	N/S	N/S	N/S	N/S
Scoliosis PA	N/S	N/S	Without	Without	Without	N/S
Scoliosis Lat	Without	Without	Without	Without	Without	N/S
Skull AP	With	With	With	With	With	N/S
Skull PA	With	With	With	With	With	N/S
Skull Lat	With	With	With	With	With	N/S

PA, postero-anterior; AP, antero-posterior; Lat, lateral (right and left); Obl, oblique; N/S, not simulated.

Table 4.2. Details of the methodology used for the Monte Carlo calculations of the dose coefficients.

Radiation transport	For the adult reference phantoms: EGSnrc (Kawrakow et al., 2009) For the paediatric reference phantoms: Geant4 (Agostinelli et al., 2003), Version 10.7 (Patch 03)
Cross-section library	EGSnrc: XCOM database (Berger and Hubbell, 1987) Geant4: Livermore data library (Perkins et al., 1991b; Cullen et al., 1997)
Particles considered	Photons and secondary electrons
Energy range considered	Adult: 3–150 keV Paediatric: 1–120 keV Calculations were performed for 1-keV bins
Source	Idealised, isotropic point, collimated beams Collimation is idealised Phantoms placed in vacuum
Anthropomorphic phantoms	ICRP reference voxel phantoms, adult (ICRP, 2009) and newborn, 1-, 5-, 10-, and 15-year-old (ICRP, 2020a) Note: newborn, 1-, 5-, and 10-year-old male and female phantoms have identical anatomy, besides the gonads and the urinary bladder
Position of arms of phantoms	See Table 4.1
Projections (examinations) simulated	See Tables 3.2 and 3.3; most frequent radiographic examinations, good radiographic practice, standard conditions
Skeletal dosimetry	Use of dose response functions (ICRP, 2010, 2023b). See Annex C
Organs and tissues for which dose coefficients are given	Red bone marrow, colon, lung, stomach, breast, ovaries or testes (gonads), bladder, oesophagus, liver, thyroid, skeletal endosteum, brain, salivary glands, skin, and remainder tissues (i.e. adrenals, extrathoracic region, gallbladder, heart, kidneys, lymphatic nodes, muscle, oral mucosa, pancreas, prostate, small intestine, spleen, thymus, uterus/cervix, and lens of the eye)
Dose coefficients	Given for each phantom, male and female, separately
Effective dose	The contribution of the male and female phantoms is given separately
Normalisation quantity	Dose per fluence at 1 m from the source
Purpose of dose coefficients	To be used for convolution with an x-ray spectral distribution to obtain spectral dose coefficients

Geant4, Geometry and Tracking, Version 4; EGSnrc, Electron Gamma Shower National Research Council of Canada.

4.1.1. EGSnrc

(48) Organ absorbed and effective dose coefficients for the adult reference phantoms were calculated at the Helmholz Zentrum Munich and Federal Office for Radiation Protection, Germany, using a code developed specifically for organ dose calculations (Schlattl et al., 2012) employing the electron-gamma-shower code system EGSnrc Version v4-2-3-1 (Kawrakow et al., 2009). EGSnrc is an extended and improved version of EGS4 (Nelson et al., 1985), maintained by the National Research Council of Canada (NRC). The transport of photons and

electrons can be simulated for particle kinetic energies from a few keV up to several hundred GeV.

(49) For photon transport, bound Compton scattering and secondary photoelectrons from K, L, and M shells are considered for all energies. In both cases, resulting fluorescence or Auger and Coster-Kronig electrons are followed. The input data for photon cross-sections agree with those of the XCOM database (Berger and Hubbell, 1987).

(50) For the calculations performed for this publication, photon transport is terminated when the photon energy is $\leq$2 keV. Secondary electrons are followed until their kinetic energy is $<$20 keV.

(51) Simulations were made only for photons in the energy range of 3–150 keV. The number of histories followed was 10 million for each single energy (in steps of 1 keV), resulting in coefficients of variance $<$1–2% for most organs in the field (i.e. receiving primary radiation). For small organs outside the beam, the coefficients of variance per single photon energy could amount to several tens of percent.

(52) Organ absorbed dose coefficients calculated for monoenergetic values for all defined organs/tissues, including all those noted explicitly in the definition of effective dose and all remainder tissues, are given as absorbed dose per fluence at the reference plane. As this publication refers to x-ray beams, and the radiation weighting factor, w_R, of photons equals unity, the equivalent dose coefficients are numerically equivalent to their corresponding absorbed dose coefficients.

(53) The computational methods for determining the absorbed dose to active marrow and skeletal endosteum for the adult reference phantoms are described in Annex D of *Publication 116* (ICRP, 2010). They made use of the DRFs described therein. For paediatric DRFs see *Publication 155* (ICRP, 2023b). Additionally, Annex C of the present publication explains their implementation to the Monte Carlo codes.

4.1.2. Geant4 code

(54) Organ absorbed and effective dose coefficients for the paediatric reference phantoms were calculated in Yonsei University, South Korea using Geant4, an open source, general-purpose Monte Carlo code, which was developed in the C++ programming language, exploiting software engineering and object-oriented technology (Agostinelli et al., 2003). Since the first public release in 1998, the Geant4 code has been improved and maintained by the Geant4 collaboration of various international research groups (http://geant4.cern.ch/). The Geant4 code can simulate a large set of particles, covering a wide energy range from 100 eV to 10 TeV or, for some particles, 10 PeV (Allison et al., 2016). It is widely used in various applications including radiation dosimetry, medical application, space science, and accelerator physics.

(55) Geant4 Version 10.7 (Patch 03) was used in the calculations of the dose coefficients for the paediatric reference voxel phantoms. These were implemented in the Geant4 code using the G4VNestedParameterisation class, which has shown the best performance in computation speed and memory usage against other Geant4

classes for implementation of voxel geometry (Schümann et al., 2012). The physics library of the G4EmLivermorePhysics, including EPDL97 (Cullen et al., 1997), EEDL (Perkins et al., 1991b), and EADL (Perkins et al., 1991a), was used to simulate photons and secondary electrons. To secure precision of simulation, a cut-off range of 1 μm was applied for secondary electrons, instead of the default cut-off range of 1 mm. Thus, if the energy of a secondary electron in the medium corresponds to <1 μm, the secondary electron is 'killed' and its kinetic energy is deposited at the position where the electron is produced, whereas the primary photon is transported until its kinetic energy is zero.

(56) Organ absorbed dose coefficients for monoenergetic photons were calculated for each projection and phantom in 1-keV energy bins across the range of 1–80 keV for the newborn, 1–90 keV for the 1-year-old, 1–100 keV for the 5-year-old, 1–110 keV for the 10-year-old, and 1–120 keV for the 15-year-old. Absorbed doses to all organs for which tissue weighting factors are assigned (ICRP, 2007b) were calculated using the G4PSEnergyDeposit class, except for the skeletal tissues (red bone marrow and endosteum) and the lens of the eye.

(57) The absorbed doses to the skeletal tissues were calculated following the skeletal dosimetry method based on the paediatric DRFs (ICRP, 2023b), as described in Annex C of the present publication. For these calculations, the scoring class derived from the G4VPrimitiveScorer class was used to convert the photon fluence values in the regions of the spongiosa or medullary cavity at a given photon energy to the corresponding skeletal doses via a log-log interpolation of the DRF values (Yeom et al., 2016). 10^8 primary photons were transported so that the coefficients of variance of the calculated organ absorbed doses were <5% for the organs within the beam fields, while relatively large errors, even >20%, could occur for organs outside the beam fields, particularly small organs.

(58) Fig. 4.2 shows, as examples, the absorbed dose per fluence at 1 m from the source plotted against energy for red bone marrow, colon, stomach, and lungs for scoliosis right lateral projection and male paediatric reference phantoms.

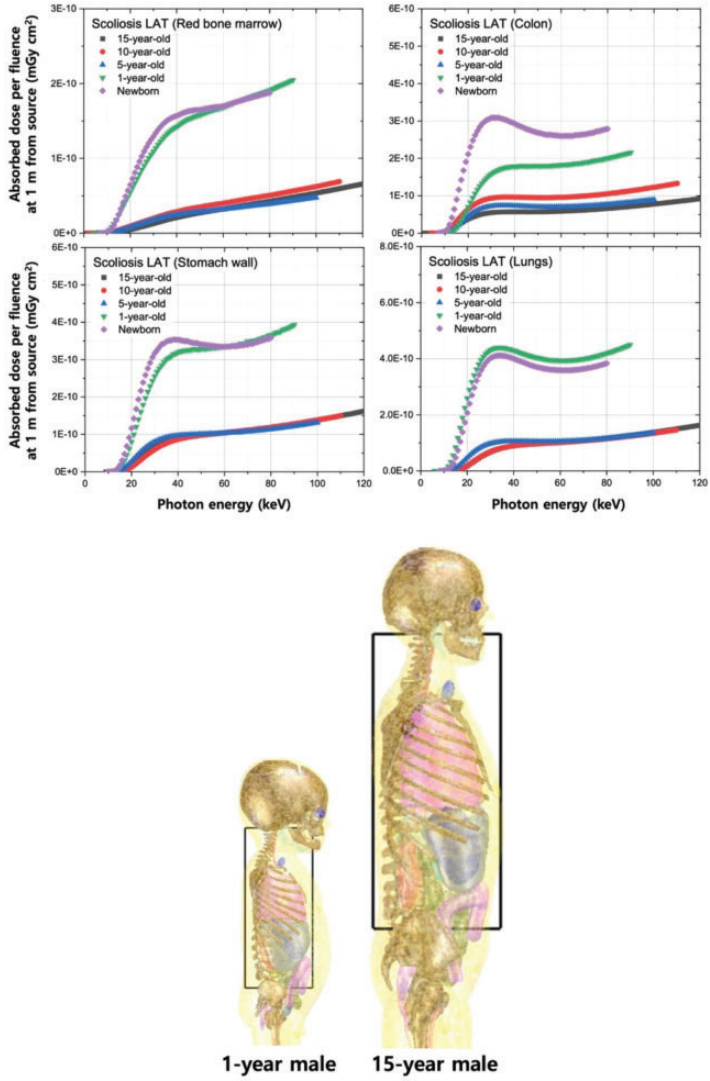


Fig. 4.2. (Top) Absorbed dose per fluence at 1 m from the source for red bone marrow, colon, stomach, and lungs as a function of photon energy for scoliosis right lateral (LAT) projection and male paediatric reference phantoms. (Bottom) Schematic representation of the irradiation fields showing anatomical landmarks on the 1- and 15-year-old male phantoms.

4.1.3. Geant4 code used for spot checks

(59) Spot checks on organ absorbed dose coefficients for the paediatric reference phantoms and some selected projections were calculated at the Department of Engineering Physics, Tsinghua University, China using a code developed specifically for phantom dose calculations employing Geant4 Version 10.4 (http://geant4.cern.ch/).

(60) The physics library of G4EmStandardPhysics_option3 was used to simulate the transport of photons and secondary electrons through paediatric reference phantoms. To balance calculation accuracy and computational efficiency, a cut-off range of 0.1 mm was used for secondary electrons in every medium.

(61) For the spot checks, organ absorbed dose coefficients for monoenergetic photons were calculated for each projection and phantom with 5-keV energy bins across the range of 1–80 keV for the newborn, 1–90 keV for the 1-year-old, 1–100 keV for the 5-year-old, 1–110 keV for the 10-year-old, and 1–120 keV for the 15-year-old. In each case, 10^8 histories were calculated. Coefficients of variance for organs inside the exposure field were $<5\%$. However, for organs outside the exposure field, especially small organs, the coefficients of variance were approximately 20%.

(62) Absorbed doses to all organs, except skeletal tissues, were calculated by counting the energy deposition directly using the GetTotalEnergyDeposit function of the G4Step class (see http://geant4.cern.ch/). For the calculations of the averaged absorbed doses for red bone marrow and endosteum, the skeletal dosimetry method based on the DRFs described in Annex C of the present publication was applied. The track length in a unit volume was used to obtain the photon fluence, which was then converted to the corresponding skeletal doses via interpolation of the DRF values for a given photon energy.

4.1.4. Quality assurance of data

(63) For verification of the calculations, a number of spot-check calculations were performed by different calculators. In the spot checks, Monte Carlo simulations were performed in different centres using a range of projections and including all the phantoms to derive selected monoenergetic coefficients. The same input parameters were used to characterise the radiation fields, but different Monte Carlo codes were employed. In addition, in some cases, the same Monte Carlo code was used by at least two different centres to check that dosimetric methods such as phantom definition and skeletal dosimetry were implemented correctly. The spot checks demonstrated very good agreement between centres, as exemplified in Figs 4.3–4.5. Fig. 4.3 compares the dose coefficients at 100 keV obtained in two different centres for a pelvic antero-posterior examination simulated with both EGSnrc and Geant4 using the adult female phantom. The comparison is between the data provided in this publication (primary) and that performed by a spot check. Fig. 4.4 shows a similar comparison for a chest antero-posterior examination simulated with Geant4

at 50 keV using the male newborn phantom. Fig. 4.5 shows the absorbed dose to the urinary bladder (left) and thyroid (right) as a function of photon energy for a lumbar spine left lateral examination simulated with Geant4 using the 10-year-old female phantom. As can be seen from Figs 4.3–4.5, there is very good agreement between primary and spot-check data.

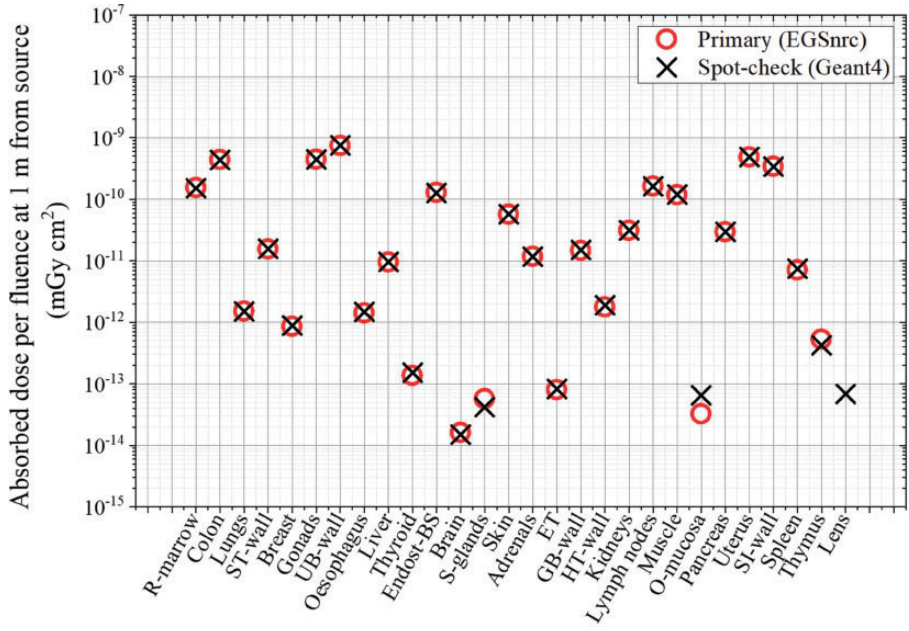


Fig. 4.3. Organ absorbed dose coefficients for a pelvic antero-posterior examination on the adult female reference phantom for 100keV photons, as calculated with EGSnrc and Geant4 codes. The red circles show the data of the primary calculations and the black crosses show the spot-check data (in most cases, the two are superimposed). R-marrow, red bone marrow; ST-wall, stomach wall; UB-wall, urinary bladder wall; Endost-BS, endosteum; S-glands, salivary glands; ET, extrathoracic region; GB-wall, gallbladder wall; HT-wall, heart wall; O-mucosa, oral mucosa; SI-wall, small intestine wall.

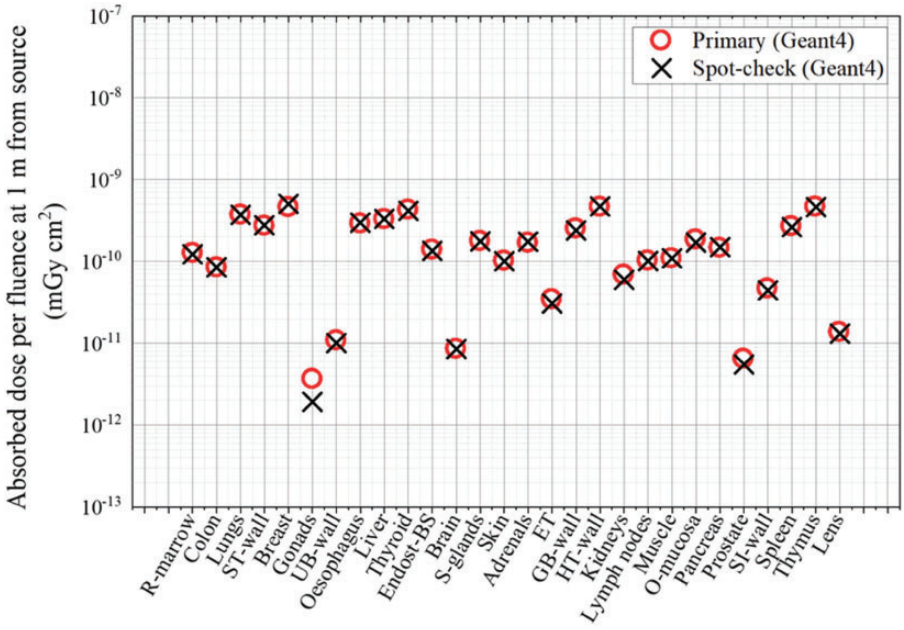


Fig. 4.4. Organ absorbed dose coefficients for a chest antero-posterior examination on the newborn male reference phantom and 50keV photons, calculated with the Geant4 code by two different calculators. The red circles show the data of the primary calculations and the black crosses show the spot-check data (in most cases, the two are superimposed). R-marrow, red bone marrow; ST-wall, stomach wall; UB-wall, urinary bladder wall; Endost-BS, endosteum; S-glands, salivary glands; ET, extrathoracic region; GB-wall, gallbladder wall; HT-wall, heart wall; O-mucosa, oral mucosa; SI-wall, small intestine wall.

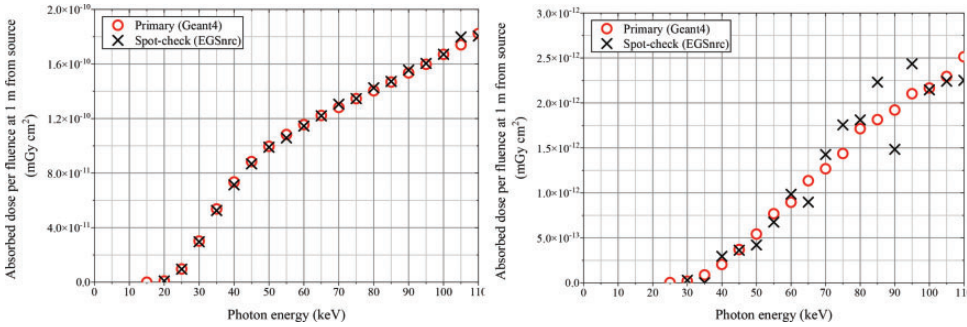


Fig. 4.5. Urinary bladder (left) and thyroid (right) absorbed dose coefficients for a lumbar spine left lateral examination on the 10-year-old female reference phantom as a function of photon energy, calculated with the Geant4 code and EGSnrc by two different calculators. The red circles show the data of the primary calculations and the black crosses show the spot-check data. Note that the urinary bladder lies inside the beam, whereas the thyroid lies outside the beam.

4.2. Dose coefficients for monoenergetic photons at 1-keV intervals – description of tables

(64) Organ absorbed doses have been calculated for each projection and phantom with Monte Carlo methods in 1-keV energy bins across the range 1 or 3 to 80, 100, 120, or 150 keV, depending on the examination and phantom (i.e. age represented). This generic method enables any beam quality to be generated using a convolution. The dose coefficients for the monoenergetic energy bins are given normalised to fluence at the reference plane, which has been defined at 1 m from the source. Note that the fluence at the reference plane is always estimated in the absence of the phantom. The air kerma free-in-air per fluence at the reference plane for the respective energy beam is also given. These values were taken from Seltzer (1993), and interpolation was used for energies not included explicitly in the above data set. Dose coefficients for a spectral beam can then be generated in terms of x-ray fluence spectrum and air kerma per fluence per energy bin (see Section 5).

(65) Organ absorbed dose coefficients normalised to fluence at 1-keV intervals for all projections considered and all 12 phantoms are given in the electronic supplement. In addition, the coefficients for $\sum w_T \times H_T^M$ and $\sum w_T \times H_T^F$ (denoted $E_{103,\#\#M}$ and $E_{103,\#\#F}$ as described in Para. 20 and Annex A) are provided, again at 1-keV intervals.

5. DOSE COEFFICIENTS FOR EXAMPLE X-RAY SPECTRA

(66) In the case of adult radiography, organ absorbed dose coefficients are provided in the electronic supplement for bremsstrahlung spectra at 60, 80, 100, and 120 kV with total filtration of 2.5, 3.0, and 3.5 mm Al. For paediatric radiography, the coefficients are provided for selected spectra in the range of 65–100 kV, depending on the projection and age of the child. As in the case of adult radiography, the spectra have total filtration between 2.5 and 3.5 mm Al. Moreover, as additional copper filtration is often recommended for paediatric radiography, dose coefficients for spectra with additional filtration (i.e. 3.0 mm Al + 0.1 mm Cu) are provided. Details of the beam qualities used in paediatric radiography can be seen in Table 5.1. The effect of alternative beam qualities on the dose coefficients can be explored easily using the monoenergetic data, as described in Section 6.

(67) The dose coefficients are reported as organ absorbed doses per kerma-area product, per fluence and air kerma free-in-air at 1 m from the source. The dose coefficients for the selected spectra are also available as part of the ICRP dose viewer, which is an application freely available via the Apple App Store (https://www.apple.com/app-store/) and the Google Play Store (https://play.google.com/store/apps).

Table 5.1. Details of beam qualities considered for paediatric examinations for which dose coefficients are provided.

	Tube potential (kV)					Filtration	
Examination	00MF	01MF	05MF	10MF	15MF	mm Al	mm Al + Cu
Chest PA	N/A	N/A	70	70	100	2.5, 3.0, 3.5	3.0 Al + 0.1 Cu
Chest AP	65	65	70	70	100	2.5, 3.0, 3.5	3.0 Al + 0.1 Cu
Pelvis AP	65	65	70	75	80	2.5, 3.0, 3.5	3.0 Al + 0.1 Cu
Abdomen AP	65	65	70	75	75	2.5, 3.0, 3.5	3.0 Al + 0.1 Cu
Lumbar spine AP	65	70	70	75	80	2.5, 3.0, 3.5	3.0 Al + 0.1 Cu
Lumbar spine Lat	65	70	70	75	80	2.5, 3.0, 3.5	3.0 Al + 0.1 Cu
Thoracolumbar spine Lat	60	60	N/A	N/A	N/A	2.5, 3.0, 3.5	3.0 Al + 0.1 Cu
Scoliosis AP	65	70	N/A	N/A	N/A	2.5, 3.0, 3.5	3.0 Al + 0.1 Cu
Scoliosis PA	N/A	N/A	75	75	80	2.5, 3.0, 3.5	3.0 Al + 0.1 Cu
Scoliosis Lat	65	70	75	80	85	2.5, 3.0, 3.5	3.0 Al + 0.1 Cu
Skull AP	70	70	75	75	75	2.5, 3.0, 3.5	3.0 Al + 0.1 Cu
Skull PA	70	70	75	75	75	2.5, 3.0, 3.5	3.0 Al + 0.1 Cu
Skull Lat	70	70	75	75	75	2.5, 3.0, 3.5	3.0 Al + 0.1 Cu

00MF, newborn male and female; 01MF, 1-year-old male and female; 05MF, 5-year-old male and female; 10MF, 10-year-old male and female; 15MF, 15-year-old male and female; PA, postero-anterior; AP, antero-posterior; Lat, lateral; N/A, not applicable.

(68) These spectra have been generated using SpekCalc (Poludniowski, 2007; Poludniowski and Evans, 2007; Poludniowski et al., 2009). SpekCalc is a software program based on theoretical bremsstrahlung cross-sections, and has been

developed for the calculation of x-ray spectra from tungsten anode x-ray tubes, primarily for medical applications. The software simulates the x-ray source as a point source, and models a wide range of tube potentials (40–300 kV) and anode angles. The heel effect is not considered; the implications are discussed in Section 7.3. Filtration can be applied for seven materials (air, water, Be, Al, Cu, Sn, and W). For all spectra considered, an anode angle of 13° has been employed, as well as vacuum as the medium.

(69) To derive the dose coefficients $(D/K_a)_S$ for each of the example x-ray spectra S, the monoenergetic dose coefficients were convolved with the specified spectra. As air kerma per fluence, K_a/ϕ, is energy-dependent (Seltzer, 1993), it has to be summed up fluence-wise, as do the organ absorbed dose coefficients. The fluence-weighted sum of the organ absorbed dose coefficients per fluence, D/ϕ, is then divided by the fluence-weighted sum of K_a/ϕ, resulting in an overall coefficient of organ absorbed dose per air kerma. This is described mathematically in Eq. 5.1 where $\phi_S(E)$ denotes the fluence of spectrum S at energy E. For further details, see Section 6.2.

$$\left(\frac{D_T}{K_a}\right)_S = \frac{\int \frac{D}{\phi}(E)\ \phi_S(E)\ \mathrm{d}E}{\int \frac{K_a}{\phi}(E)\ \phi_S(E)\ \mathrm{d}E} \tag{5.1}$$

(70) The procedure has been validated by comparing the dose coefficients obtained with the convolution method and those calculated with a Monte Carlo code using a spectral source as an input. Fig. 5.1 shows such a comparison for an abdominal antero-posterior examination using the adult male phantom, and Fig. 5.2 shows the comparison for the same examination using the 1-year-old male phantom. It can be seen that the agreement is very good – deviations are <2% for all organs within the exposure field – which validates the convolution procedure used.

(71) The overall organ absorbed dose coefficients for the projections considered and all male and female phantoms are given for each of the example spectra in the electronic supplement. As with monoenergetic data described in Section 4, coefficients for $\sum w_T \times H_T^M$ and $\sum w_T \times H_T^F$ (again denoted $E_{103,\#\#M}$ and $E_{103,\#\#F}$ as described in Para. 20 and Annex A) are provided for each spectra. For convenience, values of an effective dose coefficient $(\sum w_T \times H_T^M + \sum w_T \times H_T^F)/2$ are also provided, should the user wish to use the same value of normalisation quantity for male and female examinations.

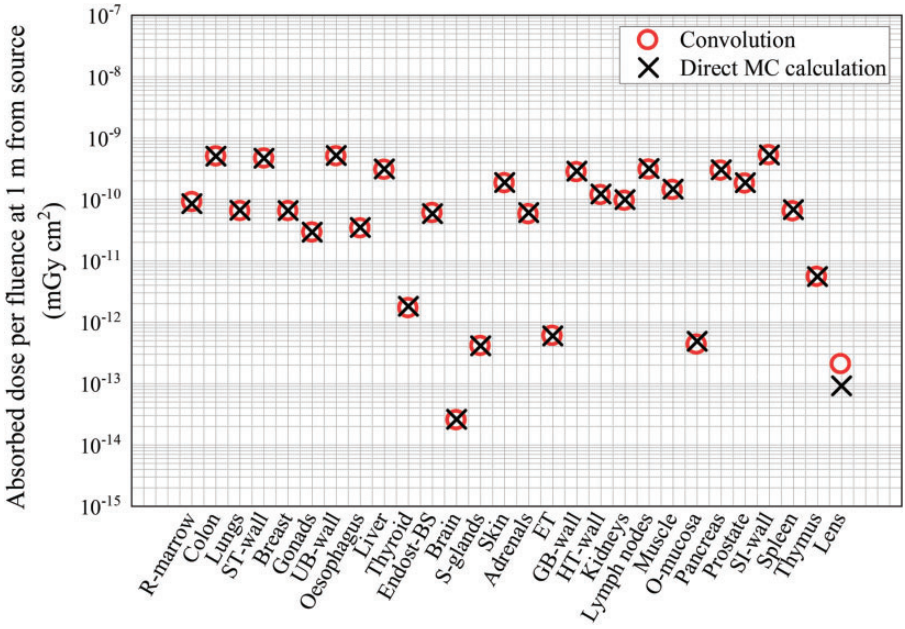


Fig. 5.1. Comparison of organ absorbed dose coefficients for abdominal antero-posterior examination of the adult male phantom, for 60 kV tube potential and additional filtration of 3.5 mm Al, evaluated using the monoenergetic data and convolution method, and direct Monte Carlo (MC) calculation employing spectral input data. R-marrow, red bone marrow; ST-wall, stomach wall; UB-wall, urinary bladder wall; Endost-BS, endosteum; S-glands, salivary glands; ET, extrathoracic region; GB-wall, gallbladder wall; HT-wall, heart wall; O-mucosa, oral mucosa; SI-wall, small intestine wall.

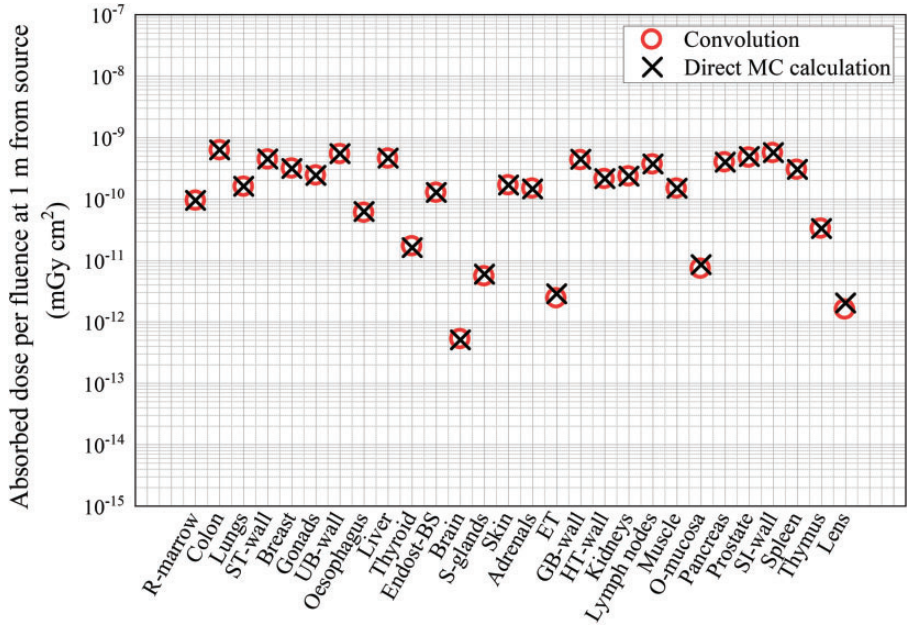


Fig. 5.2. Comparison of organ absorbed dose coefficients for abdominal antero-posterior examination of the 1-year-old male phantom (01M), for 65 kV tube potential and additional filtration of 3.0 mm Al + 0.1 mm Cu, evaluated using the monoenergetic data and convolution method and direct Monte Carlo (MC) calculation employing spectral input data. R-marrow, red bone marrow; ST-wall, stomach wall; UB-wall, urinary bladder wall; Endost-BS, endosteum; S-glands, salivary glands; ET, extrathoracic region; GB-wall, gallbladder wall; HT-wall, heart wall; O-mucosa, oral mucosa; SI-wall, small intestine wall.

6. APPLICATION OF DOSE COEFFICIENTS

6.1. Using the bremsstrahlung spectral data to estimate organ and effective doses

(72) As outlined previously, for those users who do not need or wish to fine-tune their organ absorbed dose estimates using convolution, organ absorbed dose coefficients are provided for selected bremsstrahlung spectra between 50 and 120 kV with total filtration of 2.5–3.5 mm Al and an additional 0.1 mm Cu in the case of paediatric projections. The coefficients are normalised to kerma-area product [KAP (P_{KA})], air kerma free-in-air, and fluence, all determined at 1 m from the x-ray tube. They can be used in those situations where the beam qualities encountered in the field are the same as or close to those in the tables. For simplicity, in this section, KAP (P_{KA}) will be indicated as KAP.

(73) To use the spectrum data provided in this publication, it is necessary to have as a minimum:

(i) a measurement of kerma free-in-air on the central axis of the beam at 1 m from the tube focus or KAP for a specific radiographic view;
(ii) the tube potential in kV used for the exposure; and
(iii) knowledge of the total filtration.

(74) Kerma free-in-air can be obtained from measurements made using a radiation detector, and KAP will generally be displayed on a (properly calibrated) x-ray set, but can also be calculated from kerma free-in-air and the field size. Ideally, the total filtration of the tube assembly will be known, in terms of mm Al plus any additional Cu. If the exact filtration is not known, it is suggested, on the basis of clinical experience, that 3.0 mm Al should be chosen for both adult and paediatric cases.

(75) For example, to determine the colon dose for an abdominal antero-posterior examination performed on an adult male, at a tube potential of 80 kV and KAP of 2.5 Gy cm^2 and with total filtration of 3.5 mm Al, the user should:

(i) open the file 'AbdomenAP_Spectral.xlsx' in the electronic annex;
(ii) select the tab 'AM', indicating Adult Male; and
(iii) select Row 69, which corresponds to 80 kV.

Organ absorbed dose coefficients are given in mGy Gy^{-1} cm^{-2} from Column F onwards. Thus, the colon absorbed dose (Column G, for the coefficient) is found to be 4.26×10^{-1} mGy Gy^{-1} cm^{-2} $\times$ 2.5 Gy cm^2 = 1.1 mGy.

(76) Effective dose, E, can be determined by multiplying each of the dose coefficients $E_{103,AM}$ and $E_{103,AF}$ by the KAP used in the male and female examinations, respectively, and then dividing the sum by 2. In this example, it is assumed that the

KAP for the abdominal antero-posterior examination of the female is the same as that of the male (i.e. 2.5 Gy cm^2). The dose coefficient for a male is 1.76×10^{-1} mSv Gy^{-1} cm^{-2} and the dose coefficient for a female is 2.57×10^{-1} mSv Gy^{-1} cm^{-2} (Column AH), so E for this examination is given by $E = (1.76 \times 10^{-1}$ mSv Gy^{-1} $cm^{-2} + 2.57 \times 10^{-1}$ mSv Gy^{-1} $cm^{-2})/2 \times 2.5$ Gy $cm^2 = 2.16 \times 10^{-1}$ mSv Gy^{-1} $cm^{-2} \times 2.5$ Gy $cm^2 = 0.54$ mSv. For convenience, Column AJ provides the averaged dose coefficient for those who wish to take this approach.

(77) A similar process can be followed if air kerma at the reference plane is used as the normalising quantity. In the above example, if air kerma at the reference plane is 1 mGy, the user should:

(i) open the file 'AbdomenAP_Spectral.xlsx' in the electronic annex;
(ii) select the tab 'AM', indicating Adult Male; and
(iii) select Row 52, which corresponds to 80 kV.

Organ absorbed dose coefficients are given in mGy Gy^{-1} from Column F onwards. Thus, the colon absorbed dose (Column G, for the coefficient) is again found to be 1.1×10^3 $Gy^{-1} \times 10^{-3}$ Gy = 1.1 mGy.

(78) If the filtration is known exactly and is 2.5–3.5 mm Al, simple linear interpolation can be used to obtain a more precise value for any dose coefficient.

(79) Note that in the case of paediatric examinations, the selection of tube potential is limited to a smaller set of values to represent clinical practice. Also, there is only one option for copper filtration.

6.2. Using the convolution process for a user-defined spectrum

(80) Dose coefficients for a user-defined spectrum can be obtained by convolution using the monoenergetic dose coefficients provided in this publication, as described in Eq. 5.1. By way of example, consider again the case of the abdominal antero-posterior projection performed on the adult male phantom. For this example, SpekCalc was used to generate an 80kV bremsstrahlung spectrum from a tungsten target with a 13° anode angle and 3.5 mm Al total filtration. This is the same example as in Section 6.1, but will use the convolution approach, rather than the results provided explicitly.

(81) The flow diagram (Fig. 6.1) shows how the convolution process is carried out using data from the electronic supplement, in conjunction with the calculated spectrum. The diagram references Table 6.1 which uses as an example the data file 'AbdomenAP_mono.xlsx' from the appropriate folder of the electronic supplement and, in particular, the dose coefficients of the colon.

(82) The first two columns of Table 6.1 show the energy and fluence data from the spectrum program. For simplicity, only the first 14 and last four energy intervals are shown (note, values are only provided where the fluence is not zero). The third column (relative number of photons per energy interval) is calculated as shown.

The fourth column (organ absorbed dose per fluence per energy interval) is taken from the electronic supplement (file 'AbdomenAP_mono.xlsx'), and the fifth column (fluence-weighted organ dose) is calculated as shown. The sixth column (air kerma per fluence in the reference plane) is also taken from the file 'AbdomenAP_mono.xlsx', and the final column (weighted air kerma per energy interval in the reference plane) is calculated as shown. The dose per KAP and per kerma at the reference plane are calculated as shown.

(83) Table 6.1 shows that the colon dose per KAP, estimated via convolution, is 0.43 mGy Gy^{-1} cm^{-2}, which is, as expected, the same value as that obtained using the file 'AbdomenAP_Spectra.xlsx' as described in Section 6.1 above. By using this method, one can obtain the effective dose coefficient.

(84) Use of other spectrum programs – for example, using data obtained from the IPEM Report 78 program (Sutton et al., 2015) – will produce slightly different results. For example, in the calculation performed here, use of the IPEM Report 78 software to generate the x-ray spectrum results in an estimate of colon dose per KAP of 0.45 mGy Gy^{-1} cm^{-2}.

(85) If spectra are not available at 1-keV intervals, they must be amended to be so. This applies, for example, in the IPEM Report 78 software (Sutton et al., 2015), which produces output at 0.5-keV intervals. In this case, each consecutive pair of fluence values should be summed prior to performing the convolution.

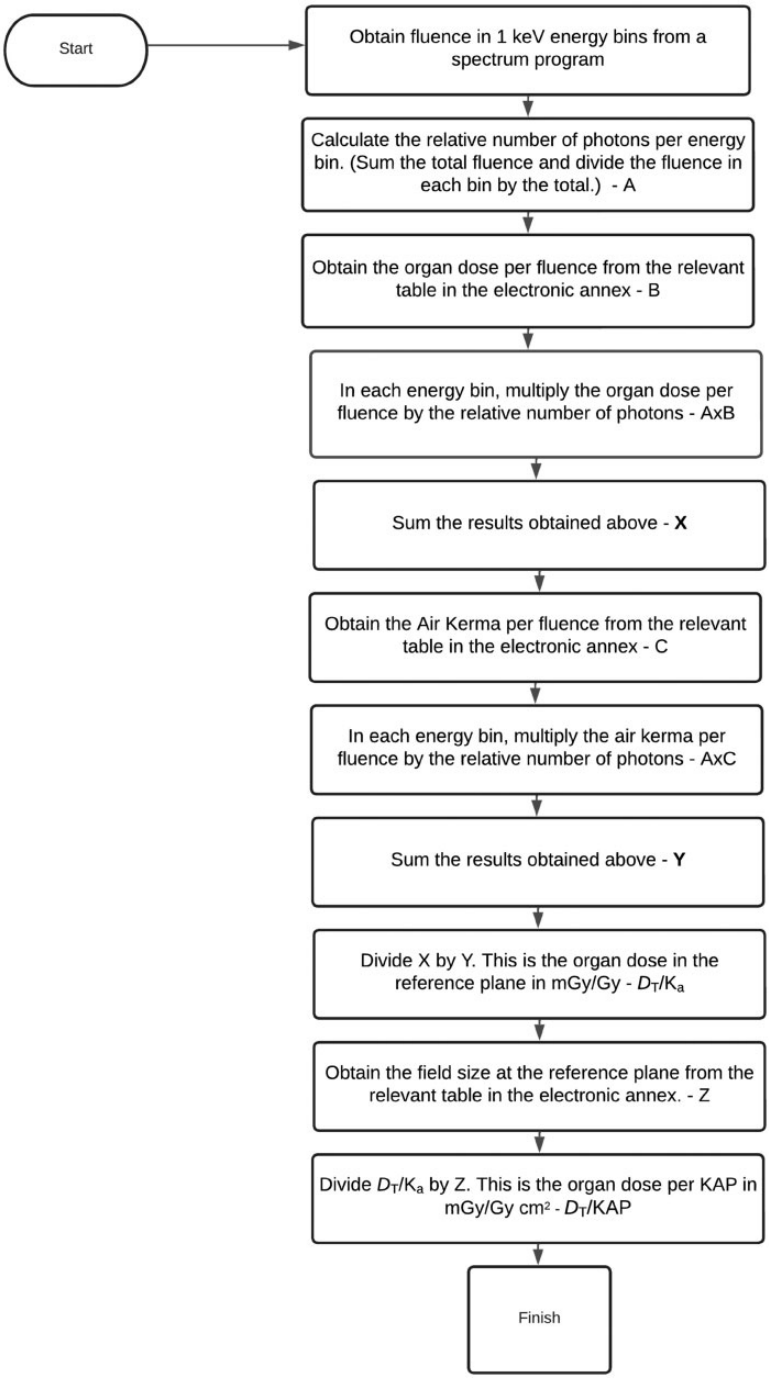


Fig. 6.1. Flow chart demonstrating the convolution process.

Table 6.1. Example of convolution for abdominal antero-posterior (AP) examination on the adult male phantom. The example employs the monoenergetic data given in the electronic supplement (AbdomenAP_mono.xlsx), and demonstrates the method for estimating the absorbed dose coefficients of the colon[¶]. Capital letters (A, B, C, X, Y, Z) refer to those used in Fig. 6.1.

Tube voltage (keV)	Fluence[*] (cm^{-2})	Relative number of photons[†] (cm^{-2}) (A)	Colon dose, D, per fluence[+] (mGy cm^2) (B)	Fluence weighted colon dose, D, (mGy) (A × B)	Air kerma per fluence in the reference plane[§] (Gy cm^2) (C)	Weighted air kerma in the reference plane (mGy) (A × C)
1	0	0	0	0	5.77×10^{-10}	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮
8	7.53×10^{-14}	7.15×10^{-22}	2.47×10^{-14}	1.76×10^{-35}	1.21×10^{-11}	8.65×10^{-33}
9	4.48×10^{-8}	4.26×10^{-16}	1.02×10^{-13}	4.35×10^{-29}	9.48×10^{-12}	4.04×10^{-27}
10	3.71×10^{-4}	3.53×10^{-12}	4.08×10^{-13}	1.44×10^{-24}	7.60×10^{-12}	2.68×10^{-23}
11	8.09×10^{-2}	7.68×10^{-10}	1.40×10^{-12}	1.08×10^{-21}	6.21×10^{-12}	4.77×10^{-21}
12	4.65	4.41×10^{-8}	3.36×10^{-12}	1.48×10^{-19}	5.16×10^{-12}	2.28×10^{-19}
13	9.00×10^{1}	8.55×10^{-7}	7.69×10^{-12}	6.58×10^{-18}	4.35×10^{-12}	3.72×10^{-18}
14	8.83×10^{2}	8.39×10^{-6}	1.62×10^{-11}	1.36×10^{-16}	3.72×10^{-12}	3.12×10^{-17}
⋮	⋮	⋮	⋮	⋮	⋮	⋮
77	3.14×10^{5}	2.99×10^{-3}	7.01×10^{-10}	2.09×10^{-12}	3.03×10^{-13}	9.05×10^{-16}
78	2.47×10^{5}	2.34×10^{-3}	7.04×10^{-10}	1.65×10^{-12}	3.05×10^{-13}	7.15×10^{-16}
79	1.37×10^{5}	1.30×10^{-3}	7.08×10^{-10}	9.19×10^{-13}	3.07×10^{-13}	3.98×10^{-16}
80	0.00	0	7.16×10^{-10}	0.00	3.09×10^{-13}	0.00
Sum	1.05×10^{8}	1.0	–	5.52×10^{-10} (X)	–	4.98×10^{-13} (Y)

[*]From spectrum program.

[†]Fluence per energy bin divided by total fluence.

[+]From file 'AbdomenAP_mono.xlsx'.

[§]From file 'AbdomenAP_mono.xlsx'.

[¶]The organ (i.e. colon) dose per air kerma at the reference plane is therefore given by $D_T = X/Y = 5.52 \times 10^{-10}/4.98 \times 10^{-10} = 1108$ mGy Gy^{-1}. From 'AbdomenAP_mono.xlsx', the field size at the reference plane (Z) is $42.4 \times 61.3 = 2599.12\,cm^2$. Thus, the colon dose per kerma-area product is expressed as $D_T/Z = 1108$ mGy $Gy^{-1}/2599.12\,cm^2 = 0.426$ mGy Gy^{-1} cm^{-2}.

6.3. Possible application of the dose coefficients

6.3.1. Using beam qualities different to those presented in the tables of coefficients

(86) Users may wish to align the dose coefficients more closely to the equipment available in their environment by reproducing the beam qualities actually available to them. Alternatively, they may wish to determine the effect of utilising new equipment with different beam qualities. Either can be done easily using the approach outlined in Section 6.2. A simple example would be a paediatric radiology department in which abdominal radiographs are performed at 70 kV using a total filtration of 3.0 mm Al + 0.2 mm Cu. The examples provided in the tables only contain coefficients for beam qualities with 3.0 mm Al + 0.1 mm Cu total filtration at 70 kV, so the convolution approach must be adopted.

(87) Take the example of an abdominal examination performed on a 5-year-old male. A fluence spectrum must first be generated using the user's preferred software. Next, using the methodology outlined in Section 6.1, the file 'AbdomenAP_mono.xlsx' is chosen and the 05M tab is selected. If the spectrum is calculated at 70 kV using SpekCalc with a 13° anode angle, the colon dose coefficient is calculated to be 1.83 mGy Gy^{-1} cm^{-2} and the effective dose coefficient $[(E_{103,05M} + E_{103,05F})/2]$ is 8.59×10^{-1} mSv Gy^{-1} cm^{-2}, assuming that KAP is the same for both a male and female examination. Alternatively, the radiology department may choose to obtain abdominal radiographs at 65 kV with a beam quality of 3 mm Al + 0.1 mm Cu. Use of the convolution approach shows that the colon dose coefficient is 1.6 mGy Gy^{-1} cm^{-2} for a 5-year-old male. The same approach can be used if a different tube potential is utilised. Table 6.2 shows how the dose coefficients vary with tube potential and filtration for the examples described. The second column of the table shows the colon dose for a beam quality of 70 kV with 3.0 mm Al + 0.1 mm Cu total filtration, and presents the results implemented using convolution and also using the spectral table 'AbdomenAP_Spectral.xlsx'. As expected, and discussed in Section 6.2, in those situations where results can be obtained using either convolution or table, there is no difference between them.

Table 6.2. Variation of the colon absorbed and effective dose coefficients due to different beam filtration for a 5-year-old male.

	Colon absorbed dose (mGy Gy^{-1} cm^{-2})		$E_{103,05F}$ (mSv Gy^{-1} cm^{-2})		$E_{103,05M}$ (mSv Gy^{-1} cm^{-2})		E_{103} (mSv Gy^{-1} cm^{-2})	
	From table	Using convolution	From table	Using convolution	From table	Using convolution	From table	Using convolution
70 kV, 3.0 mm Al +0.1 mm Cu	1.65	1.65	0.75	0.75	0.78	0.78	0.76	0.76
70 kV, 3.0 mm Al +0.2 mm Cu	N/A	1.83	N/A	0.85	N/A	0.87	N/A	0.86
65 kV, 3.0 mm Al +0.1 mm Cu	N/A	1.6	N/A	0.72	N/A	0.75	N/A	0.73

N/A, not applicable; $E_{103,05F}$, coefficient of the quantity $\sum w_T \times H_T^F$ resulting from an examination on a 5-year-old female; $E_{103,05M}$, coefficient $\sum w_T \times H_T^M$ of a male phantom; E_{103}, effective dose as sex averaged ['103' indicates that the definitions are according to *Publication 103* (ICRP, 2007b)].

6.3.2. An optimisation example

(88) Use of additional copper filtration has the advantage of lowering the dose if the automatic exposure control device is set up properly, and is often recommended for units used for paediatric examinations. Due to its larger photoelectric cross-section, copper absorbs more low-energy photons in the 20–50keV range than aluminum so, for example, inclusion of a 0.2-mm-thick copper filter in radiographic

units at tube potentials of 70–80 kV can reduce KAP by 50% for the same exit dose from the patient. The corresponding reduction in effective dose can be up to 40% (Martin et al., 1999; ICRP, 2024b). By way of example, the images shown in Fig. 6.2 below depict radiographs taken with and without copper filtration, and show little change in image quality but substantial reductions in KAP. It should be noted that these images are from a cadaver study. A disadvantage of using copper filters is that an increased tube output is required to maintain the same level of quantum noise at the receptor. There will also be a reduction in image contrast resulting from the enhanced absorption of low-energy photons.

(89) If any additional filtration is incorporated into a system, the trade-off between image quality and organ absorbed or effective dose, as well as the impact on tube loading, should be evaluated thoroughly before the system is introduced into clinical practice to ensure that the diagnostic quality of the images is not compromised.

(90) The data presented in this publication can be used to assist in that process, and also inform practitioners of potential dose-saving methods. To show the effects of filtration, consider the four pelvic radiographs shown in Fig. 6.2. A typical KAP value for a pelvic radiograph is 2.2 Gy cm^2. The beam qualities shown in Table 6.3 have been used to determine effective dose at 81 kV for a pelvic x-ray, as defined in Annex B, using the male and female projections to generate $E = (E_{103,\mathrm{AM}} + E_{103,\mathrm{AF}})/2$ for KAP values of 2.2, 1.43, 0.88, and 0.66 Gy cm^2, respectively (obtained using the ratios shown in the legend to Fig 6.2). Table 6.3 shows that effective dose can be reduced by a factor of approximately 2 by including an additional 0.2 mm copper filtration, which will result in an additional load on the tube of approximately 50%.



Fig 6.2. Pelvic radiographs of a cadaver taken at 81 kV with a Siemens Axiom Aristos FX x-ray set showing the effect of additional copper filters. Exposures from left to right were taken with the following thicknesses of copper: 0.0 mm (a), 0.1 mm (b), 0.2 mm (c), and 0.3 mm (d), and the ratio of kerma-area product values to those obtained without copper filtration were 1.0, 0.65, 0.4, and 0.3, respectively (ICRP, 2024b).

(91) The data have also been used to assess the effect of additional filtration on the dose to the uterus under the same conditions, demonstrating a similar dose reduction, as shown in Table 6.3, for a pelvic examination of an adult female. This reduction has potential implications in, for example, the justification and

optimisation of examinations of patients who may be or who are in the early stages of pregnancy. Practitioners can use this type of calculation to determine how to image such patients.

(92) Table 6.3 also gives an indication of the additional tube loading required. This was determined from the exit spectra from an additional 200 mm of polymethyl methacrylate used to mimic a patient. Spectra were generated using the software described in IPEM Report 78 (Sutton et al., 2015), for an anode angle of 13° and no voltage ripple. The doses were evaluated using the dose coefficients calculated for monoenergetic photons in conjunction with the convolution method described above.

Table 6.3. Reduction in effective dose and uterine dose in pelvic examination of an adult female caused by the addition of filtration.

Beam quality at 81 kV	KAP (Gy cm^2)	Effective dose* (mSv)	Uterus absorbed dose (mGy)	Indicative increase in tube current
3.0 mm Al + 0.0 mm Cu	2.2	0.27	0.92	0%
3.0 mm Al + 0.1 mm Cu	1.43	0.18	0.68	30%
3.0 mm Al + 0.2 mm Cu	0.88	0.12	0.45	60%
3.0 mm Al + 0.3 mm Cu	0.66	0.09	0.17	100%

KAP, kerma-area product.
*$E = (E_{103,AM} + E_{103,AF})/2$.

7. UNCERTAINTIES AND LIMITATIONS

7.1. Uncertainties of dose coefficients calculated with Monte Carlo methods

(93) The purpose of patient dosimetry for medical x-ray imaging is to check standards of good practice, contribute to the optimisation process (ICRP, 2023a), and assist in assessing detriment. For the assessment of risk, detailed knowledge is required of organ absorbed doses and the age and sex of the patients concerned. Monte Carlo methods are commonly used to derive dose coefficients relating normalisation quantities such as KAP (P_{KA}), air kerma free-in-air, or fluence to the organ absorbed dose.

(94) The dose coefficients of this publication were calculated for the reference phantoms at reference ages (newborn, 1-year-old, 5-year-old, 10-year-old, 15-year-old, and adult) (ICRP, 1995). Reference values are fixed and specified with no uncertainty, independent of the fact that the basis of these values may include uncertainties. In the present case, the accuracy of the dose coefficients provided is limited by uncertainties in the physical input parameters, such as interaction cross-sections, and deviations of the particle transport simulation from the real exposure situation (e.g. source geometry).

(95) The relative statistical uncertainties of the Monte Carlo calculations for organ absorbed dose coefficients, expressed as coefficients of variance, depend on several parameters such as organ size, location in the human body, particle energy, irradiation geometry, and number of source photons simulated. For organs inside the irradiation field, organ absorbed doses were determined with coefficients of variance (1 standard deviation) $<2\%$ for adult projections and $<5\%$ for paediatric projections (see Sections 4.1.1 and 4.1.2). In contrast, for organs outside the primary beam and for energies <10 keV, the coefficients of variance for small organs can be $\geq 20\%$, as only a few photons are able to reach such organs.

(96) Other sources of uncertainty in the Monte Carlo calculations are uncertainties in cross-sections, elemental compositions, and densities of the organs and tissues, and inadequacies in the description of the x-ray source. For example, the uncertainties in the determination of cross-sections have been estimated to be of the order of 5% below 5 keV and 2% up to 10 MeV (Hubbell and Seltzer, 1995).

7.2. Influence of phantom and field size on the dose coefficients

(97) The dose coefficients of this publication are calculated for the reference phantoms at reference ages (newborn, 1-year-old, 5-year-old, 10-year-old, 15-year-old, and adult) (ICRP, 1995), and coefficients calculated using other phantoms and other exposure conditions may differ. Petoussi-Henss et al. (2005, 2006) investigated the effect of body size and anatomy by simulating various radiographic examinations and employing, besides the two ICRP adult reference phantoms, four

non-reference voxel phantoms constructed from image data of real individuals. In these studies, individual setting parameters (radiation quality, field size, focus-image receptor, and focus-skin distance) were varied to examine their effect on the organ doses. In general, it was shown that the use of different individual voxel phantoms leads, in some cases, to differences in organ dose coefficients of up to a factor of 3. These differences were attributed to the anatomical differences between individuals, and therefore to different amounts of shielding of organs, depending both on the phantom size and topology of the organs. For those organs that lie entirely in the primary beam, it is also generally true that the dose coefficients per fluence or per air kerma decrease with increasing patient diameter. This trend is also observed for the age-dependent dose coefficients of this publication.

(98) In the same work, organ doses were found to be dependent on the exposure conditions; in general, as field size increases, organ dose coefficients per incident air kerma also increase. While this effect is small for organs that lie entirely in the field, it is most pronounced for organs that lie only partially in the primary beam, or outside the field but close to its edge.

(99) Previous comparisons of coefficients calculated at different institutes showed that differences are small (i.e. within the statistical errors and generally $<10\%$ when the exposure conditions and computational phantoms used are similar). Larger differences occur due to differences in the computational phantoms simulating the patient (e.g. due to the use of MIRD or voxel type phantoms) (ICRU, 2005). Similar conclusions can be drawn from comparisons of measured and calculated organ dose coefficients. Larger errors (i.e. well in excess of 10%) occur for coefficients concerning organs far away from the primary beam. This is not a practical problem as their contribution to the radiation risk is much less than that of organs receiving the highest doses (ICRU, 2005).

(100) The ICRP reference phantoms have reference masses as per *Publication 89* (ICRP, 2002), but as they are based on specific individuals, the position of the organs is based on the anatomy of the individual whose image data were used for the construction of the phantom. The internal anatomy is individual-specific and depends on posture, breathing, and the state of digestion. It is known, for example, that breathing motion can displace internal organs by up to several cm (Madore et al., 2022). No attempt to determine a reference organ location is made in *Publications 110* or *143* (ICRP, 2009, 2020a), or in the present work.

(101) The influence of the posture of the phantoms is discussed in Para. 28. The ICRP reference phantoms have been constructed from CT data of real patients, and therefore represent their anatomy in the supine position. For some examinations, however, the patient is in a standing or sitting position. A study on the effects of posture on organ dose coefficients from some typical idealised external exposures for a person in a supine position and an upright position showed agreement within 2–20% for photon energies >50 keV (Sato et al., 2008). For energies <50 keV, some organs could exhibit larger differences of up to a factor of approximately 2,

depending on the projection geometry (i.e. antero-posterior, postero-anterior, lateral, etc.) and the potential shift of the organ in question when the posture changes.

(102) Studies have shown that the position of the organs in relation to the field (e.g. of the colon in examinations of the abdomen where some parts may lie inside or outside the x-ray field) could result in different values of the colon dose. One study of simulated CT examinations which employed a library of different sizes of (non-reference) phantoms, based on the scaling of a limited number of size-adjusted individuals, revealed significant differences of up to 50% in the colon dose (Martin et al., 2021).

(103) It should be noted that variations in doses to small organs that are considered sensitive to radiation can have a significant effect on the calculation of effective dose. Such organs can lie close to the edge of the field of an examination, examples being the case of the thyroid for chest examinations, or the testes for examinations of the pelvis. In such cases, where positioning on the actual patient is not optimal (i.e. not representative of good practice), the uncertainty in the ensuing dose estimate may be considerably larger than reported here.

7.3. Uncertainties in dose estimates resulting from applications of the coefficients

(104) The coefficients tabulated in this publication were developed using the ICRP methodology, a well-defined framework, following documented procedures using reference computational phantoms, critically evaluated, verified for accuracy, and judged by the Commission to serve their intended purpose. The exposure conditions (i.e. field sizes, tube potential, and filtration of the x-ray beam) were selected to represent standard, good radiological practice. Within the scope of their intended use, the coefficients, as issued by ICRP as an international authority, are reference data and have, as such, no uncertainties, per the guidance of the Joint Committee for Guides in Metrology (JCGM, 2008).

(105) One of the largest contributions to the uncertainty in the application of the organ absorbed dose coefficients in any real-world implementation is the measurement uncertainty of the normalisation quantity, whether it be KAP (P_{KA}) or air kerma free-in-air.

(106) The influence of uncertainties associated with the physical measurements of dose is quite easily quantified and has been reported widely – examples being Martin (2007), IAEA (2007), Toroi et al. (2008a,b), Hourdakis et al. (2010), Worrall and Sutton (2015), and Malusek et al. (2016). Both IAEA (2007) and ICRU (2005) recommend that these uncertainties should be kept $<7\%$. This is relatively easily managed when ionisation chambers are used, but is difficult to achieve when dose data are acquired with KAP meters. For example, the International Electrotechnical Commission (IEC, 2020) recommends that the overall uncertainty associated with field class KAP meter measurements in radiography should be $<25\%$. Toroi et al. (2008a) and Malusek et al. (2016) both report that it is possible to achieve an

expanded uncertainty below ±7% (coverage factor $k = 2$) if appropriate and traceable KAP calibration is carried out across the range of diagnostic beam qualities encountered in clinical practice. Sanchez et al. (2020) suggested that it should be easy for a medical physics expert to check if a dose meter introduces bias greater than ± 10%, at least in reference conditions.

(107) Another source of uncertainty is the fact that the spectra have been calculated in a vacuum whilst, in reality, the transmission medium is air. At 81 kV, which is representative of radiographic practice in general, the error introduced by this approximation is an overestimate of air kerma by 5% at most, as can be identified using SpekCalc, IPEM Report 78, or any other spectrum calculator.

(108) Further uncertainty is introduced by the fact that the present calculations did not consider the heel effect or any other field inhomogeneity. It has been estimated that field inhomogeneities result in a coefficient of variation of up to 1.5% for a 40 cm × 40 cm field at the detector plane for exposures made across the diagnostic range of tube potential (Worrall and Sutton, 2015). This uncertainty will directly propagate to the organ absorbed dose calculation. The scatter contribution from the collimation, and other x-ray set components such as the additional filtration, to the air kerma incident on a patient is not considered in the Monte Carlo simulations. There is very little published work quantifying the magnitude of such scatter. However, calculations performed as part of this work suggest that the contribution is considerably <1%. This too will be reflected in the uncertainty introduced into the organ absorbed dose calculation.

(109) Taking the above contributions into account, the lowest uncertainty that can be expected is an expanded uncertainty ($k = 2$) in the calculated dose of approximately ±12.5% if the normalising quantity is incident air kerma or the KAP meter is subject to rigorous calibration checks. However, one can reasonably expect an uncertainty of at least approximately ±20% to be encountered in practice if one is employing KAP as the normalising factor.

(110) The dose coefficients described in this publication were derived for projections considered to be representative of good practice. Petoussi-Henss et al. (2005, 2006) showed that organ doses are dependent on field size; in general, as field size increases, organ dose coefficients per incident air kerma also increase. While this effect is small for organs that lie entirely in the field, it is most pronounced for organs that lie only partially in the primary beam, or outside the field but close to its edge. Therefore, where positioning is not optimal (i.e. not representative of good practice), the uncertainty in the ensuing dose estimate may be considerably larger than reported here.

7.4. Limitations in interpretation

(111) This publication provides patient organ absorbed dose coefficients, for males and females separately, for specific x-ray medical imaging procedures and well-defined technical factors. It also provides, for each examination, a value of either

$\sum w_T \times H_T^M$ or $\sum w_T \times H_T^F$, depending on whether the examination is performed on the male or female phantom. Effective dose for an examination can be obtained by taking the average of the two. However, this approach has limitations for the adult and 15-year-old reference ages due to the normalisation quantities used [i.e. KAP (P_{KA}), fluence, and air kerma]. For these reference ages, the male phantom has a different size from the female phantom. Consequently, the KAP, fluence, and air kerma required to achieve the same image quality for any examination will be different for males and females. This is not the case for the newborn, 1-, 5-, and 10-year-old phantoms, for which male and female anatomy and size are the same, besides the sex-specific organs.

(112) For the calculation of $\sum w_T \times H_T^M$, $\sum w_T \times H_T^F$, and effective dose, it should be noted that the current ICRP System of Radiological Protection uses a simplified set of tissue weighting factors, based on sex- and age-averaged relative detriment values, and specifies only two nominal detriment values: 5.7×10^{-2} Sv^{-1} for the whole population and 4.2×10^{-2} Sv^{-1} for adults (ICRP, 2007b). Thus, recognised differences in detriment and relative detriment (the contribution of the various organs and tissues to total detriment) as a function of age at exposure are not considered, other than in the differences between the two nominal detriment values (ICRP, 1991, 2007b). Thus, differences in effective dose coefficients as a function of age shown in this publication relate only to differences in physical size and organ masses, and do not address differences in detriment per Sv. Similarly, differences in organ absorbed dose coefficients do not inform on differences in stochastic risk per Gy as a function of age at exposure. These are issues that ICRP plans to consider in the future. However, it should also be kept in mind that patients are a different group to the general population, and may have existing pathologies, the aetiology for which could be related erroneously to the use of radiation. For guidance on the use of effective dose for patients, see *Publication 147* (ICRP, 2021). Further advice on the meaningful interpretation of effective dose in medicine can be found in Martin (2020), who describes the approximations made in the derivation of effective dose for medical exposures, emphasises the uncertainties in calculating the subsequent risk, and warns not to overstate its accuracy.

8. CONCLUSIONS

(113) There is a growing demand from patients, families, medical professionals, and relevant regulatory agencies to document patient exposures for diagnostic medical imaging procedures (Council of the European Union, 2014; IAEA, 2018). Ethical and professional considerations around accountability, transparency, and patient-centredness emphasise the importance of these requirements. Decisions around justification and optimisation require prospective knowledge of patient doses. A standardised methodology to determine patient doses will be of considerable benefit in all of these instances, facilitating and possibly simplifying decisions involving optimisation and justification. The study of variations in representative patient doses resulting from differences in equipment, radiographic technique, and other factors encountered in different countries can be very instructive in radiation protection practice, and will benefit from a standardised methodology.

(114) This publication describes the ICRP approach to developing a standardised method for the determination of organ absorbed and effective dose coefficients for radiographic imaging for a range of examinations commonly performed on adults and children. Radiographic examinations for which coefficients have been developed were primarily chosen based on their frequency. The projections used in the simulations were defined in conjunction with expert clinical input as described in Section 3 and Annex B. The coefficients have been produced for the adult male and female ICRP reference voxel phantoms, as well as for the suite of ICRP reference paediatric voxel phantoms. These dose coefficients are therefore not intended to be, and should not be, used for individual patient dosimetry.

(115) This is the first set of such coefficients published by the Commission, and the publication itself is the first of a series reporting organ absorbed and effective doses for various diagnostic x-ray modalities. Future publications will consider CT, fluoroscopically guided interventional radiology, and diagnostic paediatric fluoroscopy. The work will also be extended to consider the case of pregnant and non-reference patients, when mesh rather than voxel format phantoms will be utilised.

(116) Organ absorbed dose coefficients are provided for both male and female phantoms for well-defined exposure fields. Effective dose is a sex-averaged quantity, so this publication also provides, for each examination considered, a value of the coefficient for either $\sum w_T \times H_T^M$ or $\sum w_T \times H_T^F$, depending on whether the simulation of the examination is performed on the male or female phantom. As described in previous sections, the effective dose coefficient for an examination can be obtained by taking the average of the two.

(117) This publication is aimed at all those who have an interest in patient dosimetry and optimisation, including national authorities responsible for dose calculations, medical physicists, research scientists, radiologists, radiographers, and other radiological practitioners. For those readers who prefer an 'off-the-shelf' solution, dose coefficients are provided for a range of examinations and projections for a selection of common beam qualities. The dose coefficients are normalised to fluence,

air kerma at 1 m, and KAP (P_{KA}). They can be accessed easily using the dose viewer or from the electronic annex, as described in Annex D and Section 6 which provides worked examples. For those users who prefer to either fine-tune their results or explore the impact of novel beam qualities, monoenergetic coefficients are provided at 1-keV intervals in the electronic annex. These coefficients can be convolved with any x-ray spectrum the user chooses to derive the appropriate dose coefficients. Step-by-step instructions on this process are given in Section 6.2.

(118) Uncertainties in the dose coefficients resulting from Monte Carlo calculations performed for the reference exposure conditions (i.e. reference phantoms, defined exposures for good radiographic practice of a reference-size individual) are <5% for organs inside the beam, but can be up to 20% for small organs outside the beam and energies <10 keV (see Sections 4.1 and 7.1). However, the coefficients as issued by ICRP are reference data and have, as such, no uncertainties, per the guidance of the Joint Committee for Guides in Metrology (JCGM, 2008).

(119) The variations of the reference dose coefficients to those for real situations involving a real individual rather than a reference person or differing exposure conditions than those simulated for this work, can be large. For example, the use of different voxel phantoms, representing different individuals of different stature, can lead to differences in organ dose coefficients of up to a factor of 3, depending on the exposure conditions and the position of the organ in respect to the field. The influence of factors such as the measurement of normalising quantities on dose estimates resulting from application of the coefficients have been discussed in Section 7, as have limitations in the interpretation of results.

(120) Taking into account the age at exposure, which introduces an approximate variation of a factor of 3, effective dose can be considered an approximate indicator of the stochastic risk. This approximation is sufficient to inform judgements and the assessment of possible risks in the medical uses of radiation and communication with patients (ICRP, 2021).

(121) It is important to realise that these uncertainties and limitations cannot be ignored, and should always be considered when using the methodology to evaluate patient doses. A good understanding of the underlying principles is essential to avoid assigning a degree of precision to the dose estimates that is misleadingly high.

REFERENCES

Agostinelli, S., Allison, J., Amako, K., et al., 2003. Geant4 – a simulation toolkit. Nucl. Instrum. Methods Phys. Res. A. 506, 250–303.

Allison, J., Amako, K., Apostolakis, J., et al., 2016. Recent developments in Geant4. Nucl. Instrum. Methods Phys. Res. A. 835, 186–225.

Berger, M.J., Hubbell, J.H., 1987. XCOM: Photon Cross Sections on a Personal Computer. NBSIR 87-3597. National Bureau of Standards, Gaithersburg, MD.

Bolch, W., Lee, C., Wayson, M., Johnson, P., 2010. Hybrid computational phantoms for medical dose reconstruction. Radiat. Environ. Biophys. 49, 155–168.

Cassola, V.F., de Melo Lima, V.J., Kramer, R., et al., 2010. FASH and MASH: female and male adult human phantoms based on polygon mesh surfaces: I. Development of the anatomy. Phys. Med. Biol. 55, 133–162.

Committee to Assess Health Risks from Exposure to Low Levels of Ionizing Radiation, 2006. Health Risks from Exposure to Low Levels of Ionizing Radiation: BEIR VII Phase 2. National Academies Press, Washington, DC.

Council of the European Union, 2014. Council Directive 2013/59/Euratom of 5 December 2013 laying down basic safety standards for protection against the dangers arising from exposure to ionising radiation, and repealing Directives 89/618/Euratom, 90/641/Euratom, 96/29/Euratom, 97/43/Euratom and 2003/122/Euratom. Off. J. Eur. Union L 013, 1–73.

Cristy, M., Eckerman, K.F., 1987. Specific Absorbed Fractions of Energy at Various Ages from Internal Photon Sources, Part I–VII. ORNL Report TM-8381/V1-TM-8381/V7. Oak Ridge National Laboratory, Oak Ridge, TN.

Cullen, D.H., Hubbell, J., Kissel, L., 1997. EPDL97: the Evaluated Photo Data Library '97 Version. Lawrence Livermore National Laboratory, Livermore, CA.

Dorfman, A.L., Fazel, R., Einstein, A.J., et al., 2011. Use of medical imaging procedures with ionizing radiation in children: a population-based study. Arch. Pediatr. Adolesc. Med. 165, 458–464.

Drexler, G., Panzer, W., Widenmann, L., et al., 1985. Die Bestimmung von Organdosen in der Röntgendiagnostik. H. Hoffmann Verlag, Berlin.

Drexler, G., Panzer, W., Widenmann, L., et al., 1990. The Calculation of Dose from External Photon Exposures Using Reference Human Phantoms and Monte Carlo Methods, Part III: Organ Doses in X-ray Diagnosis. GSF-Report 11/90. GSF – National Research Center for Environment and Health, Neuherberg.

Drexler, G., Panzer, W., Stieve, F-E., et al., 1993. Die Bestimmung von Organdosen in der Röntgendiagnostik. H. Hoffmann Verlag, Berlin.

European Commission, 2015. Medical Radiation Exposure of the European Population; Part 1/2. European Commission, Brussels.

European Commission, 2018. European Guidelines on Diagnostic Reference Levels for Paediatric Imaging. Report 185. European Commission, Brussels.

Hart, D., Jones, D.G., Wall, B.F., 1994a. Normalised Organ Doses for Medical X-ray Examinations Calculated Using Monte Carlo Techniques. NRPB-SR-262. National Radiological Protection Board, Chilton.

Hart, D., Jones, D.G., Wall, B.F., 1994b. Estimation of Effective Dose in Diagnostic Radiology from Entrance Surface Dose and Dose-area Product Measurements. NRPB-R-262. National Radiological Protection Board, Chilton.

Hart, D., Jones, D.G., Wall, B.F., 1996a. Normalised Organ Doses for Paediatric X-ray Examinations Calculated Using Monte Carlo Techniques. NRPB-SR-279. National Radiological Protection Board, Chilton.

Hart, D., Jones, D.G., Wall, B.F., 1996b. Coefficients for Estimating Effective Doses from Paediatric X-ray Examinations. NRPB-R-279. National Radiological Protection Board, Chilton.

Hart, D., Wall, B.F., Hillier, M.C., Shrimpton, P.C., 2010. Frequency and Collective Doses for Medical and Dental X-ray Examinations in the UK, 2008. HPA-CRCE-012. Health Protection Agency, Chilton. Available at: https://www.gov.uk/government/publications/medical-and-dental-x-rays-frequency-and-collective-doses-in-the-uk (last accessed 9 February 2024).

Hough, M., Johnson, P., Rajon, D., et al., 2011. An image-based skeletal dosimetry model for the ICRP reference adult male – internal electron sources. Phys. Med. Biol. 56, 2309–2346.

Hourdakis, C.J., Boziari, A., Manetou, A., 2010. Performance evaluation of diagnostic radiology dosimeters in clinical and calibration x-ray beams. Health Phys. 98, 704–716.

Hubbell, J.H., Seltzer, S.M., 1995. Tables of X-ray Mass Attenuation Coefficients and Mass Energy-Absorption Coefficients 1 keV to 20 MeV for Elements $Z = 1$ to 92 and 48 Additional Substances of Dosimetric Interest. NISTIR 5632. US Department of Commerce, National Institute of Standards and Technology, Gaithersburg, MD.

IAEA, 2007. Dosimetry in Diagnostic Radiology: an International Code of Practice. Technical Reports Series No. 457. International Atomic Energy Agency, Vienna.

IAEA, 2018. Radiation Protection and Safety in Medical Uses of Ionizing Radiation. IAEA Safety Standards Series No. SSG-46. International Atomic Energy Agency, Vienna.

ICRP, 1975. Report on the Task Group on Reference Man. ICRP Publication 23. Pergamon Press, Oxford.

ICRP, 1988. Radiation dose to patients from radiopharmaceuticals. ICRP Publication 53. Ann. ICRP18(1-4).

ICRP, 1991. 1990 Recommendations of the International Commission on Radiological Protection. ICRP Publication 60. Ann. ICRP 21(1–3).

ICRP, 1992a. Addendum 1 to Publication 53–Radiation dose to patients from radiopharmaceuticals. In: ICRP Publication 62-Radiological protection in Biomedical Research. Ann. ICRP 22 (3), pp. 23–50.

ICRP, 1992b. Radiological protection in biomedical research. ICRP Publication 62. Ann. ICRP 22(3).

ICRP, 1995. Basic anatomical and physiological data for use in radiological protection: the skeleton. ICRP Publication 70. Ann. ICRP 25(2).

ICRP, 1998. Radiation dose to patients from radiopharmaceuticals. ICRP Publication 80. Ann. ICRP 28(3).

ICRP, 2002. Basic anatomical and physiological data for use in radiological protection: reference values. ICRP Publication 89. Ann. ICRP 32(3/4).

ICRP, 2007a. Managing patient dose in multi-detector computed tomography (MDCT). ICRP Publication 102. Ann. ICRP 37(1).

ICRP, 2007b. The 2007 Recommendations of the International Commission on Radiological Protection. ICRP Publication 103. Ann. ICRP 37(2–4).

ICRP, 2008. Radiation dose to patients from radiopharmaceuticals. ICRP Publication 106. Ann. ICRP 38(1/2).

ICRP, 2009. Adult reference computational phantoms. ICRP Publication 110. Ann. ICRP 39(2).
ICRP, 2010. Conversion coefficients for radiological protection quantities for external radiation exposures. ICRP Publication 116. Ann. ICRP 40(2–5).
ICRP, 2013. Assessment of radiation exposure of astronauts in space. ICRP Publication 123. Ann. ICRP 42(4).
ICRP, 2015a. Radiation dose to patients from radiopharmaceuticals: a compendium of current information related to frequently used substances. ICRP Publication 128. Ann. ICRP 44(2S).
ICRP, 2015b. Occupational intakes of radionuclides: Part 1. ICRP Publication 130. Ann. ICRP 44(2).
ICRP, 2016a. The ICRP computational framework for internal dose assessment for reference adults: specific absorbed fractions. ICRP Publication 133. Ann. ICRP 45(2).
ICRP, 2016b. Occupational intakes of radionuclides: Part 2. ICRP Publication 134. Ann. ICRP 45(3/4).
ICRP, 2017. Occupational intakes of radionuclides: Part 3. ICRP Publication 137. Ann. ICRP 46(3/4).
ICRP, 2019. Occupational intakes of radionuclides: Part 4. ICRP Publication 141. Ann. ICRP 48(2/3).
ICRP, 2020a. Paediatric computational reference phantoms. ICRP Publication 143. Ann. ICRP 49(1).
ICRP, 2020b. Dose coefficients for external exposures to environmental sources. ICRP Publication 144. Ann. ICRP 49(2).
ICRP, 2021. Use of dose quantities in radiological protection. ICRP Publication 147. Ann. ICRP 50(1).
ICRP, 2022. Occupational intakes of radionuclides: Part 5. ICRP Publication 151. Ann. ICRP 51(1/2).
ICRP, 2023a. Optimisation of radiological protection in digital radiology techniques for medical imaging. ICRP Publication 154. Ann. ICRP 52(3).
ICRP, 2023b. Specific absorbed fractions for reference paediatric individuals. ICRP Publication 155. Ann. ICRP 52(4).
ICRP, 2024a. Ethics in radiological protection for patients in diagnosis and treatment. ICRP Publication 157. Ann. ICRP 53(3).
ICRP, 2024b. Practical aspects in optimisation of radiological protection in digital radiography, fluoroscopy, and CT. ICRP Publication 160. Ann. ICRP 53(7).
ICRU, 1992. Photon, Electron, Proton and Neutron Interaction Data for Body Tissues. ICRU Report 46. International Commission on Radiation Units and Measurements, Bethesda, MD.
ICRU, 2005. Patient Dosimetry for X Rays Used in Medical Imaging. ICRU Report 74. International Commission on Radiation Units and Measurements, Bethesda, MD.
IEC, 2020. Medical Electrical Equipment – Dose Area Product Meters. IEC 60580:2020. International Electrotechnical Commission, Geneva.
JCGM, 2008. International Vocabulary of Metrology – Basic and General Concepts and Associated Terms (VIM). JCGM 200:2008. Joint Committee for Guides in Metrology, Sévres.

Johnson, P.B., Bahadori, A.A., Eckerman, K.F., et al., 2011. Response functions for computing absorbed dose to skeletal tissues from photon irradiation – an update. Phys. Med. Biol. 56, 2347–2365.

Jones, D.G., Wall, B.F., 1985. Organ Doses from Medical X-ray Examinations Calculated Using Monte Carlo Techniques. NRPB Report 186. National Radiological Protection Board, Chilton.

Kawrakow, I., Mainegra-Hing, E., Rogers, D.W.O., et al., 2009. The EGSnrc Code System: Monte Carlo Simulation of Electron and Photon Transport. PIRS Report 701. National Research Council of Canada, Ottawa.

Kramer, R., Zank, M., Williams, G., et al., 1982. The Calculation of Dose from External Photon Exposures Using Reference Human Phantoms and Monte Carlo Methods, Part I: The Male (Adam) and Female (Eva) Adult Mathematical Phantoms. GSF-Report S-885. GSF – National Research Center for Environment and Health, Neuherberg.

Kramer, R., Khoury, H.J., Vieira, J.W., et al., 2006. MAX06 and FAX06: update of two adult human phantoms for radiation protection dosimetry. Phys. Med. Biol. 51, 3331–3346.

Kramer, R., Khoury, H.J., Vieira, J.W., 2008. CALDose_X – a software tool for the assessment of organ and tissue absorbed doses, effective dose and cancer risks in diagnostic radiology. Phys. Med. Biol. 53, 6437–6459.

Kramer, R., Cassola, V.F., Khoury, H.J., et al., 2010. FASH and MASH: female and male adult human phantoms based on polygon mesh surfaces: II. Dosimetric calculations. Phys. Med. Biol. 55, 163–189.

Le Heron, J.C., 1992. Effective dose in diagnostic radiology. J. Radiol. Prot. 12, 111–114.

Lee, C., Lodwick, D., Hurtado, J., et al., 2010. The UF family of reference hybrid phantoms for computational radiation dosimetry. Phys. Med. Biol. 55, 339–363.

Lee, C., Lamart, S., Moroz, B.E., 2013. Computational lymphatic node models in pediatric and adult hybrid phantoms for radiation dosimetry. Phys. Med. Biol. 58, N59–N82.

Lee, C., 2021. A review of organ dose calculation tools for patients undergoing computed tomography scans. J. Radiat. Prot. Res. 46, 151–159.

Madore, B., Belsley, G., Cheng, C.C., et al., 2022. Ultrasound-based sensors for respiratory motion assessment in multimodality PET imaging. Phys. Med. Biol. 67.

Malusek, A., Sandborg, M., Carlsson, G.A., 2016. Accurate KAP meter calibration as a prerequisite for optimisation in projection radiography. Radiat. Prot. Dosimetry 169, 353–359.

Martin, C.J., 2007. An evaluation of semiconductor and ionization chamber detectors for diagnostic x-ray dosimetry measurements. Phys. Med. Biol. 52, 4465–4480.

Martin, C.J., 2020. Effective dose in medicine. Ann. ICRP 49(S1), 126–140.

Martin, C.J., Sutton, D.G., Sharp, P.F., 1999. Balancing patient dose and image quality. Appl. Radiat. Isot. 50, 1–19.

Martin, C.J., Abuhaimed, A., Lee, C., 2021. Dose quantities for measurement and comparison of doses to individual patients in computed tomography (CT). J. Radiol. Prot. 41, 792.

NCRP, 2019. Medical Radiation Exposure of Patients in the United States. NCRP Report 184. National Council on Radiation Protection and Measurements, Bethesda, MD.

Nelson, W.R., Hirayama, H., Rogers, D.W.O., 1985. The EGS4 Code System. SLAC Report 265. Stanford Linear Accelerator Center, Stanford, CA.

Pafundi, D.H., 2009. Image-based Skeletal Tissue and Electron Dosimetry Models for the ICRP Reference Pediatric Age Series. University of Florida, Gainesville, FL.

Perkins, S.T., Cullen, D., Seltzer, S.M., 1991a. Tables and Graphs of Atomic Subshell and Relaxation Data Derived from the LLNL Evaluated Atomic Data Library (EADL), Z=1–100. UCRL-50400-Vol. 30. Lawrence Livermore National Laboratory, Livermore, CA.

Perkins, S.T., Cullen, D., Seltzer, S.M., 1991b. Tables and Graphs of Electron-interaction Cross Sections from 10 eV to 100 GeV Derived from the LLNL Evaluated Electron Data Library (EEDL), Z = 1 to 100. UCRL-50400-Vol. 31. Lawrence Livermore National Laboratory, Livermore, CA.

Petoussi-Henss, N., Panzer, W., Zankl, M., et al., 1995. Dose-area product and body doses. Radiat. Prot. Dosimetry 57, 363–366.

Petoussi-Henss, N., Zankl, M., Fill, U., Regulla, D., 2002. The GSF family of voxel phantoms. Phys. Med. Biol. 47, 89–106.

Petoussi-Henss, N., Zankl, M., Panzer, W., 2005. Estimation of organ doses in radiology using voxel models describing different patients. Biomed. Tech. (Berl). 50, 664–665.

Petoussi-Henss, N., Zankl, M., Schlattl, H., et al., 2006. Organdosiskonversionsfaktoren für Röntgendiagnostische Untersuchungen anhand von Monte-Carlo-Simulationen. Jahrestagung der Deutschen Gesellschaft für Medizinische Physik e.V, Regensburg.

Poludniowski, G.G., 2007. Calculation of x-ray spectra emerging from an x-ray tube. Part II. X-ray production and filtration in x-ray targets. Med. Phys. 34, 2175–2186.

Poludniowski, G.G., Evans, P.M., 2007. Calculation of x-ray spectra emerging from an x-ray tube. Part I. Electron penetration characteristics in x-ray targets. Med. Phys. 34, 2164–2174.

Poludniowski, G., Landry, G., DeBlois, F., et al., 2009. SpekCalc: a program to calculate photon spectra from tungsten anode x-ray tubes. Phys. Med. Biol. 54, N433–N438.

Rannikko, S., Ermakov, I., Lampinen, J.S., et al., 1997. Computing patient doses of x-ray examinations using a patient size- and sex-adjustable phantom. Br. J. Radiol. 70, 708–718.

Rosenstein, M., 1976. Organ Doses in Diagnostic Radiology. Bureau of Radiological Health, Rockville, MD.

Rosenstein, M., 1988. Handbook of Selected Tissue Doses for Projections Common in Diagnostic Radiology. Food and Drug Administration, Rockville, MD.

Rosenstein, M., Beck, T.J., Warner, G.G., 1979. Handbook of Selected Organ Doses from Projections Common in Paediatric Radiology. Bureau of Radiological Health, Rockville, MD.

Sanchez, R., Vañó, E., Fernandez Soto, J.M., et al., 2020. Updating national diagnostic reference levels for interventional cardiology and methodological aspects. Phys. Med. 70, 169–175.

Sato, K., Endo, A., Saito, K., 2008. Dose Conversion Coefficients Calculated Using a Series of Adult Japanese Voxel Phantoms Against External Photon Exposure. JAEA-Data/Code 2008-016. Japan Atomic Energy Agency, Ibaraki.

Schlattl, H., Zankl, M., Becker, J., et al., 2012. Dose conversion coefficients for paediatric CT examinations with automatic tube current modulation. Phys. Med. Biol. 57, 6309–6326.

Schultz, F.W., Geleijns, J., Zoetelief, J., 1994. Calculation of dose conversion factors for posterior-anterior chest radiography of adults with a relatively high-energy X-ray spectrum. Br. J. Radiol. 67, 775–785.

Schümann, J., Paganetti, H., Shin, J., et al., 2012. Efficient voxel navigation for proton therapy dose calculation in TOPAS and Geant4. Phys. Med. Biol. 57, 3281–3293.

Seidenbusch, M., Rösenberger, V., Schneider, K., 2019. Radiation dose and dose reconstruction in conventional paediatric radiology. In: Seidenbusch, M., Rösenberger, V.,

Schneider, K. Imaging Practice and Radiation Protection in Pediatric Radiology: Conventional Radiography. Springer International Publishing, Cham, pp. 25–44.
Seltzer, S.M., 1993. Calculation of photon mass energy-transfer and mass energy-absorption coefficients. Radiat. Res. 136, 147–170.
Servomaa, A., Tapiovaara, M., 1998. Organ dose calculation in medical x ray examinations by the program PCXMC. Radiat. Prot. Dosimetry 80, 213–219.
Snyder, W.S., Ford, M.R., Warner, G.G., et al., 1969. Estimates of absorbed fractions for monoenergetic photon sources uniformly distributed in various organs of a heterogeneous phantom. J. Nucl. Med. 10, 7–52.
Sutton, D.G., Reilly, A.R., Cranley, A.J.K., Gilmore, B.J. et al., 2015. Catalogue of Diagnostic X-ray Spectra and Other Data: Electronic Version, second ed. Institute of Physics and Engineering in Medicine, York.
Tapiovaara, M., Siiskonen, T., 2008. PCXMC, a Monte Carlo Program for Calculating Patient Doses in Medical X-ray Examinations. Radiation and Nuclear Safety Authority in Finland, Vantaa.
Toroi, P., Komppa, T., Kosunen, A., 2008a. A tandem calibration method for kerma-area product meters. Phys. Med. Biol. 53, 4941–4958.
Toroi, P., Komppa, T., Kosunen, A., et al., 2008b. Effects of radiation quality on the calibration of kerma-area product meters in x-ray beams. Phys. Med. Biol. 53, 5207–5221.
UNSCEAR, 2010. Sources and Effects of Ionising Radiation. Volume 1, Annex A – Medial Radiation Exposures. UNSCEAR 2008 Report. United Nations Scientific Committee on the Effects of Atomic Radiation, New York.
UNSCEAR, 2022. Sources, Effects and Risks of Ionizing Radiation, United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR) 2020/2021 Report, Volume I: Report to the General Assembly, with Scientific Annex A – Evaluation of Medical Exposure to Ionizing Radiation. United Nations Scientific Committee on the Effects of Atomic Radiation, New York.
Veit, R., Zankl, M., 1992. Influence of patient size on organ doses in diagnostic radiology. Radiat. Prot. Dosimetry 43, 241–243.
Veit, R., Zankl, M., 1993. Variation of organ doses in paediatric radiology due to patient diameter, calculated with phantoms of varying voxel size. Radiat. Prot. Dosimetry 49, 353–356.
Wayson, M.B., 2012. Computational Internal Dosimetry Methods as Applied to the University of Florida Series of Hybrid Phantoms. University of Florida, Gainesville, FL.
Whitley, A.S., Jefferson, G., Holmes, K., et al., 2015. Clark's Positioning in Radiography. 13E. CRC Press, London.
Winslow, M., Huda, W., Xu, X.G., et al., 2004. Use of the VIP-Man model to calculate energy imparted and effective dose for x-ray examinations. Health Phys. 86, 174–182.
Worrall, M., Sutton, D.G., 2015. Developing a method and deriving an uncertainty budget for the internal calibration of dosemeters for radiographic equipment. J. Radiol. Prot. 35, 209–222.
Xu, X.G., Eckerman, K.F. (Eds.), 2010. Handbook of Anatomical Models for Radiation Dosimetry. CRC Press, Boca Raton, FL.
Yeom, Y.S., Wang, Z.J., Nguyen, T.T., et al., 2016. Development of skeletal system for mesh-type ICRP reference adult phantoms. Phys. Med. Biol. 61, 7054–7073.
Zankl, M., 1993. Computational models employed for dose assessment in diagnostic radiology. Radiat. Prot. Dosimetry 49, 339–344.

Zankl, M., 2010. The GSF voxel computational phantom family. In: Xu, X.G., Eckerman, K. F. (Eds.), Handbook of Anatomical Models for Radiation Dosimetry. Taylor & Francis, Boca Raton, FL.

Zankl, M., Wittmann, A., 2001. The adult male voxel model "Golem" segmented from whole body CT patient data. Radiat. Environ. Biophys. 40, 153–162.

Zankl, M., Fill, U., Petoussi-Henss, N., et al., 2002. Organ dose conversion coefficients for external photon irradiation of male and female voxel models. Phys. Med. Biol. 47, 2367–2385.

Zankl, M., Becker, J., Fill, U., et al., 2005. GSF Male and Female Adult Voxel Models Representing ICRP Reference Man – the Present Status. Monte Carlo 2005 Topical Meeting: The Monte Carlo Method: Versatility Unbounded in a Dynamic Computing World, 17–21 April 2005, Chattanooga, TN, USA.

ANNEX A. DOSIMETRIC QUANTITIES USED IN PATIENT DOSIMETRY IN RADIOGRAPHY

A.1. Organ absorbed dose and equivalent dose

(A 1) The mean absorbed dose averaged over the volume of organs and tissues is the primary scientific quantity from which effective dose is calculated. Absorbed dose, D, is defined as the quotient of mean energy imparted, $\mathrm{d}\bar{\varepsilon}$, by ionising radiation in a volume element and the mass, $\mathrm{d}m$, of the matter in that volume:

$$D = \frac{\mathrm{d}\bar{\varepsilon}}{\mathrm{d}m} \tag{A.1}$$

(A 2) The SI unit of absorbed dose is J kg^{-1} and its special name is gray (Gy). Absorbed dose is derived from the mean value of the stochastic quantity of energy imparted, ε, and does not reflect the random fluctuations of the interaction events in tissue. While it is defined at any point in matter, its value is obtained as an average over a mass element, $\mathrm{d}m$, and hence over many atoms or molecules of matter.

(A 3) When using the quantity 'absorbed dose' in radiological protection, doses are averaged over tissue volumes. It is assumed that for low doses, the mean value of absorbed dose averaged over a specific organ or tissue can be correlated with radiation detriment for stochastic effects in that tissue with an accuracy sufficient for the purposes of radiological protection. The averaging of absorbed dose is carried out over the volume of a specified organ (e.g. liver) or tissue (e.g. active bone marrow) or the sensitive region of a tissue (e.g. endosteal surfaces of the skeleton).

(A 4) Equivalent dose, H_{T}, to a tissue or organ is defined as:

$$H_{\mathrm{T}} = \sum_{R} w_{\mathrm{R}} D_{\mathrm{T,R}} \tag{A.2}$$

where w_{R} is the radiation weighting factor for radiation type R, and $D_{\mathrm{T,R}}$ is the mean absorbed dose from the incident radiation type R on the body for a tissue or organ T of the age-specified reference male or female. As w_{R} is dimensionless, the SI unit for the equivalent dose is the same as for absorbed dose, J kg^{-1}, and its special name is sievert (Sv). Values of w_{R} are shown in Table A.1 and are taken from *Publication 103* (ICRP, 2007).

Table A.1. International Commission on Radiological Protection radiation weighting factors (ICRP, 2007).

Radiation type	Radiation weighting factor, w_R
Photons	1
Electrons and muons	1
Protons and charged pions	2
Alpha particles, fission fragments, heavy ions	20
Neutrons	Continuous function of neutron energy See Eq. 4.3 of *Publication 103* (ICRP, 2007)

A.2. Effective dose

(A 5) Effective dose, E, introduced in *Publication 60* (ICRP, 1991), is the risk-related quantity in radiological protection, and is defined as a weighted sum of organ equivalent doses. In accordance with the definition of effective dose in *Publication 103* (ICRP, 2007), E is computed as:

$$E = \sum w_T \left[\frac{H_T^M + H_T^F}{2} \right] \tag{A.3}$$

where H_T^M and H_T^F are the equivalent doses to the tissues or organs T of the reference male and female, respectively, and w_T is the tissue weighting factor for target tissue T, with $\Sigma w_T = 1$. The sum is performed over all organs and tissues of the human body considered to be sensitive to the induction of stochastic effects. Values of w_T are given in Table A.2 (ICRP, 2007). As w_R and w_T are dimensionless, the SI unit for effective dose is the same as for absorbed and equivalent dose, J kg^{-1}, and its special name is sievert (Sv).

(A 6) Eq. A.3 is numerically the same as:

$$E = \frac{E_M + E_F}{2} \tag{A.4}$$

where $E_M = \sum w_T \times H_T^M$ and $E_F = \sum w_T \times H_T^F$, respectively. In this publication, these contributions are annotated as $E_{103,AM}$ and $E_{103,AF}$ for the adult male and female phantoms, respectively; $E_{103,15M}$ and $E_{103,15F}$ for the 15-year-old male and female phantoms, respectively; $E_{103,10M}$ and $E_{103,10F}$ for the 10-year-old male and female phantoms, respectively, and so on, and are given in the tables of the electronic supplement. Effective dose can be obtained by taking the average of the two values.

(A 7) Effective dose was originally introduced for the control of occupational exposures to external and internal sources of radiation. While the concept has remained essentially unchanged through *Publication 60* (ICRP, 1991) to

Publication 103 (ICRP, 2007), its use has been extended to members of the public of all ages, including in-utero exposures of the fetus (ICRP, 2001, 2004, 2006). ICRP provides effective dose coefficients for situations of external and internal exposures of workers and members of the public, and for radiopharmaceutical administrations to patients, as reference values for use in prospective and retrospective dose assessments.

Table A.2. International Commission on Radiological Protection tissue weighting factors (ICRP, 2007).

Tissue	w_T	$\sum w_T$
Bone marrow, breast, colon, lung, stomach, remainder tissues*	0.12	0.72
Gonads	0.08	0.08
Urinary bladder, oesophagus, liver, thyroid	0.04	0.16
Bone surface, brain, salivary glands, skin	0.01	0.04

w_T, tissue weighting factor.
*Remainder tissues: adrenals, extrathoracic regions of the respiratory tract, gallbladder, heart, kidneys, lymphatic nodes, muscle, oral mucosa, pancreas, prostate (male), small intestine, spleen, thymus, and uterus/cervix (female).

(A 8) The tissue weighting factors of Table A.2 are sex- and age-averaged values for all organs and tissues, including the male and female breast, testes, and ovaries (i.e. gonads, related to possible carcinogenic and heritable effects). This averaging implies that the application of this approach is restricted to the determination of effective dose in radiological protection (ICRP, 2007).

(A 9) Effective dose is calculated for sex-averaged reference persons at specified ages as defined in *Publication 89* (ICRP, 2002). The *Publication 103* (ICRP, 2007) definition includes the specification of reference male and female anatomical phantoms for radiation transport calculations. While exposures may relate to individuals or population groups, effective dose is calculated for reference persons exposed in the same way.

(A 10) Effective dose, in units of sievert (Sv), is accepted internationally as the central radiological protection quantity and is used for regulatory purposes worldwide, providing a risk-adjusted measure of total body dose from both external and internal sources in relation to stochastic risks of cancer and hereditary effects, expressed in terms of detriment. Considering the uncertainties associated with risk projection to low doses or low dose rates, effective dose may be considered as an approximate indicator of possible risk, recognising also that lifetime cancer risk varies with age at exposure, sex, and population group. Effective dose has proved to be a valuable and robust quantity for use in the optimisation of protection, and in the setting of control criteria such as dose limits, constraints, and reference levels for workers, research subjects, or members of the public.

(A 11) *Publication 147* (ICRP, 2021) provides additional guidance for the use of effective dose to estimate risks, particularly in evaluating exposures of patients from

medical procedures. In medical applications, estimates of effective dose can be used for comparing doses from different diagnostic and interventional imaging modalities (e.g. CT and nuclear medicine) and exposure techniques that give different spatial distributions of radiation within the body tissues. In this context, effective dose is used to provide a generic indicator for classifying different types of medical procedure into broad risk categories for the purpose of communicating risks to clinicians and patients. Effective dose is also used to inform decisions on justification of patient diagnostic and interventional procedures, planning requirements in research studies, and evaluation of unintended exposures. In each of these cases, effective dose provides an approximate indicator of possible risk. Thus, effective dose can be used prospectively as an indicator of radiation detriment in justification decisions, and when planning medical research studies involving radiation exposure, or retrospectively in assessments of accidental exposures.

(A 12) Moreover, *Publication 147* (ICRP, 2021) lists the situations for medical procedures for which effective dose is not recommended to be used: measurable quantities are used directly in applications comparing doses from particular procedures in different health centres, including the setting of diagnostic reference levels and in the maintenance of patient records. For situations in which a single organ receives the majority of the dose, such as the breast in mammography or the thyroid from therapeutic administration of radioiodine, mean absorbed doses to the tissues of interest should be used rather than effective dose.

(A 13) For communication of doses and associated health risks, the reader is referred to *Publication 147* (ICRP, 2021).

A.3. Air kerma and associated quantities

(A 14) Kerma, K, for ionising uncharged particles, is given by:

$$K = \frac{\mathrm{d}E_{\mathrm{tr}}}{\mathrm{d}m} \tag{A.4}$$

where $\mathrm{d}E_{\mathrm{tr}}$ is the mean sum of the initial kinetic energies of all the charged particles liberated in the mass, $\mathrm{d}m$, of a material by the uncharged particles incident on $\mathrm{d}m$. The unit of kerma is J kg^{-1}, and has the special name gray (Gy). The quantity ($\mathrm{d}E_{\mathrm{tr}}$) includes the kinetic energy of the charged particles emitted in the decay of excited atoms/molecules or in nuclear de-excitation or disintegration.

(A 15) The air kerma-area product is the integral of the air kerma free-in-air (i.e. in the absence of backscatter) over the area of the x-ray beam in a plane perpendicular to the beam axis. In many medical publications, and also this publication, the acronym used for this quantity is KAP (measured in mGy cm^2 in this publication). The ICRU notation for this quantity is P_{KA} (ICRU, 2005). KAP is easily measurable and can be used as a normalisation quantity to derive relevant dosimetric quantities, such as organ and effective doses, using dose coefficients.

(A 16) The air kerma free-in-air at the reference plane, K_a, is the air kerma from the incident beam on the central x-ray beam axis at the centre of a reference plane without backscatter. Air kerma free-in-air can be calculated from the x-ray tube output, where output is measured using a calibrated ionising chamber. It can be used to determine the incident air kerma.

(A 17) The entrance surface air kerma is the air kerma on the central x-ray beam axis at the point where the x-ray beam enters the patient or phantom. It includes the contribution from backscattered photons.

(A 18) The incident air kerma is the air kerma from the incident beam on the central x-ray beam axis at the skin entrance plane. Only the primary radiation incident on the patient or phantom is included, and not the backscattered radiation (ICRU, 2005).

A.4. References

ICRP, 1991. 1990 Recommendations of the International Commission on Radiological Protection. ICRP Publication 60. Ann. ICRP 21(1–3).

ICRP, 1991. 1990 Recommendations of the International Commission on Radiological Protection. ICRP Publication 60. Ann. ICRP 21(1–3).

ICRP, 2001. Doses to the embryo and fetus from intakes of radionuclides by the mother. ICRP Publication 88. Ann. ICRP 31(1–3).

ICRP, 2002. Basic Anatomical and Physiological Data for Use in Radiological Protection Reference Values. ICRP Publication 89. Ann. ICRP 32(3–4).

ICRP, 2004. Doses to infants from ingestion of radionuclides in mothers' milk. ICRP Publication 95. Ann. ICRP 34(3/4).

ICRP, 2006. Assessing dose of the representative person for the purpose of the radiation protection of the public. ICRP Publication 101a. Ann. ICRP 36(3).

ICRP, 2007. The 2007 Recommendations of the International Commission on Radiological Protection. ICRP Publication 103. Ann. ICRP 37(2–4).

ICRP, 2021. Use of dose quantities in radiological protection. ICRP Publication 147. Ann. ICRP 50(1).

ICRU, 2005. Patient dosimetry for x rays used in medical imaging. ICRU Report 74. International Commission on Radiation Units and Measurements, Bethesda, MD.

ANNEX B. DEFINITIONS OF FIELDS FOR THE PROJECTIONS CONSIDERED

(B 1) In the following, the anatomical landmarks for field definition are shown for each examination. The fields are shown with their sizes at the centre of the phantoms.

(B 2) As described in Section 3.3, the projections used in the simulations were defined using an iterative process as shown in Fig. 3.1. Typical field sizes and positioning for each projection were obtained initially from a standard textbook of radiography, intended to teach student radiographers the proper way to obtain radiographs (Whitley et al., 2015). The information was then used, in conjunction with clinical input, to define initial positioning on the phantom.

(B 3) For defining the field borders with respect to the phantoms, information on the rectangular prisms containing the organs under consideration was used (i.e. the minimum and maximum columns, rows, and slabs occupied by the respective organs), together with the voxel dimensions. If specified in the geometry description, additional margins were added. For example, for a position '1–2 cm above dome of diaphragm', 1.5 cm was added to the topmost coordinate of the liver. In this way, the minimum and maximum z coordinates to be covered by the field were evaluated, and thus the field height and centre, where the latter marks the z coordinate of the radiation source. Next, the maximum extensions of the slab were assessed where the source z coordinate is located. For antero-posterior and postero-anterior projections, the minimum and maximum rows of that slab were assessed, and for lateral projections, the minimum and maximum columns were assessed. These mark the entrance and exit points of the central beam with respect to the body. If use of an anti-scatter grid was assumed, the image receptor plane was assumed to be 10 cm away from the beam exit point; if no grid was to be considered, the exit point marked the location of the image receptor plane. The source coordinate was then assumed at the focus-to-receptor distance on the opposite side of the body. For the lateral field borders, the same method was applied as for the superior and inferior borders (i.e. use of the rectangular prism containing the organs considered and adding margins if needed). Thus, the field width and centre were assessed, where the centre marks the third source coordinate. The field size determined was assumed to be at the centre of the phantom, in a plane between the beam entrance and exit points into and from the body. Similarly, the irradiation fields shown on the phantoms in this annex are at the centre of the phantoms.

(B 4) The position of each projection was then assessed by two experienced radiologists, and positioning and field definition were amended until there was agreement between them that both positioning and field definition were representative of good technique.

B.1. Adult radiography

(B 5) In adult radiography, anti-scatter grids are used for all examinations considered, thus causing a skin-to-detector distance of 10 cm.

B.1.1. Chest postero-anterior examination

Field height extension:	Extends from approx. 1 cm above clavicles and scapulae to lateral bottom of the rib cage (10th rib, costophrenic angles)
Field lateral extension:	Includes at least skin margins (in non-obese reference phantoms) and medial part (half) of the humeral head
Focus-to-detector distance:	180 cm
Skin-to-detector distance:	10 cm (grid)

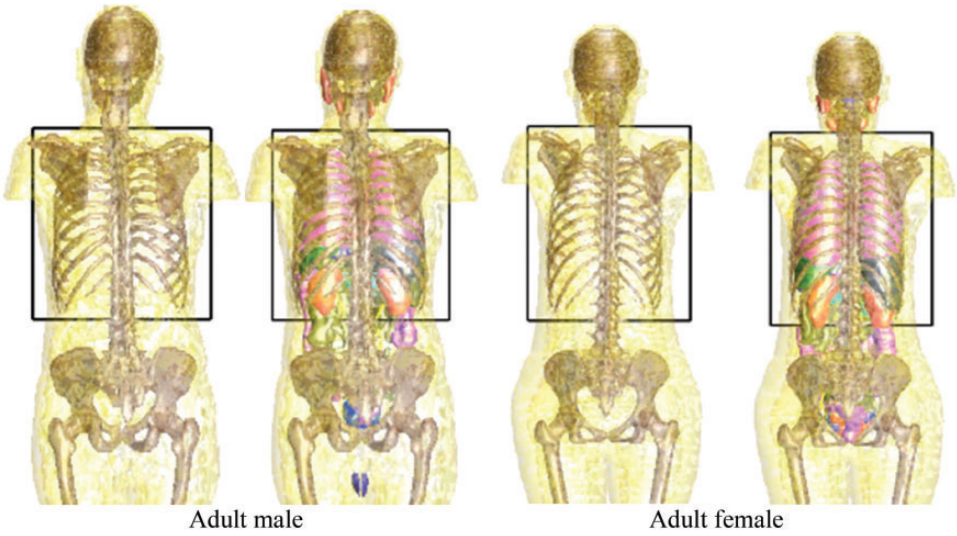


Fig. B.1. Coronal images at the centre of the adult male and female phantoms showing the anatomical landmarks for chest postero-anterior examination. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.1.2. Chest lateral examination

Field height extension:	Extends from approx. 1 cm above clavicles and scapulae to lateral bottom of the rib cage (10th rib, costophrenic angles)
Field lateral extension:	Should include at least skin margins (in non-obese reference phantoms)
Focus-to-detector distance:	100 cm
Skin-to-detector distance:	10 cm (grid)

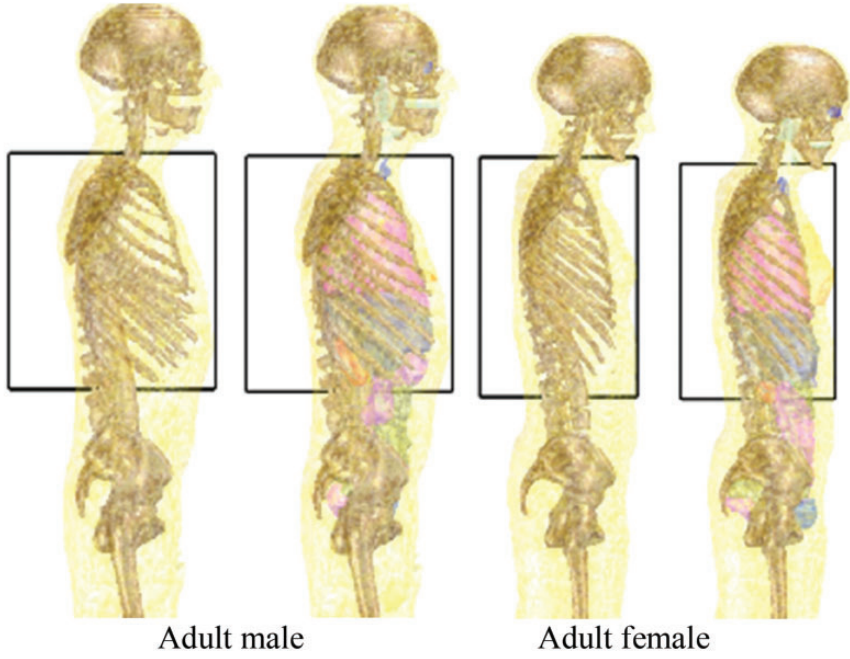


Fig. B.2. Sagittal images at the centre of the adult male and female phantoms showing the anatomical landmarks for chest lateral examination. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.1.3. Chest antero-posterior examination

Field height extension:	Extends from approx. 1 cm above clavicles and scapulae to lateral bottom of the rib cage (10th rib, costophrenic angles)
Field lateral extension:	Includes at least skin margins (in non-obese reference phantoms) and medial part (half) of the humeral head
Focus-to-detector distance:	132.6 and 130.4 cm for the adult male and female, respectively
Skin-to-detector distance:	10 cm (grid)

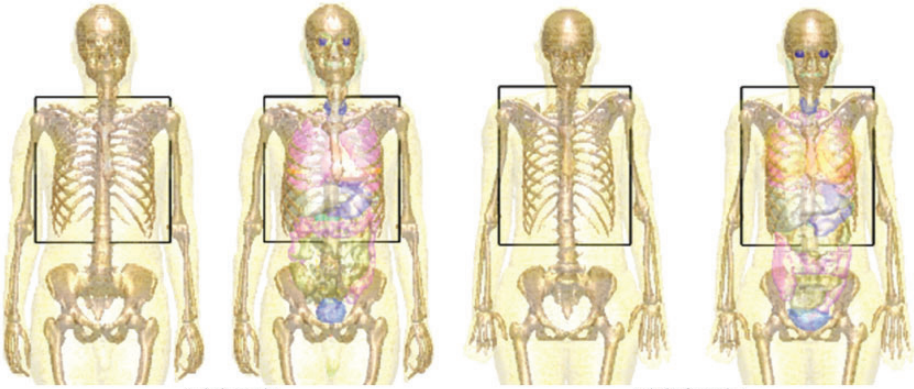


Fig. B.3. Coronal images at the centre of the adult male and female phantoms showing the anatomical landmarks for chest antero-posterior examination. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.1.4. Pelvic antero-posterior examination

Field height extension:	From 3 cm above iliac crest to lesser trochanter
Field lateral extension:	Should include the hips
Focus-to-detector distance:	115 cm
Skin-to-detector distance:	10 cm (grid)

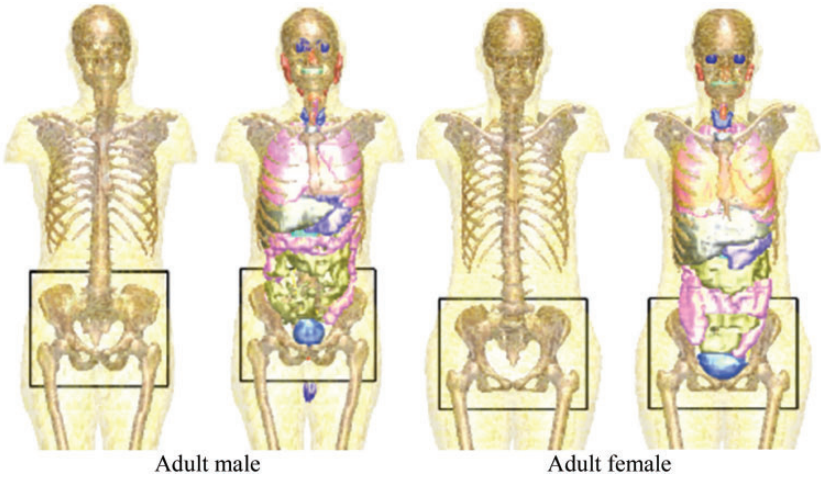


Fig. B.4. Coronal images at the centre of the adult male and female phantoms showing the anatomical landmarks for pelvic antero-posterior examination. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.1.5. Abdominal antero-posterior examination

Field height extension:	From 1–2 cm above dome of diaphragm to 1–2 cm below symphysis pubis
Field lateral extension:	Should extend approx. 2 cm beyond pelvis on either side (include skin margin except where widest)
Focus-to-detector distance:	100 cm
Skin-to-detector distance:	10 cm (grid)

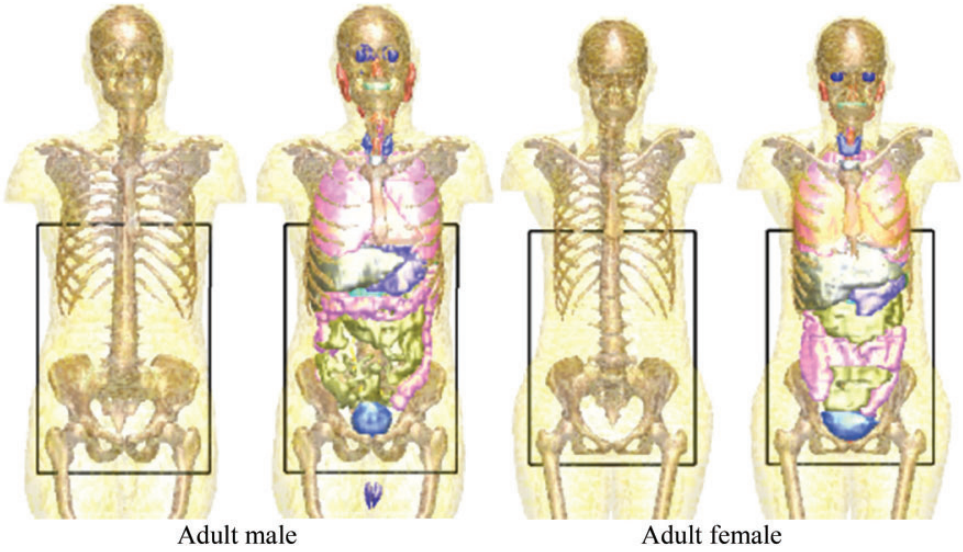


Fig. B.5. Coronal images at the centre of the adult male and female phantoms showing the anatomical landmarks for abdominal antero-posterior examination. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.1.6. Lumbar spine antero-posterior examination

Field height extension:	From T11 or approx. mid T10 to half sacrum (above coccyx)
Field lateral extension:	Should include at least all of the sacro-iliac joints
Focus-to-detector distance:	115 cm
Skin-to-detector distance:	10 cm (grid)

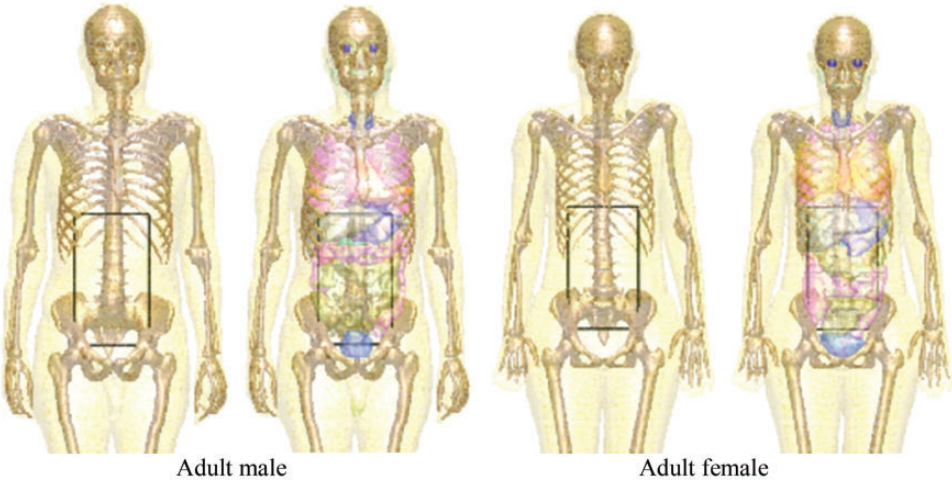


Fig. B.6. Coronal images at the centre of the adult male and female phantoms showing the anatomical landmarks for lumbar spine antero-posterior examination. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.1.7. Lumbar spine lateral examination

Field height extension:	From T11 or approx. mid T10 to half sacrum (above coccyx)
Field lateral extension:	Should extend approx. 2 cm beyond the bone on either side
Focus-to-detector distance:	115 cm
Skin-to-detector distance:	10 cm (grid)

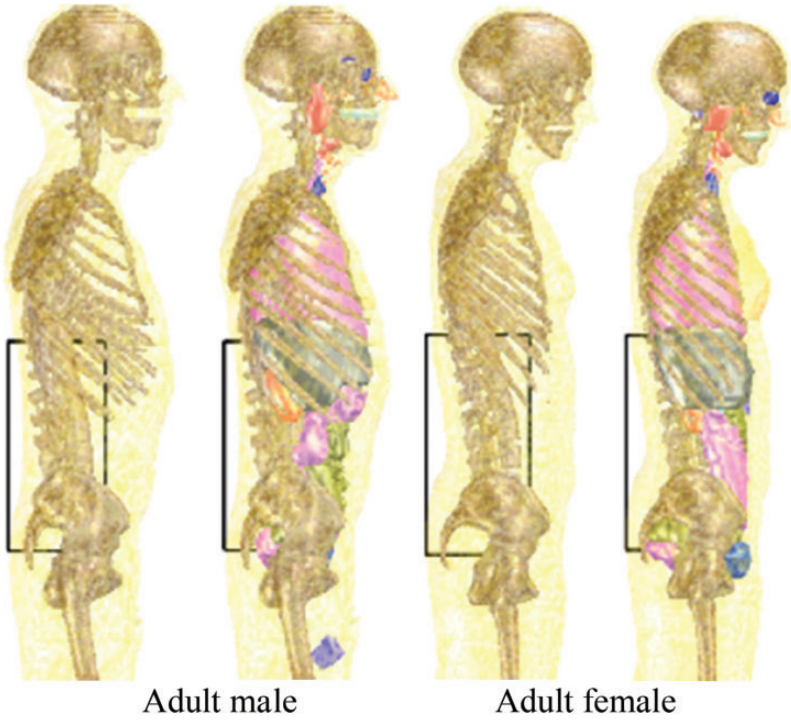


Fig. B.7. Sagittal images at the centre of the adult male and female phantoms showing the anatomical landmarks for lumbar spine lateral examination. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.1.8. Lumbar spine oblique examination (15° angulation)

Field height extension:	From T11 or approx. mid T10 to half sacrum (above coccyx)
Field lateral extension:	Should include at least all of the sacro-iliac joints
Focus-to-detector distance:	100 cm
Skin-to-detector distance:	10 cm (grid)

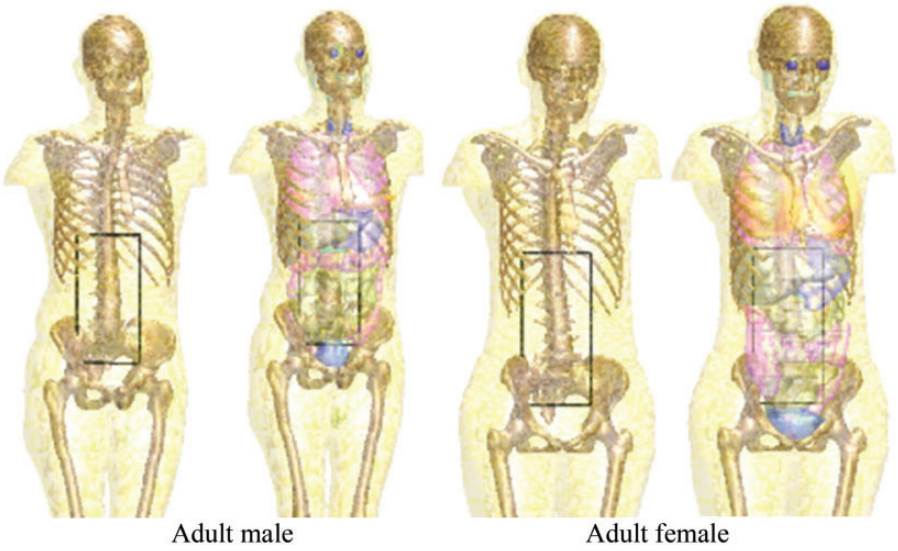


Fig. B.8. Coronal images at the centre of the adult male and female phantoms showing the anatomical landmarks for lumbar spine oblique (15°) examination. Left to right: the first image shows the left skeletal landmarks alone, and the second image adds the organs of interest. Note: the analogous right lumber spine oblique examination is not shown as the field positioning and size are similar.

B.1.9. Cervical spine antero-posterior examination (10° cranial angulation)

Field height extension:	From slightly above C1 to T2 – superiorly avoid orbit
Field lateral extension:	Should include all soft tissues of the neck
Focus-to-detector distance:	100 cm
Skin-to-detector distance:	10 cm (grid)

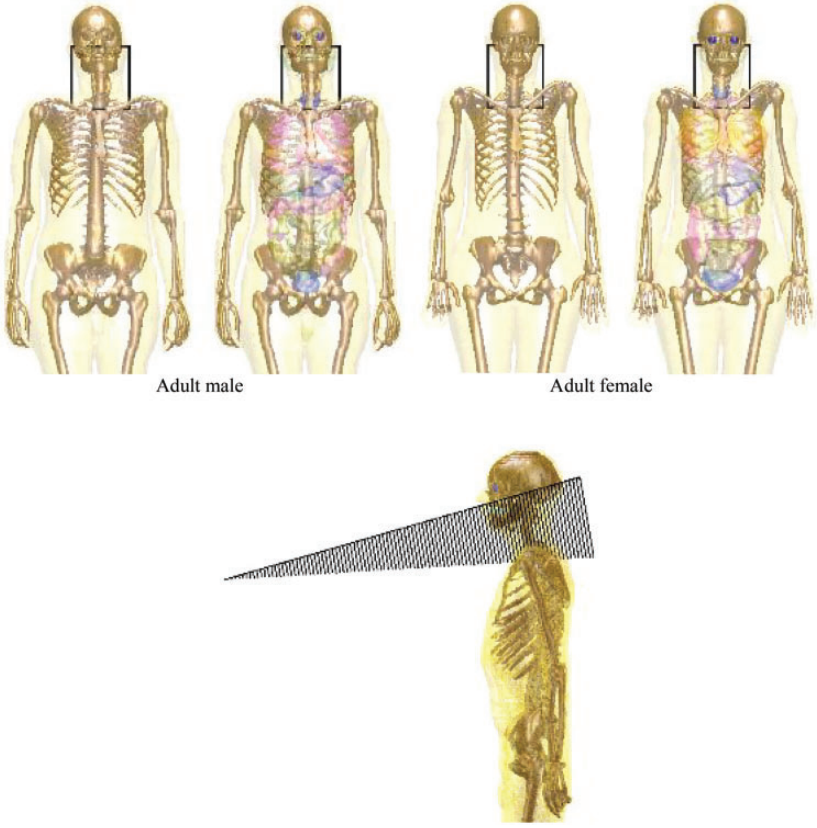


Fig. B.9. (Top) Coronal images at the centre of the adult male and female phantoms showing the anatomical landmarks for cervical spine anterior posterior (15° cranial angulation) examination. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest. (Bottom) Image demonstrating the beam angulation on the male phantom.

B.1.10. Cervical spine lateral examination

Field height extension:	From slightly above C1 to T1
Field lateral extension:	Should include all soft tissues of the neck
Focus-to-detector distance:	180 cm
Skin-to-detector distance:	10 cm (grid)

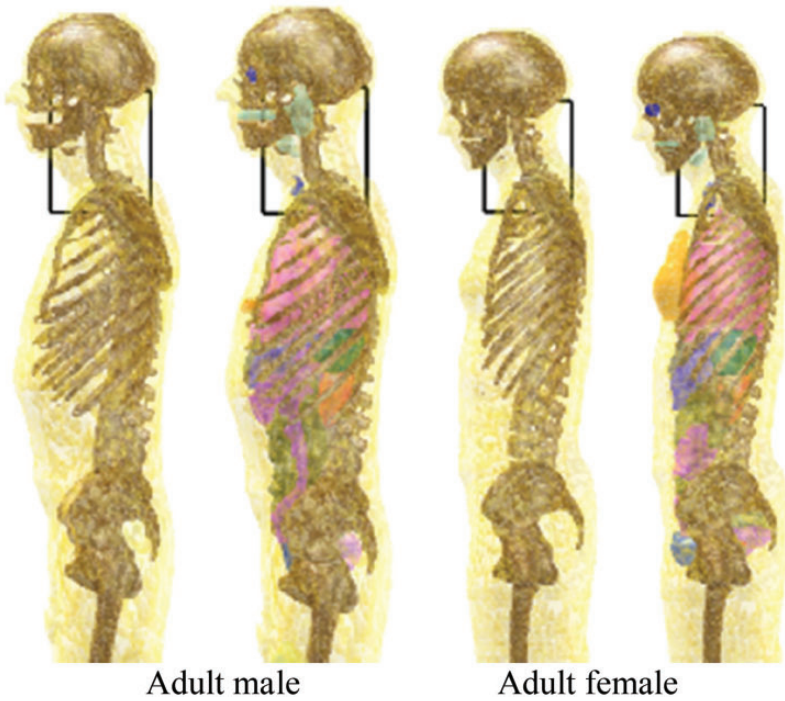


Fig. B.10. Sagittal images at the centre of the adult male and female phantoms showing the anatomical landmarks for cervical spine lateral examination. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.1.11. Thoracic spine antero-posterior examination

Field height extension:	From slightly above C7 to slightly below L2
Field lateral extension:	Should include approx. 5-cm margins on either side
Focus-to-detector distance:	100 cm
Skin-to-detector distance:	10 cm (grid)

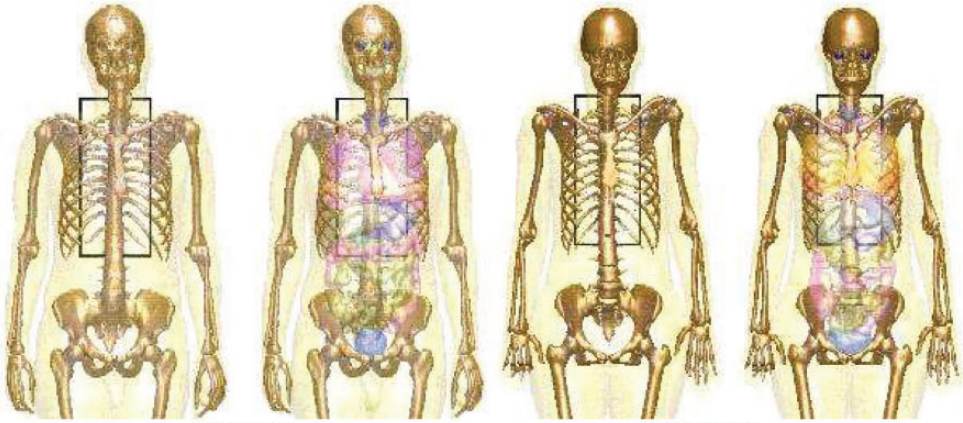


Fig. B.11. Coronal images at the centre of the adult male and female phantoms showing the anatomical landmarks for thoracic spine antero-posterior examination. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.1.12. Thoracic spine lateral examination

Field height extension:	Superior margin includes C7, inferior margin includes L1–2
Field lateral extension:	Field centre should be approx. 5 cm anterior to spinous process of T6–7 Anterior margin includes all vertebral bodies, and posterior margin includes spinous processes
Focus-to-detector distance:	100 cm
Skin-to-detector distance:	10 cm (grid)

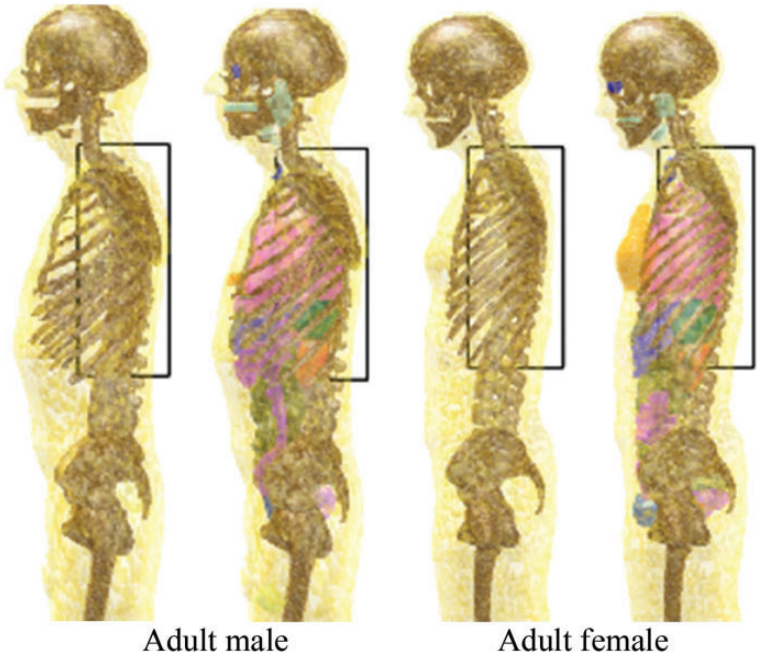


Fig. B.12. Sagittal images at the centre of the adult male and female phantoms showing the anatomical landmarks for thoracic spine lateral examination. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.2. Paediatric radiography

(B 6) For many examinations in paediatric radiography, anti-scatter grids are only used if the patient has at least a certain thickness (Knight, 2014). Hence, the skin-to-detector distance depends on the age of the reference computational phantoms.

(B 7) The male and female newborn, 1-, 5-, and 10-year-old phantoms are anatomically identical, except for their gender-specific organs and urinary bladder. Hence, for these ages, the male phantoms alone are shown in the following illustrations.

B.2.1. Chest antero-posterior examination

Field height extension:	Extends from approx. 1 cm above clavicles and scapulae to lateral bottom of the rib cage (10th rib, costophrenic angles)
Field lateral extension:	Includes at least skin margins (in non-obese reference phantoms) and medial part (half) of the humeral head
Focus-to-detector distance:	100 cm
Skin-to-detector distance:	Anti-scatter grid for patient thickness of at least 20 cm (Knight, 2014) 0 cm (no grid) for newborn, 1-, and 5-year-old 10 cm (grid) for 10- and 15-year-old

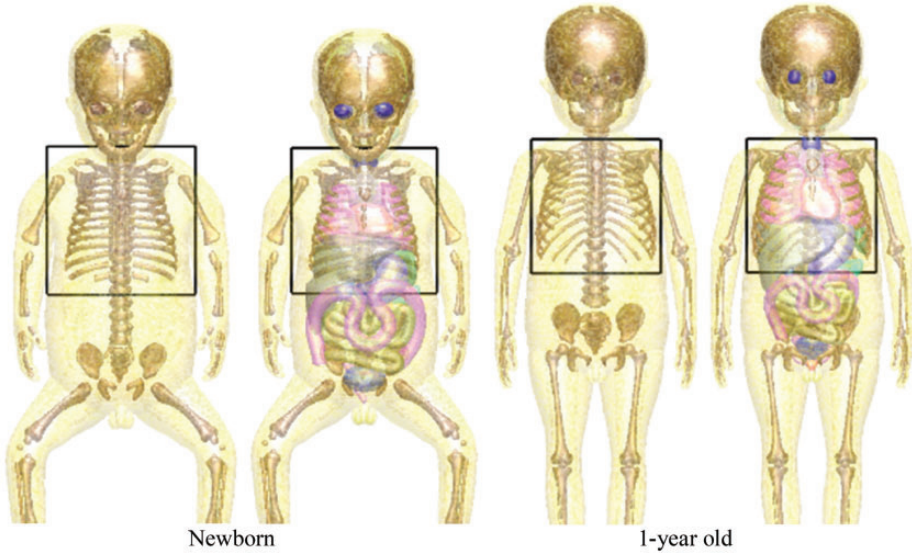


Fig. B.13. Coronal images at the centre of the paediatric reference phantoms showing the anatomical landmarks for chest antero-posterior examination. For newborn, 1-, 5-, and 10-year-old, the images for the male phantoms are shown. For 15-year-old (third row), images for the male and female are shown. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest (*continued on next page*).

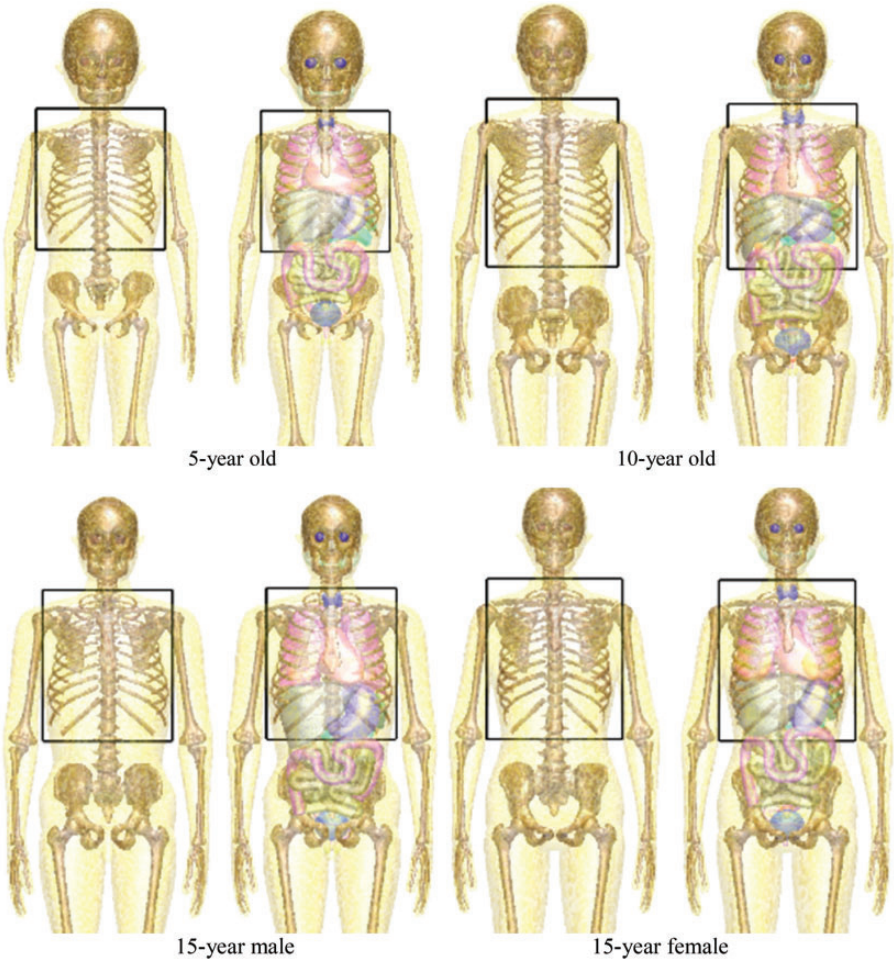


Fig. B.13. (*continued*).

B.2.2. Chest postero-anterior examination (5-, 10-, and 15-year-old)

Field height extension:	Extends from approx. 1 cm above clavicles and scapulae to lateral bottom of the rib cage (10th rib, costophrenic angles)
Field lateral extension:	Includes at least skin margins (in non-obese reference phantoms) and medial part (half) of the humeral head
Focus-to-detector distance:	180 cm
Skin-to-detector distance:	Anti-scatter grid for patient thickness of at least 20 cm (Knight, 2014) 0 cm (no grid) for 5-year-old 10 cm (grid) for 10- and 15-year-old

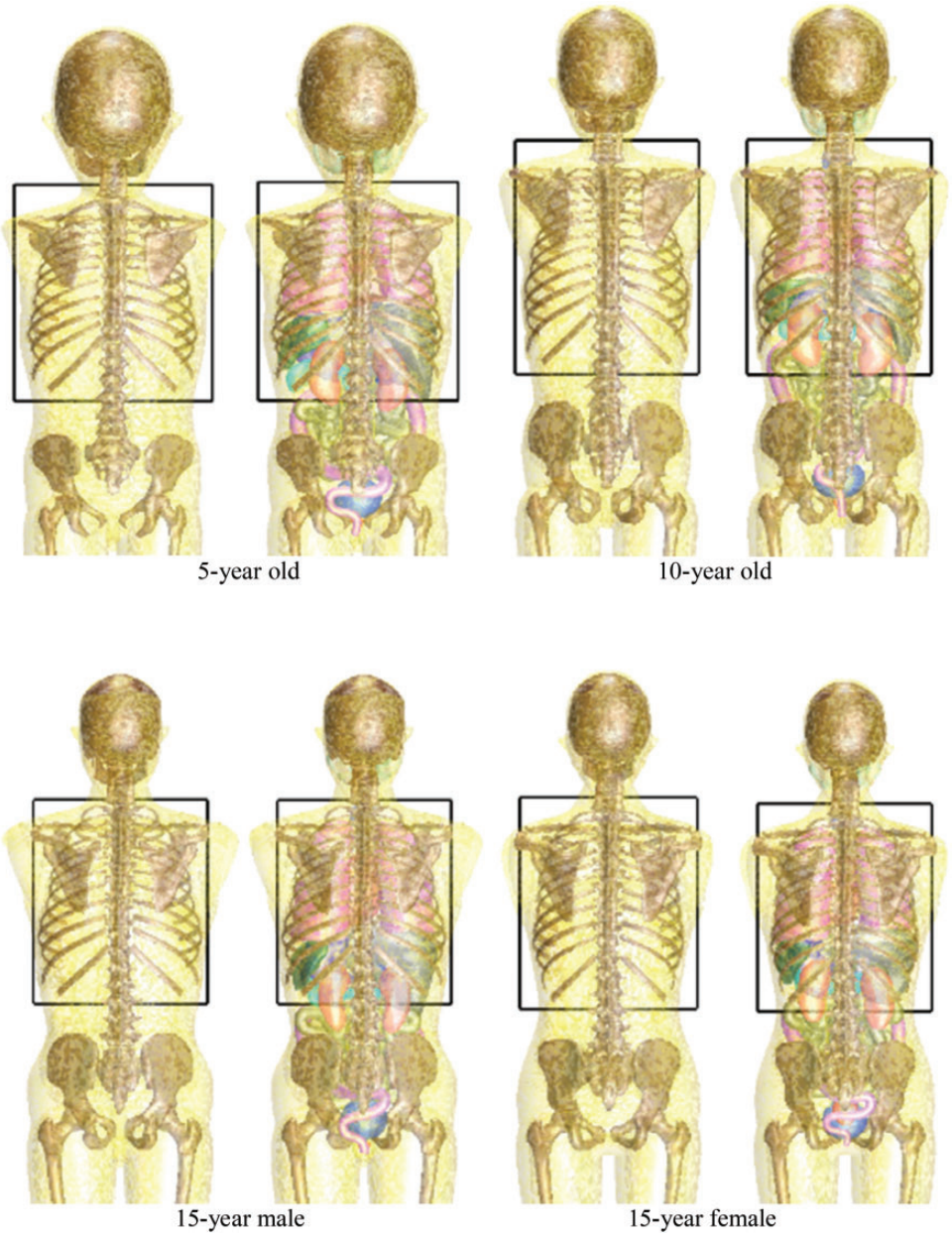


Fig. B.14. Coronal images at the centre of the paediatric reference phantoms showing the anatomical landmarks for chest postero-anterior examination. For 5- and 10-year-old phantoms, images for the male alone are shown. For the 15-year-old phantoms (third row), images for both the male and female are shown. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.2.3. Pelvic antero-posterior examination

Field height extension:	Extends from 3 cm above iliac crest to slightly below lesser trochanter
Field lateral extension:	Should include the hips
Focus-to-detector distance:	100 cm for newborn and 1-year-old 115 cm for 5-, 10-, and 15-year-old
Skin-to-detector distance:	Anti-scatter grid for patient thickness of at least 13–14 cm (Knight, 2014) 0 cm (no grid) for newborn 10 cm (grid) for 1-, 5-, 10-, and 15-year-old

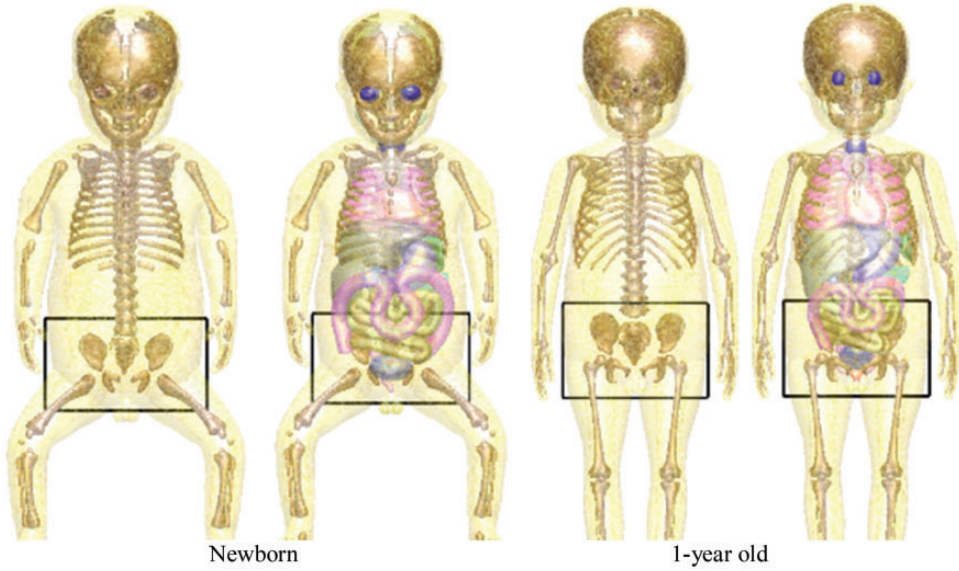


Fig. B.15. Coronal images at the centre of the paediatric reference phantoms showing the anatomical landmarks for pelvic antero-posterior examination. For newborn, 1-, 5-, and 10-year-old phantoms, images for the male alone are shown. For 15-year-old phantoms (third row), images for both the male and female are shown. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest (*continued on next page*).

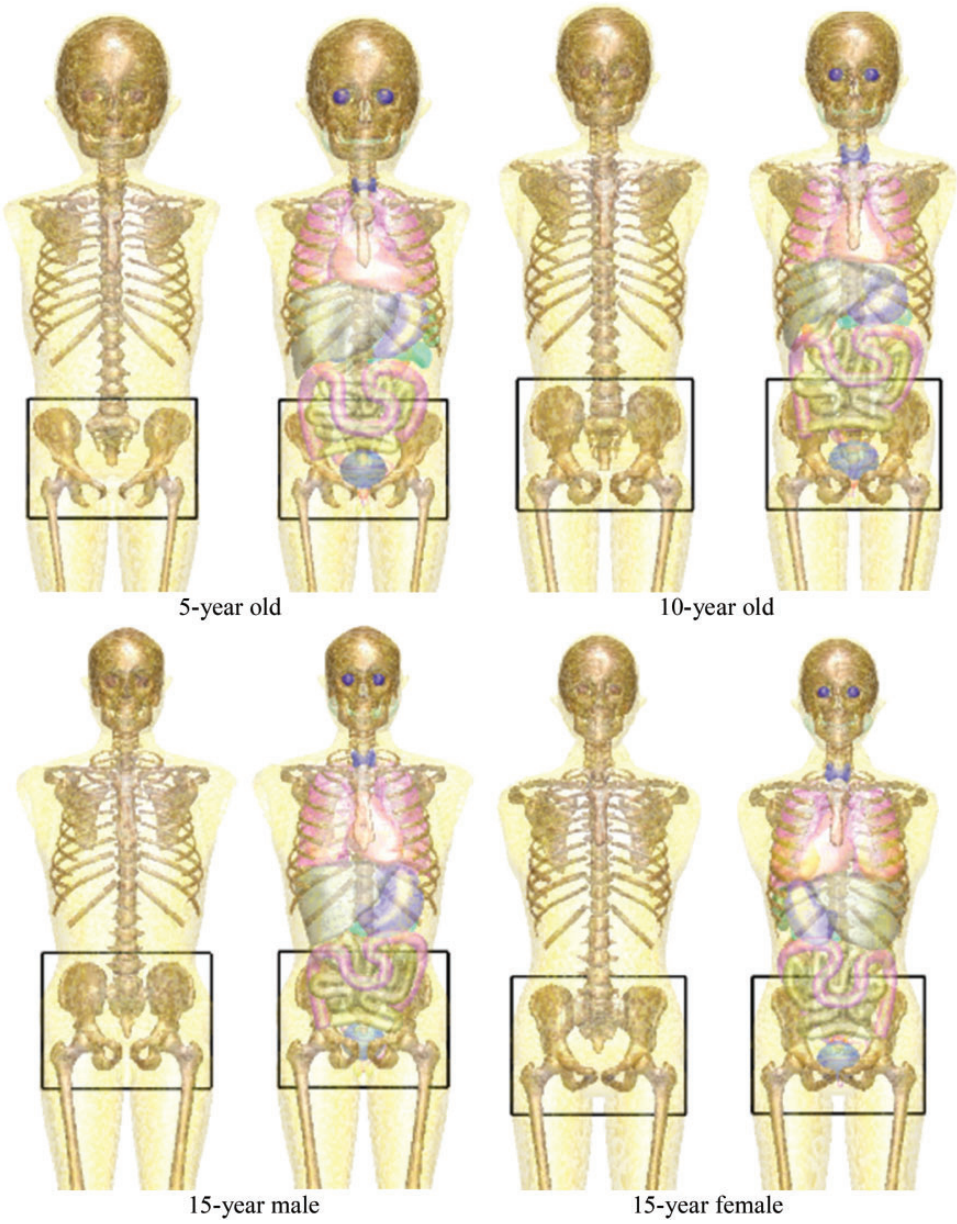


Fig. B.15. (*continued*).

B.2.4. Abdominal antero-posterior examination

Field height extension:	From 1–2 cm above dome of diaphragm to approx. 1 cm below symphysis pubis
Field lateral extension:	Should extend approx. 2 cm beyond pelvis on either side
Focus-to-detector distance:	100 cm
Skin-to-detector distance:	Anti-scatter grid for patient thickness of at least 13–14 cm (Knight, 2014) 0 cm (no grid) for newborn 10 cm (grid) for 1-, 5-, 10-, and 15-year-old

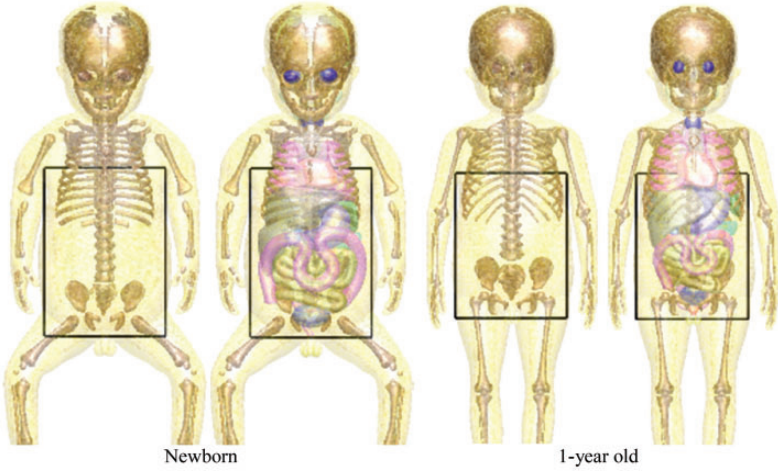


Fig. B.16. Coronal images at the centre of the paediatric reference phantoms showing the anatomical landmarks for abdominal antero-posterior examination. For newborn, 1-, 5-, and 10-year-old phantoms, images for the male alone are shown. For 15-year-old phantoms (third row), images for the male and female are shown. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest (*continued on next page*).

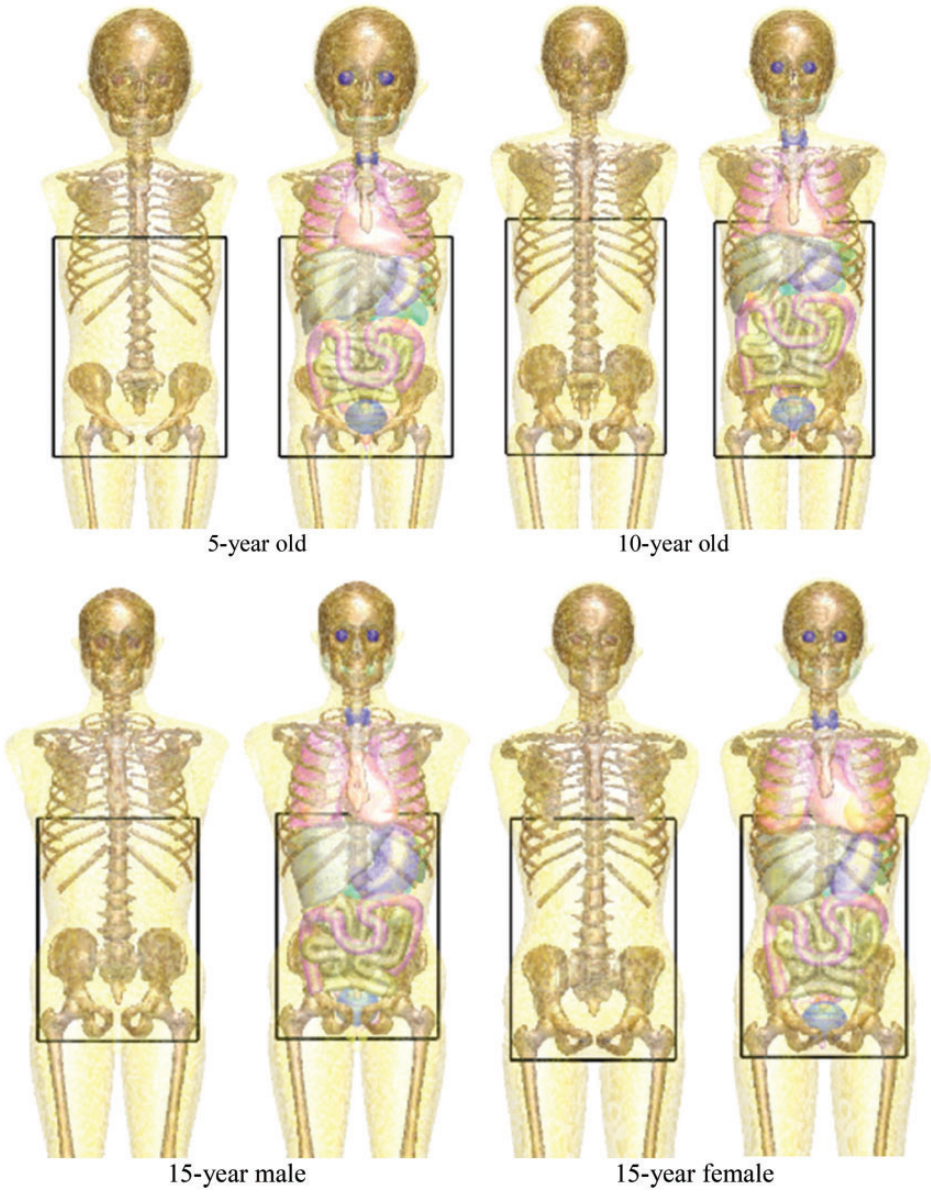


Fig. B.16. (*continued*).

B.2.5. Lumbar spine antero-posterior examination

Field height extension:	From T11 or approx. mid T10 to half sacrum (above coccyx)
Field lateral extension:	Should include at least all of the sacro-iliac joints
Focus-to-detector distance:	100 cm for newborn and 1-year-old 115 cm for 5-, 10-, and 15-year-old
Skin-to-detector distance:	Anti-scatter grid for patient thickness of at least 13–14 cm (Knight, 2014) 0 cm (no grid) for newborn 10 cm (grid) for 1-, 5-, 10-, and 15-year-old

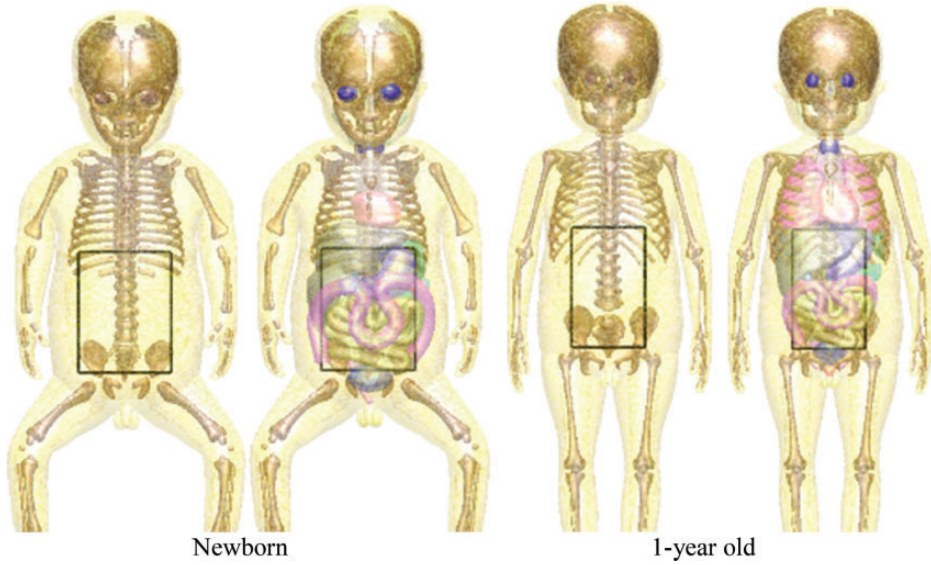


Fig. B.17. Coronal images at the centre of the paediatric reference phantoms showing the anatomical landmarks for lumbar spine antero-posterior examination. For newborn, 1-, 5-, and 10-year-old phantoms, the images for the male alone are shown. For 15-year-old phantoms (third row), images for the male and female are shown. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest (*continued on next page*).

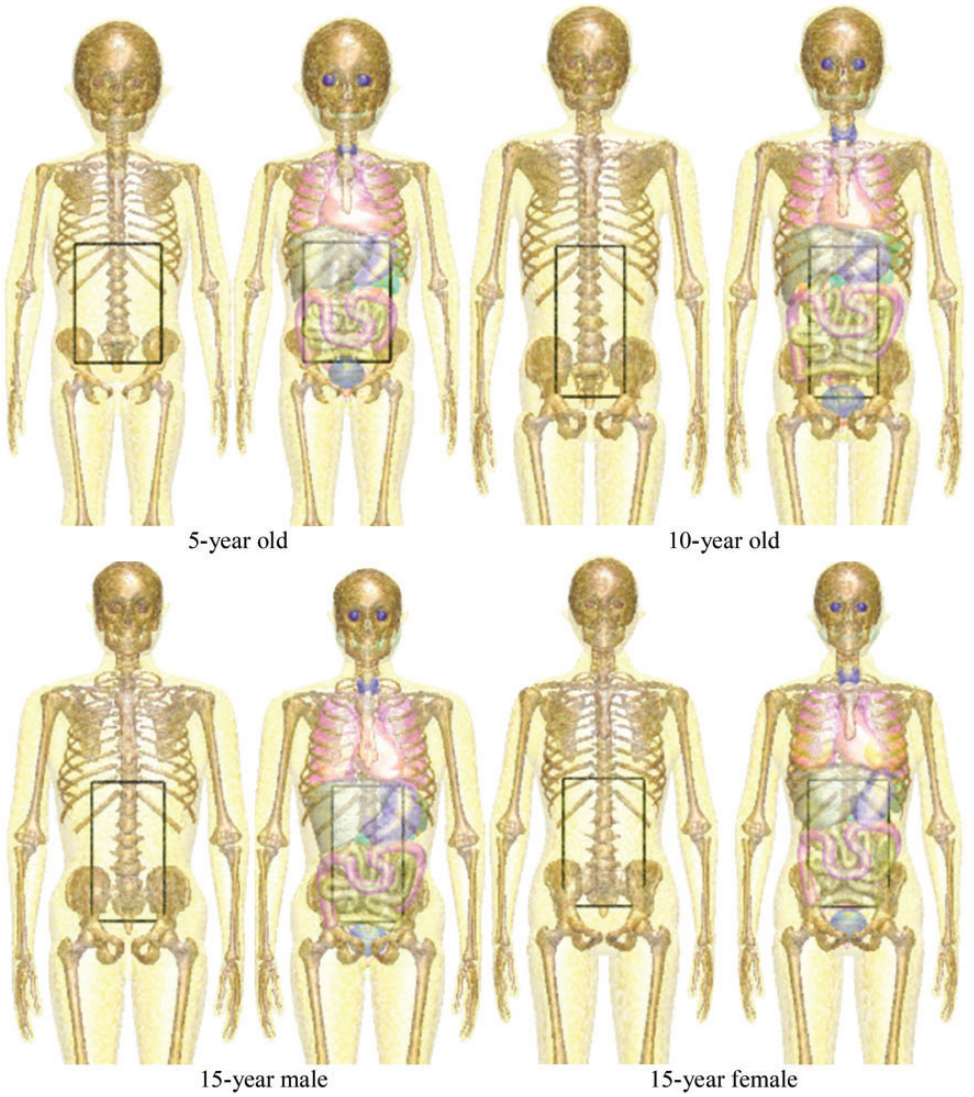


Fig. B.17. (*continued*).

B.2.6. Lumbar spine lateral examination

Field height extension:	From T11 or approx. mid T10 to half sacrum (above coccyx)
Field lateral extension:	Should extend approx. 2 cm beyond the bone on either side
Focus-to-detector distance:	100 cm for newborn and 1-year-old 110 cm for 5-, 10-, and 15-year-old
Skin-to-detector distance:	10 cm (grid) for all ages (Knight, 2014)

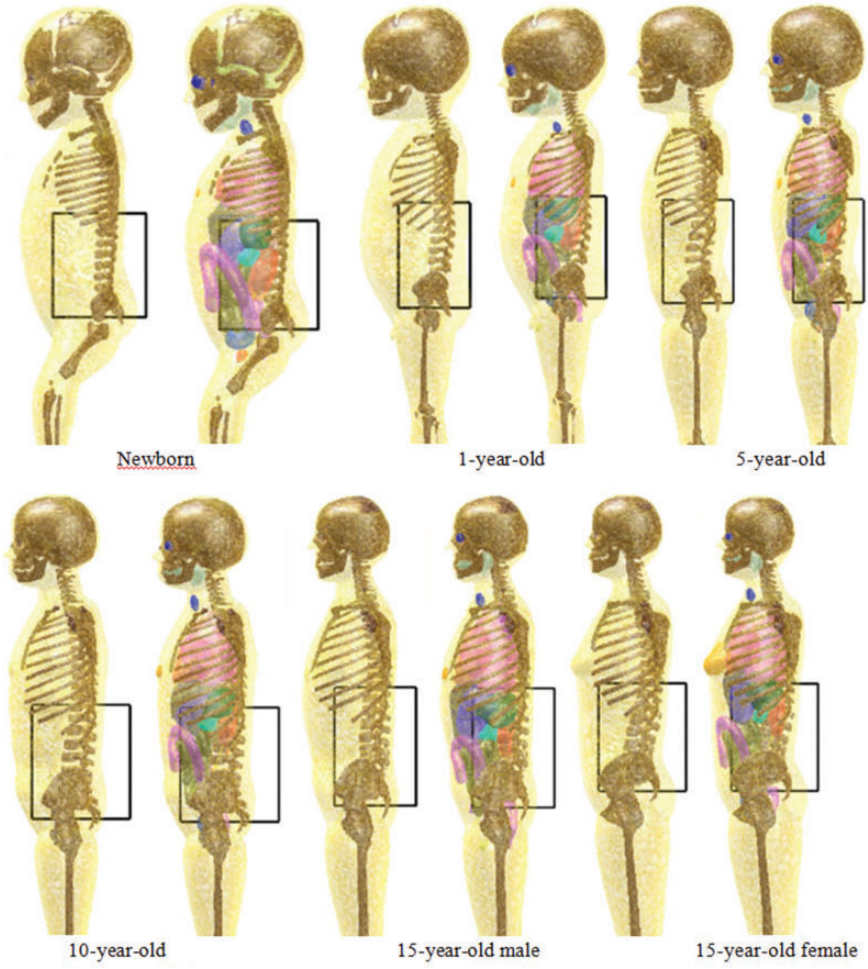


Fig. B.18. Sagittal images at the centre of the paediatric reference phantoms showing the anatomical landmarks for lumbar spine lateral examination. For newborn, 1-, 5-, and 10-year-old phantoms, images for the male alone are shown. For 15-year-old phantoms, images for the male and female are shown. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.2.7. Thoracolumbar spine lateral examination (newborn and 1-year-old)

Field height extension:	From C7 to half sacrum (above coccyx); for newborn, include cervical spine
Field lateral extension:	Should extend approx. 2 and 4 cm beyond the bone on the front and back, respectively
Focus-to-detector distance:	100 cm
Skin-to-detector distance:	10 cm (grid) (Knight, 2014)

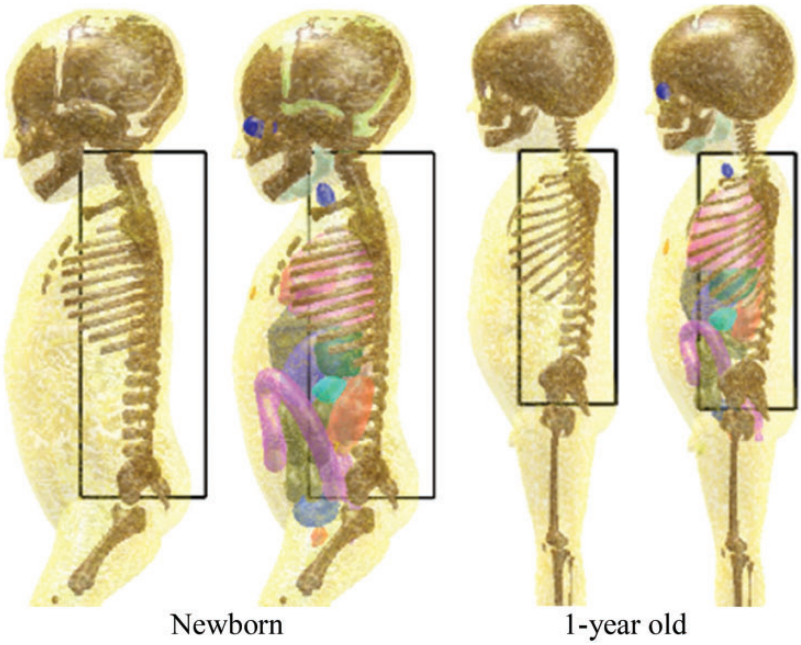


Fig. B.19. Sagittal images at the centre of the newborn and 1-year-old male phantoms showing the anatomical landmarks for thoracolumbar spine lateral examination. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.2.8. Scoliosis antero-posterior examination (newborn and 1-year-old)

Field height extension:	From skull base to proximal femurs
Field lateral extension	Should include all skeletal anatomy (without arms)
Focus-to-detector distance:	110 cm
Skin-to-detector distance:	0 cm (no grid) for the newborn 10 cm (grid) (Knight, 2014) for 1-year-old

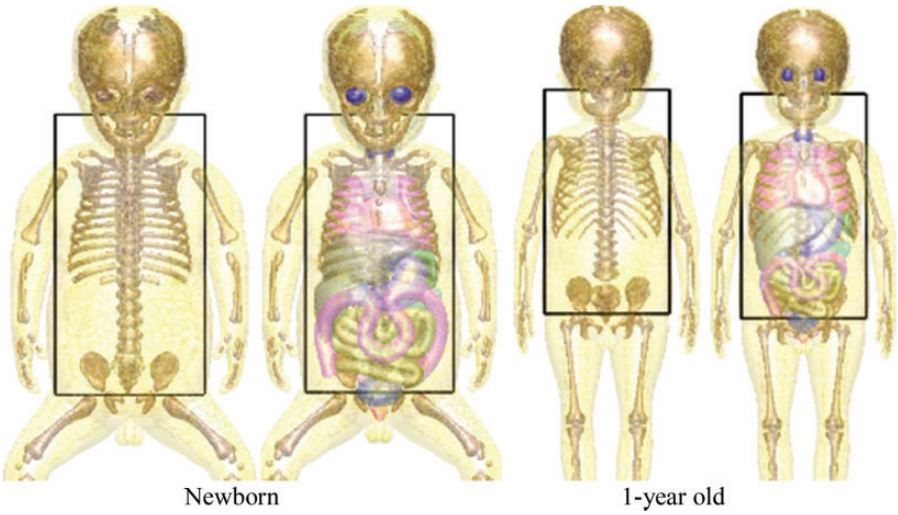


Fig. B.20. Coronal images at the centre of the newborn and 1-year-old male phantoms showing the anatomical landmarks for scoliosis antero-posterior examination. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.2.9. Scoliosis postero-anterior examination (5-, 10- and 15-year-old)

Field height extension:	From skull base to proximal femurs
Field lateral extension:	Should include all skeletal anatomy (without arms)
Focus-to-detector distance:	180 cm
Skin-to-detector distance:	10 cm (grid) for all ages (Knight, 2014)

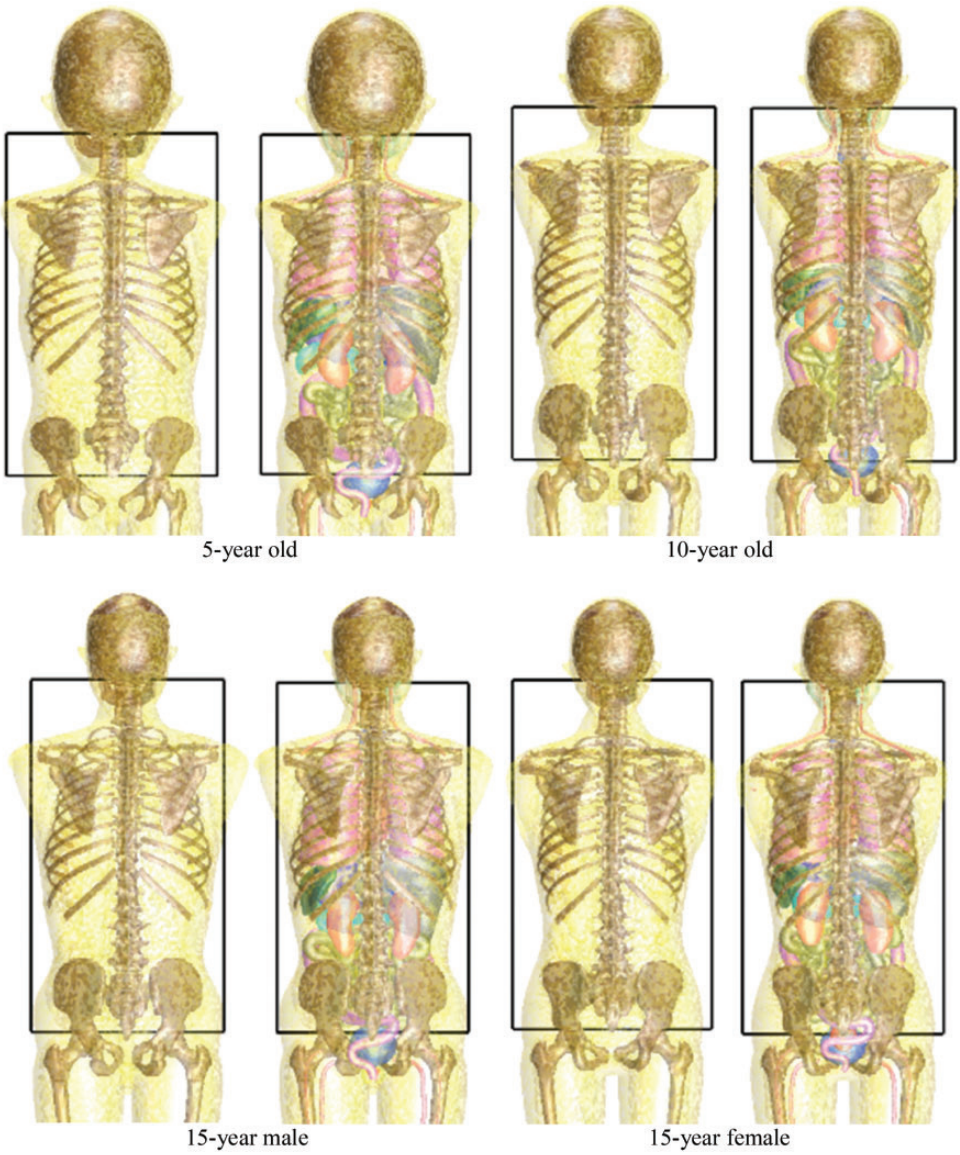


Fig. B.21. Coronal images at the centre of the paediatric reference phantoms showing the anatomical landmarks for scoliosis postero-anterior examination. For 5- and 10-year-old phantoms, images for the male alone are shown. For 15-year-old phantoms, images for the male and female are shown. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.2.10. Scoliosis lateral examination

Field height extension:	From skull base to proximal femurs
Field lateral extension:	Should include all skeletal anatomy (without arms)
Focus-to-detector distance:	110 cm for newborn and 1-year-old 180 cm for 5-, 10-, and 15-year-old
Skin-to-detector distance:	10 cm (grid) for all ages (Knight, 2014)

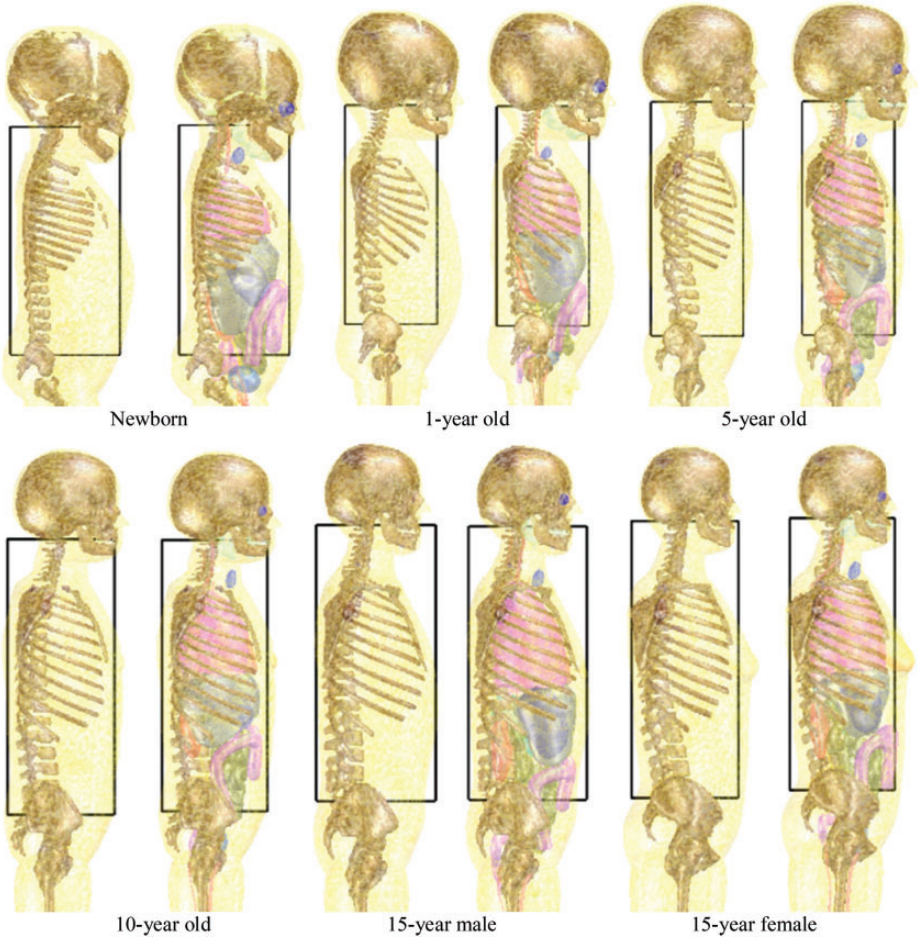


Fig. B.22. Sagittal images at the centre of the paediatric reference phantoms showing the anatomical landmarks for scoliosis lateral examination. For newborn, 1-, 5-, and 10-year-old phantoms, images for the male alone are shown. For 15-year-old phantoms, images for the male and female are shown. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.2.11. Skull antero-posterior examination

Field height extension:	Includes all skull and mandible; for newborn and 1-year-old, also cervical spine
Field lateral extension:	Should include all skull and mandible
Focus-to-detector distance:	110 cm
Skin-to-detector distance:	10 cm (grid) for all ages (Knight, 2014)

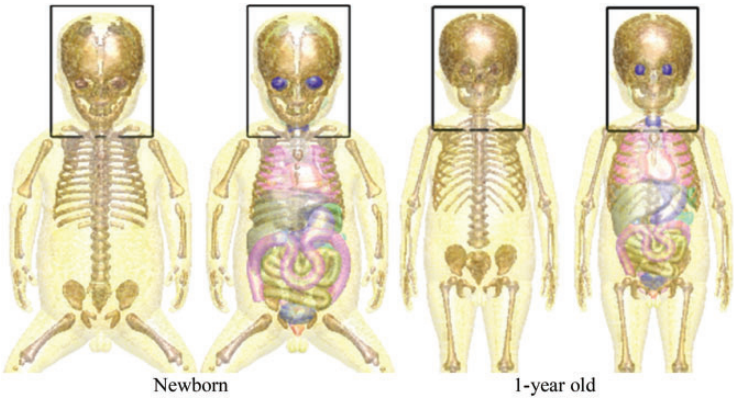


Fig. B.23. Coronal images at the centre of the paediatric reference phantoms showing the anatomical landmarks for skull antero-posterior examination. For newborn, 1-, 5-, and 10-year-old phantoms, images for the male alone are shown. For 15-year-old phantoms (third row), images for the male and female are shown. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

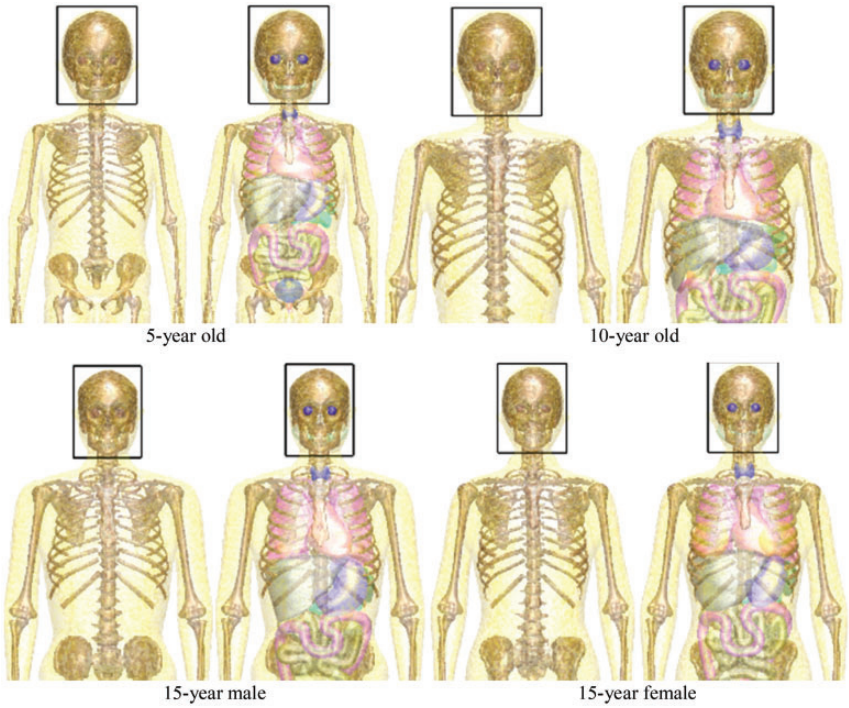


Fig. B.23. (*continued*).

B.2.12. Skull lateral examination

Field height extension:	Includes all skull and mandible; for neonate and 1-year-old, also cervical spine
Field lateral extension:	Should include all skull and mandible
Focus-to-detector distance:	110 cm
Skin-to-detector distance:	10 cm (grid) for all ages (Knight, 2014)

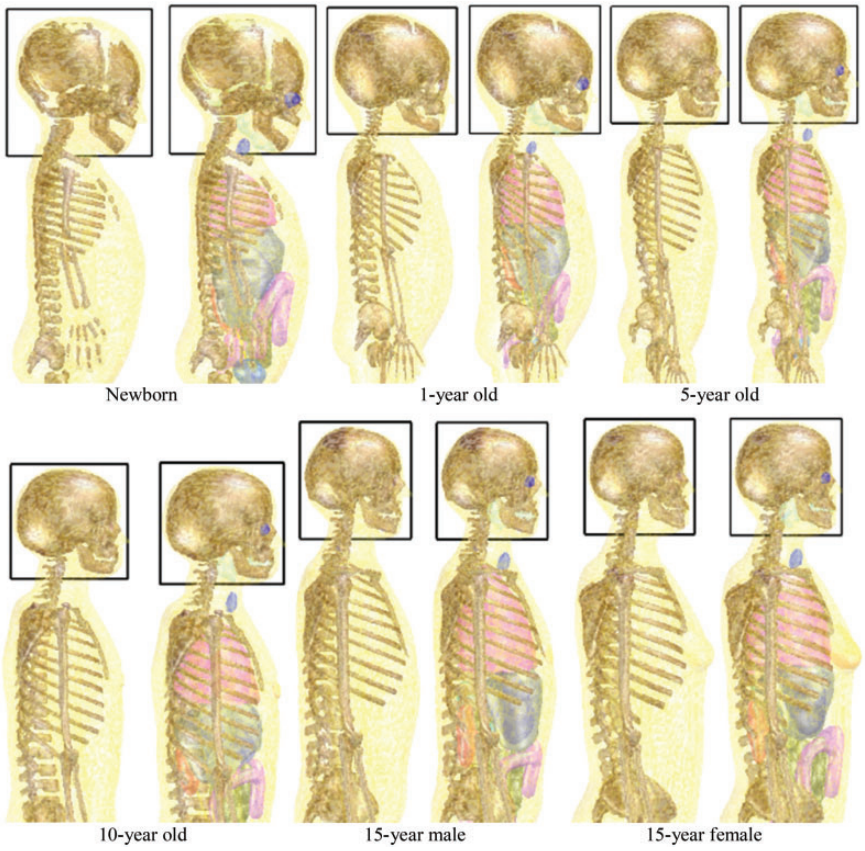


Fig. B.24. Sagittal images at the centre of the paediatric reference phantoms showing the anatomical landmarks for skull lateral examination. For newborn, 1-, 5-, and 10-year-old phantoms, images for the male alone are shown. For 15-year-old phantoms (third row), images for the male and female are shown. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest.

B.2.13. Skull postero-anterior examination (15° caudal angulation)

Field height extension:	Includes all skull and mandible; for neonate and 1-year-old, also cervical spine
Field lateral extension:	Should include all skull and mandible
Focus-to-detector distance:	110 cm
Skin-to-detector distance:	10 cm (grid) for all ages (Knight, 2014)

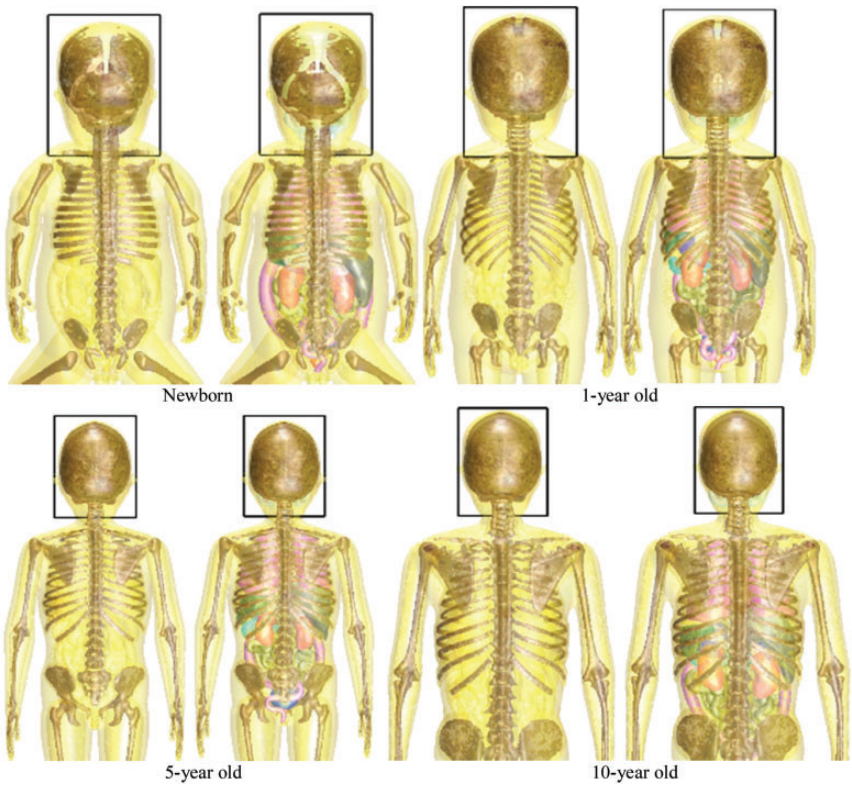


Fig. B.25. Coronal images at the centre of the paediatric reference phantoms showing the anatomical landmarks for skull postero-anterior (15° caudal angulation) examination. For newborn, 1-, 5-, and 10-year-old phantoms, images for the male alone are shown. For 15-year-old phantoms (third row), images for the male and female are shown. Left to right: the first image shows the skeletal landmarks alone, and the second image adds the organs of interest. For the 15-year-old female phantom (fourth row), the angulation of the x-ray tube for this exposure is demonstrated graphically. This is similar for all other ages (*continued on next page*).

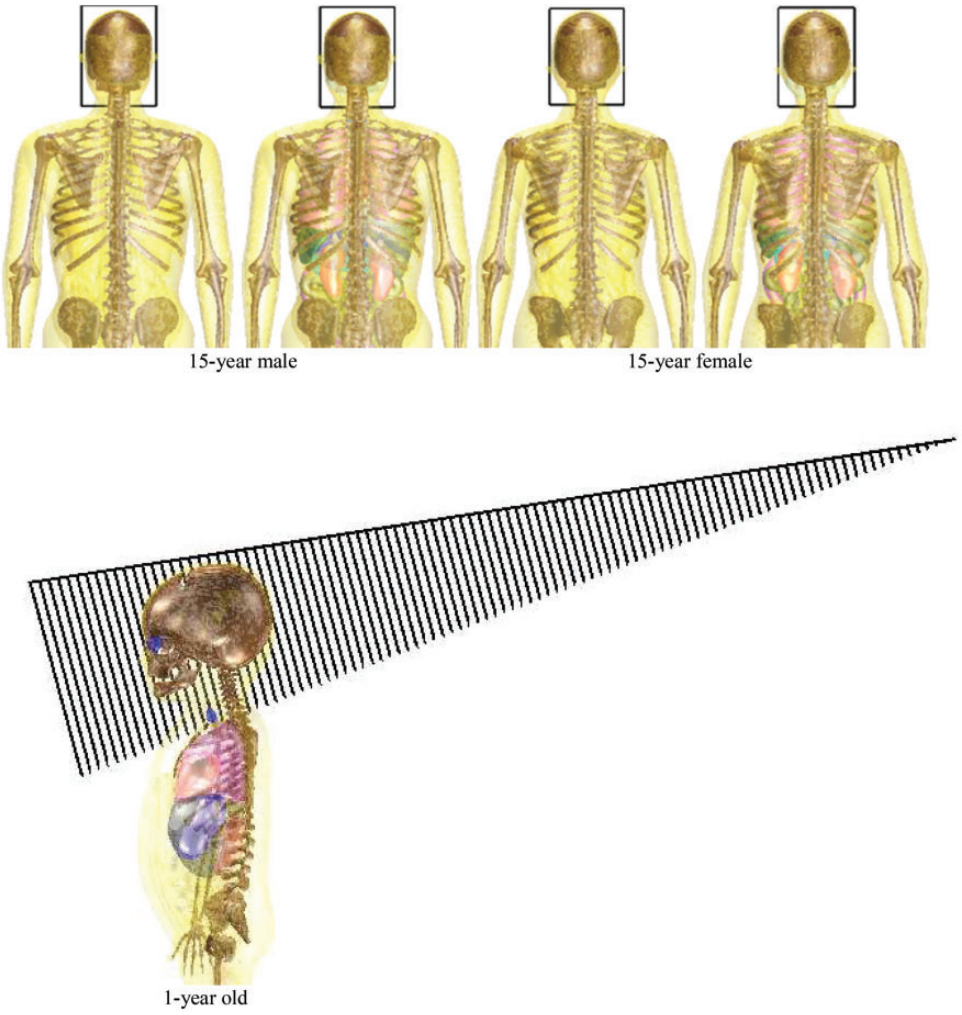


Fig. B.25. (*continued*).

B.3. References

Knight, S.P., 2014. A paediatric x-ray exposure chart. J. Med. Radiat. Sci. 61, 191–201.
Whitley, A.S., Jefferson, G., Holmes, K., et al., 2015. Clark's Positioning in Radiography 13E. CRC Press, London.

ANNEX C. APPLICATION OF SKELETAL FLUENCE-TO-DOSE RESPONSE FUNCTIONS FOR PHOTONS IN MONTE CARLO RADIATION TRANSPORT CODES

C.1. Skeletal dosimetry method of *Publication 116*

(C 1) Bone dosimetry is different from dosimetry of other organs, for which the average absorbed dose in the whole organ is estimated. For bone dosimetry, the active marrow (AM) dose, associated with the risk of ionising radiation-induced leukaemia, and the endosteum (TM_{50}) dose, associated with the risk of ionising-radiation-induced bone cancer, are needed. These bone tissues reside in the skeletal spongiosa and medullary cavity. The spongiosa bone compartment consists of AM, inactive marrow, and trabecular bone in a spongious structure with dimensions in the range of tens to hundreds of micrometres (Hough and Bolch, 2007; Hough et al., 2011).

(C 2) According to the current ICRP recommendations (ICRP, 2007), the absorbed dose to different target tissues has to be assessed in the computation of effective dose. These include the haematopoietically active bone marrow and the skeletal endosteum. The former target region is taken as the non-adipose region of the bone marrow cavities within both spongiosa and medullary marrow cavities of the skeleton, while the latter target region is taken to be the total marrow, localised within 50 μm of the bone trabeculae surfaces and along the interior surfaces of the long bone medullary cavities. As described in *Publications 110* and *116* (ICRP, 2009, 2010), the bone trabeculae and marrow cavities are tissue structures of the order of tens to hundreds of micrometres of thickness and extent, and thus cannot be modelled fully within the voxel resolution of either the reference adult or paediatric phantoms.

(C 3) The marrow dose is enhanced because there are more photoelectric events in the trabecular bone (due to their higher atomic number Z) than in the marrow (lower Z), increasing the overall electron fluence from the trabecular bone into the marrow cavity, which consequently increases the marrow absorbed dose, compared with the situation of assumed secondary electron equilibrium. To address this, Johnson et al. (2011) calculated dose response functions (DRFs) and dose enhancement factors (DEFs) for an adult person using radiation transport methods in various bones, reconstructed by three-dimensional micro-CT images.

(C 4) Consequently, absorbed dose to these two target tissues is determined by employing the concept of the DRF or DEF for photons, as described in Annex D of *Publication 116* (ICRP, 2010). Further detail on skeletal dosimetry can be found in *Publication 144* (ICRP, 2020b).

(C 5) *Publication 116* (ICRP, 2010) includes two annexes on skeletal dosimetry of the adult male and female phantoms for photons and neutrons, respectively. The annex dealing with photons briefly explains the method of calculation of DRFs and consequently of DEFs, and provides tables of these values. Similar techniques for

application to the paediatric reference phantoms have been developed, and the respective DRF values can be found in *Publication 155* (ICRP, 2023) as part of the electronic supplement and are based on the work of Pafundi et al. (2010) and Wayson et al. (2012). The DRFs of the adult phantoms applied for the calculations of the present publication are those from *Publication 116* [based on Johnson et al. (2011)], with some slight revisions which will be described briefly below.

(C 6) The main aim of the present annex is to help the user to implement the DRFs or DEFs in the Monte Carlo code by including code-specific instructions.

(C 7) In a Monte Carlo program, the various regions are defined by their shape, position, material, and density. Normally, the medium of a region of interest is distributed homogeneously through the region's volume, and the average dose to the whole volume is scored to estimate absorbed dose. Monte Carlo programs transporting photons alone (i.e. without electron transport) enable the user to calculate the dose to the volume of interest using a track length or collision density estimator as described in Eqs D.4b and D.4a of *Publication 116* (ICRP, 2010). For dose calculations using voxel phantoms, the track length estimator is preferred over the collision density estimator, as the number of voxels through which a photon is going will be larger than the number of voxels where a photon collision is taking place.

(C 8) The spongiosa of a bone in the ICRP reference phantoms is composed of AM, inactive marrow, trabecular bone, cartilage, and miscellaneous tissues in a non-homogenous way. The targets of interest in the spongiosa region are AM and TM_{50}, a mixture of AM and inactive marrow. The mixture of AM and inactive marrow is dependent on the bone site and age of the ICRP reference phantom. For adult reference phantoms, this can be derived from the information listed in *Publication 110*, Table 4.2 (ICRP, 2009) based on the data of Cristy (1981). When the range of a secondary electron is much smaller than the smallest size of the marrow-filled cavities, and the latter are much smaller than the size of the volume, the fluence-to-dose function can be used. For a bone compartment, x, assuming secondary electron equilibrium, the AM dose can be then calculated (ICRU, 2011) as:

$$D_{\mathrm{AM,x}} = \left(\frac{\mu_{\mathrm{en}}}{\rho}\right)_{\mathrm{AM}} E\ \Phi \tag{C.1}$$

where $D_{\mathrm{AM,x}}$ is the absorbed dose to the AM in bone compartment x, $\left(\frac{\mu_{\mathrm{en}}}{\rho}\right)_{\mathrm{AM}}$ is the mass energy attenuation coefficient for AM, E is the photon energy, and Φ is the photon fluence in this compartment. Note that both the mass energy attenuation coefficient and fluence are dependent on the photon energy, E. In dosimetric calculations, the AM dose can be expressed in terms of the photon energy distribution, Φ_{E}, (ICRU, 2011) as:

$$D_{\mathrm{AM,x}} = \int \left(\frac{\mu_{\mathrm{en}}}{\rho}\right)_{\mathrm{AM}} E\ \Phi_{\mathrm{E}}\ \mathrm{d}E \tag{C.2}$$

(C 9) For calculation of the endosteum dose in bone compartment x, it is convenient to define the marrow cellularity factor, CF_x (Cristy, 1981) of this bone compartment as:

$$CF_x = \frac{V_{AM,x}}{V_{AM,x} + V_{IM,x}} \tag{C.3}$$

where $V_{AM,x}$ is the volume of AM in bone compartment x, and $V_{IM,x}$ is the volume of inactive marrow in bone compartment x. It is assumed that the cellularity, CF_x as defined by Cristy (1981), for the whole bone compartment x, remains valid within 50 μm of the bone trabeculae surfaces and along the interior surfaces of the long bone medullary cavities.

(C 10) Eq. C.4 expresses the cellularity factor, CF_x, using the target mass, which is the more relevant quantity for dosimetry:

$$CF_x = \frac{\frac{M_{AM,x}}{\rho_{AM}}}{\frac{M_{AM,x}}{\rho_{AM}} + \frac{M_{IM,x}}{\rho_{IM}}} \tag{C.4}$$

where $M_{AM,x}$ is the mass of AM in bone compartment x (in g), $M_{IM,x}$ is the mass of the inactive marrow in bone compartment x, ρ_{AM} is the density of AM (in g cm^{-3}), and ρ_{IM} is the density of inactive marrow (in g cm^{-3}). Note that the densities are assumed to be independent of compartment x, as the blood content is not considered. The endosteum dose, with the same limitations, in bone compartment x can now be calculated using the mass-weighted AM and inactive marrow dose, assuming that the cellularity is constant throughout the entire bone compartment x:

$$D_{TM50,x} = \frac{CF_x\,\rho_{AM}\,\left(\frac{\mu_{en}}{\rho}\right)_{AM} + (1 - CF_x)\rho_{IM}\left(\frac{\mu_{en}}{\rho}\right)_{IM}}{CF_x\;\rho_{AM} + (1 - CF_x)\rho_{IM}} E \cdot \Phi \tag{C.5}$$

(C 11) For convenience, one can define $\left(\frac{\mu_{en}}{\rho}\right)_{TM50}$ as:

$$\left(\frac{\mu_{en}}{\rho}\right)_{TM50} = \frac{CF_x\rho_{AM}\left(\frac{\mu_{en}}{\rho}\right)_{AM} + (1 - CF_x)\rho_{IM}\left(\frac{\mu_{en}}{\rho}\right)_{IM}}{CF_x\;\rho_{AM} + (1 - CF_x)\rho_{IM}} \tag{C.6}$$

(C 12) The above equation is valid if secondary electron equilibrium exists. If this is not the case, a DRF can be used to derive absorbed dose for AM or skeletal endosteum (TM_{50}), expressed as:

$$D_{AM,x} = \int \mathrm{DRF}(E, \mathrm{AM}, \mathrm{x})\;\Phi_E \mathrm{d}E \tag{C.7}$$

$$D_{\mathrm{TM}_{50},\mathrm{x}} = \int \mathrm{DRF}(E, \mathrm{TM}_{50}, \mathrm{x})\ \Phi_{\mathrm{E}} \mathrm{d}E \tag{C.8}$$

(C 13) An alternative method is the use of a DEF. In this case, the absorbed doses are calculated as:

$$D_{\mathrm{AM,x}} = \int \left(\frac{\mu_{\mathrm{en}}}{\rho}\right)_{\mathrm{AM}}\ \mathrm{DEF}(E, \mathrm{AM}, \mathrm{x})\ E\ \Phi_{\mathrm{E}}\ \mathrm{d}E \tag{C.9}$$

$$D_{\mathrm{TM}_{50},\mathrm{x}} = \int \left(\frac{\mu_{\mathrm{en}}}{\rho}\right)_{\mathrm{TM}_{50}}\ \mathrm{DEF}(E, \mathrm{TM}_{50}, \mathrm{x})\ E\ \Phi_{\mathrm{E}} \mathrm{d}E \tag{C.10}$$

(C 14) It should be noted that, if the blood content is added to AM and TM_{50}, its contribution to the dose calculation should be considered. However, its influence on the doses is too small and has been neglected for the calculations presented in this publication.

(C 15) If the electron range is in the range of the organ size or larger, the electron will quite likely escape the volume while other electrons will enter the volume; these electrons should, therefore, be transported and photon-only transport may no longer produce accurate results. This topic is, however, beyond the scope of this annex, as it is not relevant for the photon energy range of 10–150 keV applicable to diagnostic x-rays.

(C 16) Johnson et al. (2011) derived DRFs and consequently DEFs by Monte Carlo calculations of photon and electron transport in three-dimensional micro-CT images of various bone compartments. The results are dependent on the bone compartment, and tallies per bone compartment have to be implemented in the code. In this way, AM and endosteum dose in a bone compartment can be obtained.

(C 17) To derive the total AM and endosteum doses for an anthropomorphic phantom, the bone compartment doses are summed as weight fractions of the AM or endosteum cells, using, in the case of the adult reference phantoms, the weight fractions of Table 4.2 of *Publication 110* (ICRP, 2009), or Table 3.2 of *Publication 116* (ICRP, 2010); for the paediatric reference phantoms, Tables 3.3–3.8 of *Publication 143* (ICRP, 2020a).

C.2. Refinements of the original fluence-to-dose response function sets for both adult and paediatric reference phantoms

(C 18) This section describes the refinements of the DRFs and DEFs for adults which appear in *Publication 116* (ICRP, 2010; Johnson et al., 2011). It has been observed that the DRFs and DEFs for the same bone and photon energy are not compatible in all cases. For a very limited number of cases at very low photon energies, some DRF values erroneously indicate a negative dose enhancement

(i.e. a dose reduction). To correct this, these DEF values are set to 1.0, assuming secondary electron equilibrium. This correction has also been applied to recalculate the AM and inactive marrow kerma coefficients appearing in Table 5 of Johnson et al. (2011). Thus, the use of either the DRF or DEF method would result in the same values of AM and TM_{50} doses.

(C 19) As the elementary composition of AM and inactive marrow is age-independent, and consequently the respective kerma coefficients are also age-independent, a similar correction for negative values of dose enhancements could also be applied to the paediatric DRFs of Wayson et al. (2012). The revised set of DRFs for all phantoms/ages can be found in the electronic supplement of *Publication 155* (ICRP, 2023).

C.3. Implementation of fluence-to-dose response functions and dose enhancement factors to some Monte Carlo codes used in medical physics

(C 20) The next subsections describe examples of the implementation of DRFs and DEFs into the Monte Carlo radiation transport code systems used for calculation of the radiographic dose coefficients presented in this work, i.e. Geant4 (Geometry and Tracking, Version 4) and EGSnrc (Electron Gamma Shower National Research Council of Canada). They also describe the implementation in MCNP (Monte Carlo N-Particle Transport Code), which was employed by a further work of ICRP Task Group 113 for the calculations of dose coefficients for CT, as well as in PHITS (Particle and Heavy Ion Transport code System), which has been used for *Publication 144* (ICRP, 2020b). This section is aimed to help potential users with their implementation, and employs code-specific terms which are not explained here but are well known to the Monte Carlo code users to whom this section is addressed. Some further practical guidance and technical hints for incorporating DRFs and DEFs into various types of radiation transport codes can be found in Zankl et al. (2021).

C.3.1. MCNP

(C 21) MCNP (Pelowitz, 2013) transporting photons alone (MODE P), without electron transport, enables the user to calculate the dose in a volume with a track length estimator (Werner et al., 2017, 2018) by using tally 6 and the ID number characterising this volume or cell. If the range of the secondary electron is much smaller than the smallest volume dimension, this will give an accurate estimate:

F6:P ID_{volume}

(C 22) The MCNP output will then give the absorbed dose in volume ID_{volume} in units of MeV g^{-1} or, if multiplied by 1.602×10^{-10}, in SI units of Gy or J kg^{-1}. The F6 card denotes that the track length estimator tally is used for absorbed dose averaged over a cell or volume with ID number, ID_{volume}.

(C 23) Most Monte Carlo radiation transport codes offer convenient ways for calculating, for instance, the air dose in a cell that contains polymethyl methacrylate or water, as it is a situation that is quite common. This means that the Monte Carlo radiation transport code allows to transport radiation particles through a medium (material) X while multiplying the track length at every step with the fluence-response function for a desired dose quantity in medium Y. For MCNP, tally 4, which is a track length estimator tally like F6 but without the default dose multipliers, is employed to derive a photon fluence that can be used to calculate the AM dose with the 'FM' card (i.e. a tally multiplier) as outlined in the MCNP6.2 user manual (Werner et al., 2017, 2018), see example 5:

> *When the material is not present in the model, rads can be obtained from type 1, 2, 4, and 5 tallies by using an FM card where c is equal to the factor above times No* $\eta \times 10^{-24}/A$, *where No is Avogadro's number and* η *and A are the number of atoms/molecule and the atomic weight, respectively, of the material of interest. This value of c equals* ρ_a/ρ_g *as discussed in Section 5.3.4 of the MCNP5 Theory Manual (Ref X-5 Monte Carlo Team, 2008). The implicit assumption when the material is not present is that it does not affect the radiation transport significantly. In the reaction list on the FM card, you must enter −4 1 for neutron heating and −5 −6 for photon heating. For both F4 and F6, if a heating number from the data library is negative, it is set to zero by the code.*

This limitation, stated in the above-mentioned manual and not considered to affect radiation transport significantly, is not applicable in this case, as the material or tissue is present but in conjunction with other materials. The only assumption in this case is that the AM is distributed homogeneously through the volume. The MCNP input file should contain:

```
F4:P ID_volume
C Track length estimator (F4) for photons (:P) for cell ID_volume
FM4 1.62840E-11 999 (-5 -6)
C Tally multiplier (FM4)
C Multiplier constant: 1.62840E-11 =
C Conversion factor from (MeV/g) to (J/kg=Gy): 1.60218E-10 *
C Atomic density (Table 50/60) (atoms/barn-cm): 1.04686E-01/
C Gram density (Table 50 or 60) (g/cm**3): 1.03000E+00
C material number 999 (see also M999 card below)
C Reaction list (-5 -6)
C Photon total cross section (-5) times
C Photon heating number (-6) is the total energy deposited
M999  1000 -0.105 6000 -0.414 7000 -0.034 8000 -0.439 11000 -0.001
      12000 -0.002 15000 -0.002 16000 -0.002 26000 -0.001
C Material 999 has elemental composition of:
C 0.105 H, 0.414 C, 0.034 N, 0.439 O, 0.001 Na, 0.002 Mg,
C 0.002 P, 0.002 S, 0.001 Fe
C Active marrow without blood; Table 13.4 (ICRP, 2002) ρ=1.03 g/cm^3
```

```
 C (ICRU, 1989)
PRINT 50 60
```

(C 24) In the above MCNP input, the lines starting with a 'C' are comment lines, intended for the information of the reader but not interpreted by the code. It is easy to calculate the FM card value of 1.62840×10^{-11}, using the conversion factor for MeV g^{-1} to J kg^{-1} or Gy of 1.602×10^{-10}, times the atomic density, divided by the density of material M999 [in this case, of red bone marrow or AM (ICRU, 1989)]. This is given in g cm^{-3} in the column Gram density of Table 50 or 60 of the MCNP output generated using the 'PRINT 50 60' card. The 'M' card indicates the integer material number followed directly by a space separating the elementary composition of the material denoted in pair numbers. The first number is an integer denoting the atomic number times 1000. The second negative number expresses, as an absolute value, the mass fraction of this element to the material. The total sum of mass fractions of the elements should be 1.0 or, if this is not the case, MCNP will provide the normalisation to 1.0 and print a warning message that the material was not normalised.

(C 25) If the electron range becomes larger but is still much smaller than the smallest volume size, the assumption of electron range being much smaller than the marrow cavity size is no longer valid. This is the case for the diagnostic x-ray energy range with photon energies up to 150 keV: the photoelectrons could then have energies up to 150 keV, with a continuous slowing down approximation range in water of 281.7 μm (ICRU, 1997, Table B.1). This photoelectron range is comparable with the mean pathlengths in the marrow cavities of the various bones of an adult male which range between 154 and 1913 μm (Hough et al., 2011).

(C 26) From these DRFs, DEFs can be evaluated (Johnson et al., 2011) and these can be used with the MCNP tally 4 in order to modify the calculated doses by multiplying those with these DEFs. An example is shown below for the AM dose and cranium spongiosa bone compartment [with volume ID number of 27 (ICRP, 2009)].

```
F4:P 27
FM4 1.62840E-11 999 (-5 -6)
DE4 0.008  0.010  0.015  0.02  0.03  0.04  0.05
     0.06   0.08   0.10  0.15  0.20  0.30

DF4 1.0000 1.0107 1.0349 1.0663 1.1286 1.2059 1.2472
    1.2417 1.1686 1.1130 1.0217 1.0071 1.0000

C Active Marrow DEF for cranium; according to Table 6, (Johnson et al., 2011).
M999 1000 -0.105 6000 -0.414 7000 -0.034 8000 -0.439 11000 -0.001
     12000 -0.002 15000 -0.002 16000 -0.002 26000 -0.001
C Active marrow without blood; Table 13.4 (ICRP, 2002); ρ=1.03 g/cm3
C (ICRU, 1989)
PRINT 50 60
```

(C 27) The 'FM' card and the 'M' card are as explained above. The 'DE' card gives the photon energies in MeV units, with the DEF given as a fraction on the corresponding position on the 'DF' card. The log-log interpolation technique is used to calculate the DEF for an arbitrary photon energy, *E*. Lines beginning with a 'C' followed by spaces are comment lines. The PRINT card asks the MCNP program to display the table of the corresponding argument numbers. Tables 50 and 60 are used to calculate the first argument on the 'FM' card. If the second argument on the 'FM' card, indicating the material card number, is not used in any MCNP cell, then a virtual cell with this material should be added to the MCNP input which is far away in the 'outside world' (i.e. outside the region of interest, and thus no photon can reach it). In this way, this cell does not contribute to the Monte Carlo calculated results, but it is contained in the MCNP set-up tables.

(C 28) As not all Monte Carlo radiation transport codes have the option to tally a region for another material, and the present publication intends to provide guidance for the most used codes, the DRFs are used, as defined in Johnson et al. (2011) and *Publication 116* (ICRP, 2010). The above example for cranium and spongiosa bone compartment and the respective dose enhancement of the latter are expressed in MCNP format, using DRFs and converting the unit Gy m^2 into Gy cm^2, as required by MCNP.

```
F4:P 27
DE4 0.010     0.015     0.02      0.03      0.04      0.05
    0.06      0.08      0.10      0.15      0.2       0.3
    0.4       0.5       0.6       0.8       1.0       1.5
    2.0       3.0       4.0       5.0       6.0       8.0
    10.0
DF4 6.207E-12 2.673E-12 1.478E-12 6.755E-13 4.439E-13 3.630E-13
    3.399E-13 3.663E-13 4.386E-13 6.656E-13 9.503E-13 1.484E-12
    2.025E-12 2.543E-12 3.037E-12 3.951E-12 4.783E-12 6.563E-12
    8.076E-12 1.070E-11 1.309E-11 1.537E-11 1.764E-11 2.217E-11
    2.689E-11
C Active Marrow DRF for cranium; according to Johnson et al. (2011), C Table 1.
```

(C 29) The 'F4' card indicates a track length estimation of the cell or volume with the ID number as the first argument. The integer number following the F is the tally ID number; in this case, the number 4. The ':P' that follows this integer tally ID number indicates to use the tallying (i.e. scoring) of photons alone. The 'DE' card gives, as previously, the photon energies in MeV within the corresponding position, and the 'DF' card returns the DRF in Gy cm^2. The number directly following the 'DE' and 'DF' is the tally ID. Again, log-log interpolation is used to derive the DRF for an arbitrary photon energy, *E*.

C.3.2. PHITS

(C 30) The Particle and Heavy Ion Transport code System, PHITS (Sato et al., 2023) is a general-purpose Monte Carlo radiation transport code that can simulate the behaviour of most particle species with energies up to 1 TeV (per nucleon for ions). The code is maintained by the Japan Atomic Energy Agency, and is available free of charge upon request.

(C 31) Photon transport in PHITS is performed using the Evaluated Photon Data Library 1997 Version, EPDL97 (Cullen et al., 1997), which includes photon interaction data for all elements from atomic number 1 to 100 with photoionisation, photoexcitation, coherent and incoherent scattering, and pair-production cross-sections in the energy range from 1 keV to 1 GeV at default settings. In addition, the EGS5 model (Hirayama et al., 2005) is recommended to be used for the simulation that transports both photons and electrons.

(C 32) To compute AM and endosteum doses at each bone component using DRFs and DEFs, a calculation of photon fluence at each site is required. The ICRP reference phantom defines bone components with their volume and elemental composition, and PHITS reads this information with unique IDs for the regions and materials, which are referred to as cell ID and material ID, respectively. PHITS subsequently implements photon-transport simulation, and calculates the photon fluence at each site designated by the cell ID using a track length estimator.

(C 33) The track length estimator is defined at '[T-track]' section in the PHITS input. Within a single '[T-track]' section, multiple evaluation cites can be specified using cell IDs. The energy bins for scoring of the photons are also set in this section. If the parameter 'unit' is set to 2, photon fluence is selected as an output in units of photons MeV^{-1} cm^{-2} $source^{-1}$, which can be used as the photon fluence Φ_E in Eqs C.7–C.10 given above. The fluence is then used to compute the AM and endosteum doses via post-processing. The post-processing method has the advantage that, in case of eventual future revisions of DRF and DEF values, the skeletal doses could be re-evaluated using the photon fluence without recalculation by a Monte Carlo simulation.

(C 34) In addition, PHITS has the ability to calculate AM and endosteum doses 'on-the-fly'. The advantage of 'on-the-fly' processing is that the photon energy is known precisely with no information lost due to energy binning. DRF and DEF values as a function of photon energy are given for each material specified by material ID in the '[Multiplier]' section of the PHITS input. The method used to interpolate between data points, such as log-log or linear-linear interpolation, is also described in this section. These multiplier data are called by the multiplier subsection in the '[T-track]' section, and the photons scored by the track length estimator are multiplied by a factor corresponding to their energy.

(C 35) By using the 'on-the-fly' processing feature of PHITS and setting the photon-fluence unit to photons cm^{-2} $source^{-1}$ without energy binning in the '[T-track]' section, the track length estimator directly evaluates the doses at

the bone sites in units of Gy. The total AM and endosteum doses for the whole skeleton can then be obtained via post-processing by summing the doses at each site, considering the weight fraction of the AM and endosteum.

C.3.3. Geant4

(C 36) While the MCNP and PHITS codes provide specific functions (e.g. DE/DF cards in MCNP and Multiplier in PHITS) that can implement the DRF 'on-the-fly' calculations combined with the respective track length estimators (e.g. 'F4' tally in MCNP and '[T-track]' in PHITS), specific functions for DRF implementation are not provided in the Geant4 code (Agostinelli et al., 2003). Geant4 users should, therefore, develop their own functions. As an example, the approach used for the implementation of DRFs in the Geant4 code for the development of the dose coefficients presented in this publication is explained in this section.

(C 37) For implementation of the DRFs, a new Geant4 class was derived from the G4VPrimitiveScorer class, which is the abstract class of all score classes pre-defined in the Geant4 code such as G4PSTrackLength, G4PSCellFlux, G4PSDoseDeposit, etc. The main purpose of the new class is to modify the 'ProcessHits' function, which is the key member function originated from the G4VPrimitiveScorer class in order to implement the DRF 'on-the-fly' calculations. The ProcessHits function is invoked when the current step of a transported particle occurs in the volume of the reference phantom which was previously assigned as a sensitive detector volume in the DetectorConstruction class for the phantom implementation (derived from G4VUserDetectorConstruction class).

(C 38) The ProcessHits function first checks if the particle for the step is a photon by calling the GetDefinition function in the GetTrack function in the G4Step class. If this is not the case, it returns FALSE to quit the ProcessHit function and moves to the next step. If yes, the ProcessHit function subsequently checks if the step occurs in one of the spongiosa regions by calling and checking the voxel ID. If no, it returns FALSE again, but if yes, the ProcessHit function calculates a step length (the distance between the pre-step point and post-step point) by calling the GetStepLength function in the G4Step class. Next, the calculated step length is divided by the volume of the corresponding spongiosa region to derive the photon fluence. The fluence at the given photon energy is multiplied by the corresponding DRF value, determined using the log-log interpolation, to calculate the absorbed dose to the red bone marrow and endosteum, respectively. Finally, the calculated values of the skeletal doses for each step are summed using the G4THitsMap class.

(C 39) It is worth noting that the DRF values show linear characteristics under log-log fitting. Thus, it is more reasonable to get a value via a log-log interpolation, rather than a single log interpolation, especially when extrapolating beyond the provided energy bins (Lee et al., 2006).

C.3.4. EGSnrc

(C 40) The implementation of DRFs in EGSnrc (Kawrakow et al., 2009), or more specifically in its component DOSxyznrc, requires some non-trivial changes in the user code, as there is no pre-defined macro to compute the photon fluence in an organ or tissue. With user codes being very different, an approximate scheme is provided below, rather than specific code fragments.

(C 41) The basis to implement DRFs in EGSnrc is to compute the fluence in the different skeletal regions using an estimator such as the collision-density or track-length estimator described, for example, in Annex D of *Publication 116* (ICRP, 2010). The most suitable position to implement any of these estimators is using AUSGAB (an essential subroutine programmed by the user) connected with suitable IAUSFL elements set to 1.

(C 42) A major challenge in EGSnrc in order to use any of these fluence estimators is the specific implementation of bound Compton scattering. EGSnrc uses Compton cross sections for free electrons (i.e. Klein-Nishina cross sections). To account for bound electrons with discrete energy levels, sampled Compton interactions resulting in invalid energy transfers are rejected without being resampled. Therefore, these signals cannot be used without caution to determine the end of a particle track, as required by the track-length estimator, or to count the interactions, as required by the collision-density estimator. Therefore, using those flags that lead to a call of AUSGAB before an interaction is made is potentially not useful. The preferred method seems to be setting those IAUSFL elements that lead to a call of AUSGAB after an interaction takes place, as rejected Compton interactions can be identified (by NP=NPold). This requires, however, more book-keeping of the particle history (e.g. to provide the photon energy before an interaction and also after the interaction took place).

(C 43) Furthermore, the rejection method for bound Compton scattering also means that the linear-attenuation coefficients used by EGSnrc itself for the transport are different from the actual linear-attenuation coefficients, and thus cannot be used for the collision-density fluence estimator. Thus, for this fluence estimator, the actual linear-attenuation coefficients have to be obtained from other sources [e.g. XCOM/NIST (Berger et al., 1998)].

(C 44) With the fluence available from any suitable estimator, the computation of the red bone marrow and endosteum dose is performed easily by multiplying it by the appropriate DRF value.

C.4. References

Agostinelli, S., Allison, J., Amako, K., et al., 2003. Geant4 – a simulation toolkit. Nucl. Instrum. Methods Phys. Res. A. 506, 250–303.

Berger, M.J., Hubbell, J.H., Seltzer, S.M., et al., 1998. NIST Standard Reference Database 8 (XGAM). NIST, PML, Radiation Physics Division, Gaithersburg, MD.

Cristy, M., 1981. Active bone marrow distribution as a function of age in humans. Phys. Med. Biol. 26, 389–400.

Cullen, D.E., Hubbell, J.H., Kissel, L., 1997. EPDL97: the Evaluated Photon Data Library, '97 Version. Lawrence Livermore National Laboratory, Livermore, CA.

Hirayama, H., Namito, Y., Bielajew, A.F., Wilderman, S.J., Nelson, W.R., 2005. The EGS5 Code System. SLAC-R-730 and KEK Report 2005-8. High Energy Accelerator Research Organization, Ibaraki.

Hough, M., Bolch, W.E., 2007. A skeletal reference dosimetry model for the 40-year male. Health Phys. 93(S1), S46.

Hough, M., Johnson, P., Rajon, D., Jokisch, D., Lee, C., Bolch, W., 2011. An image-based skeletal dosimetry model for the ICRP reference adult male – internal electron sources. Phys. Med. Biol. 56, 2309–2346.

ICRP, 2002. Basic Anatomical and Physiological Data for Use in Radiological Protection Reference Values. ICRP Publication 89. Ann. ICRP 32 (3–4).

ICRP, 2007. The 2007 Recommendations of the International Commission on Radiological Protection. ICRP Publication 103. Ann. ICRP 37(2–4).

ICRP, 2009. Adult reference computational phantoms. ICRP Publication 110. Ann. ICRP 39(2).

ICRP, 2010. Conversion coefficients for radiological protection quantities for external radiation exposures. ICRP Publication 116. Ann. ICRP 40(2–5).

ICRP, 2020a. Paediatric computational reference phantoms. ICRP Publication 143. Ann. ICRP 49(1).

ICRP, 2020b. Dose coefficients for external exposures to environmental sources. ICRP Publication 144. Ann. ICRP 49(2).

ICRP, 2023. Specific absorbed fractions for reference paediatric individuals. ICRP Publication 155. Ann. ICRP 52(4).

ICRU, 1989. Tissue Substitutes in Radiation Dosimetry and Measurement. ICRU Report 44. International Commission on Radiation Units and Measurements, Bethesda, MD.

ICRU, 1997. Dosimetry of External Beta-rays for Radiation Protection. ICRU Report 56. International Commission on Radiation Units and Measurements, Bethesda, MD.

ICRU, 2011. Fundamental Quantities and Units for Ionizing Radiation. ICRU Report 85a. International Commission on Radiation Units and Measurements, Bethesda, MD.

Johnson, P.B., Bahadori, A.A., Eckerman, K.F., Lee, C., Bolch, W.E., 2011. Response functions for computing absorbed dose to skeletal tissues from photon irradiation – an update. Phys. Med. Biol. 56, 2347–2365.

Kawrakow, I., Mainegra-Hing, E., Rogers, D.W.O., et al., 2009. The EGSnrc Code System: Monte Carlo Simulation of Electron and Photon Transport. PIRS Report 701. National Research Council of Canada, Ottawa.

Lee, C., Lee, C., Shah, A.P., Bolch, W.E., 2006. An assessment of bone marrow and bone endosteum dosimetry methods for photon sources. Phys. Med. Biol. 51, 5391–5407.

Pafundi, D., Rajon, D., Jokisch, D., Lee, C., Bolch, W., 2010. An image-based skeletal dosimetry model for the ICRP reference newborn: internal electron sources. Phys. Med. Biol. 55, 1785–1814.

Pelowitz, D.B. (ed.), 2013. Initial MCNP6 Release Overview – MCNP6 version 1.0. LA-UR-13-22934, Los Alamos National Laboratory, Los Alamos, NM.

Sato, T., Iwamoto, Y., Hashimoto, S., et al., 2023. Recent improvements of the particle and heavy ion transport code system – PHITS version 3.33. J. Nucl. Sci. Technol. 61, 127–135.

Wayson, M.B., Lee, C., Sgouros, G., Treves, S.T., Frey, E., Bolch, W.E., 2012. Internal photon and electron dosimetry of the newborn patient – a hybrid computational phantom study. Phys. Med. Biol. 57, 1433–1457.
Werner, C.J., Armstrong, J.C., Brown, F.B., et al., 2017. MCNP User's Manual – Code Version 6.2. Los Alamos National Laboratory, Los Alamos, NM.
Werner, C.J., Bull, J.S., Solomon, C.J., et al., 2018. MCNP Version 6.2 Release Notes. Los Alamos National Laboratory, Los Alamos, NM.
Zankl, M., Eakins, J., Gómez Ros, J-M., Huet, C., 2021. The ICRP recommended methods of red bone marrow dosimetry. Radiat. Measur. 146, 106611.

ANNEX D. DESCRIPTION OF ELECTRONIC SUPPLEMENT

(D 1) The electronic supplement, which is available for download with this publication, is organised into 26 main folders, each for a projection considered (see Section 3 for projections). Each main folder has two subfolders: one tabulates the dose coefficients in 1-keV bins as dose per fluence at 1 m from the source; and the other provides dose coefficients for selected bremsstrahlung spectra (see Section 5 for spectra considered). These coefficients are reported as doses per KAP (P_{KA}), per air kerma free-in-air at 1-m distance from the source, and per fluence at 1-m distance from the source. All data are given in Microsoft Excel spreadsheets as well as in ASCII files.

(D 2) Data are given separately for each age group and for male and female phantoms. Organ absorbed dose coefficients are given for the following organs: red bone marrow, colon, lung, stomach, breast, ovaries or testes (gonads), bladder, oesophagus, liver, thyroid, skeletal endosteum, brain, salivary glands, skin, and remainder tissues (adrenals, extrathoracic region, gallbladder, heart, kidneys, lymphatic nodes, muscle, oral mucosa, pancreas, prostate, small intestine, spleen, thymus, uterus/cervix, and lens of the eye). The lens of the eye is included as it is an organ of interest, being sensitive to radiation with a risk of causing cataracts.

(D 3) For the data of monoenergetic photons as well as the data for the selected spectra, coefficients for $\sum w_T \times H_T^M$ and $\sum w_T \times H_T^F$ are provided (denoted $E_{103,\#\#M}$ and $E_{103,\#\#F}$ as described in Para. 20 and Annex A). For the selected spectra, values of $(\sum w_T \times H_T^M + \sum w_T \times H_T^F)/2$ are also given (as an effective dose coefficient). Note that the latter are compatible with the definition of effective dose as per *Publication 103* (ICRP, 2007) if the user chooses the same value of normalisation quantity for male and female examinations.

(D 4) All dose coefficients can also be accessed via two excel-based viewers: one returns the dose coefficients in 1-keV energy bins; a second one returns the dose coefficients for the selected bremsstrahlung spectra.

(D 5) The dose coefficients for the selected bremsstrahlung spectra are also available as part of the ICRP dose viewer, which is an application freely available via the Apple App Store and the Google Play Store.

ACKNOWLEDGEMENTS

This is the first publication from ICRP Task Group 113 on Reference Organ and Effective Dose Coefficients for Common Diagnostic X-ray Imaging Examinations. Task Group 113 is a joint Task Group of ICRP Committees 2 and 3, and was established in May 2019. This publication deals with exposures in diagnostic radiography; upcoming publications of Task Group 113 will cover CT, paediatric diagnostic fluoroscopy, and fluoroscopically guided interventions. This is the first time that ICRP has published reference dose coefficients for these types of exposures.

ICRP thanks all those involved in the development of this publication for their hard work and dedication. ICRP gratefully acknowledges Martin Andersson for including the dose coefficients of the spectral beams to the ICRP dose viewer; Thomas Otto, ICRU Observer to Committee 2, for reviewing the manuscript; and Daiki Satoh for his input on skeletal dosimetry.

Task Group 113 members ***(2019–2025)***

N. Petoussi-Henss (Co-Chair)
D. Sutton (Co-Chair)
K. Applegate
W.E. Bolch
D. Borrego
J.W. Choi
K. Griffith
H. Han
J.T.M. Jansen
K. Jones
C.H. Kim
P. Kunert
J. Li
C. Lee
Y. Lee
S. Mann
E. Marshall
D.L. Miller
D. Satoh
H. Schlattl
I. Sechopoulos
W. Smither
K. Wunderle
Y.S. Yeom
M. Zankl

Task Group 113 technical secretary

B. Brevitt

Committee 2 critical reviewer

A. Giussani

Committee 3 critical reviewer

C.J. Martin

Main Commission critical reviewers

D. Laurier
W. Rühm

Editorial members

C.H. Clement (Scientific Secretary, CEO and *Annals of the ICRP* Editor-in-Chief)
T. Yasumune (Assistant Scientific Secretary and *Annals of the ICRP* Associate Editor)
K. Nakamura (Assistant Scientific Secretary and *Annals of the ICRP* Associate Editor)
Q. Fan (Assistant Scientific Secretary and *Annals of the ICRP* Associate Editor)

Committee 2 members during preparation of this publication

(2017–2021)

J.D. Harrison (Chair)
F. Paquet (Vice-Chair)
W.E. Bolch (Secretary)
V. Berkovski
E. Blanchardon
A. Giussani
D. Jokisch
C.H. Kim
R. Leggett
J. Li
M.A. Lopez
N. Petoussi-Henss
T. Sato
T. Smith
A. Ulanowski
F. Wissmann

(2021–2025)

F. Bochud (Chair)
F. Paquet (Vice-Chair)
M.A. Lopez (Secretary)
M. Andersson
V. Berkovskyy
D. de Souza Santos
A. Giussani
D. Jokisch
C.H. Kim
M.S. Kulkarni
S. Lamart
C. Lee
J. Li
J.W. Marsh
N. Petoussi-Henss
T. Sato
T. Smith
A. Ulanowski
Y.S. Yeom

Committee 2 emeritus members

K. Eckerman

Committee 3 members during the preparation of this publication

(2017–2021)

K. Applegate (Chair)
C.J. Martin (Vice-Chair)
M. Rehani (Secretary)
J. Alsuwaidi
M. Bourguignon
M.C. Cantone
S. Demeter
M. Hosono
K. Kang
R. Loose
J. Marti-Climent
Y. Niu
C. Ruebe
W. Small
D.G. Sutton
L. Van Bladel

(*2021–2025*)

K. Applegate (Chair)	A. Isambert	C. Ruebe
C.J. Martin (Vice-Chair)	M. Kortesniemi	W. Small
D.G. Sutton (Secretary)	A. Magistrelli	Å. Søvik
M.C. Cantone	M. Mahesh	I. Thierry-Chef
J. Damilakis	J. Marti-Climent	I. Williams
M. Hosono	J.C. Paeng	W. Zhuo

Committee 3 emeritus members

S. Mattsson	M.M. Rehani	M. Rosenstein

Main Commission members at the time of approval of this publication

Chair: W. Rühm, *Germany*
Vice-Chair: S. Bouffler, *UK*
Scientific Secretary and CEO: C.H. Clement, *Canada*; sci.sec@icrp.org*

K.E. Applegate, *USA*
F. Bochud, *Switzerland*
K.W. Cho, *Korea*
G. Hirth, *Australia*
M. Kai, *Japan*
D. Laurier, *France*
S. Liu, *China*
S. Romanov, *Russia*
T. Schneider, *France*

Emeritus members
C. Cousins, *UK*
J. Lochard, *France*
F.A. Mettler Jr, *USA*
R.J. Pentreath, *UK*
R.J. Preston, *USA*
E. Vañó, *Spain*

*Although formally not a Main Commission member since 1988, the Scientific Secretary is an integral part of the Main Commission.

Finally, thank you very much to all organisations and individuals who took the time to provide comments on the draft of this publication during the consultation process.